



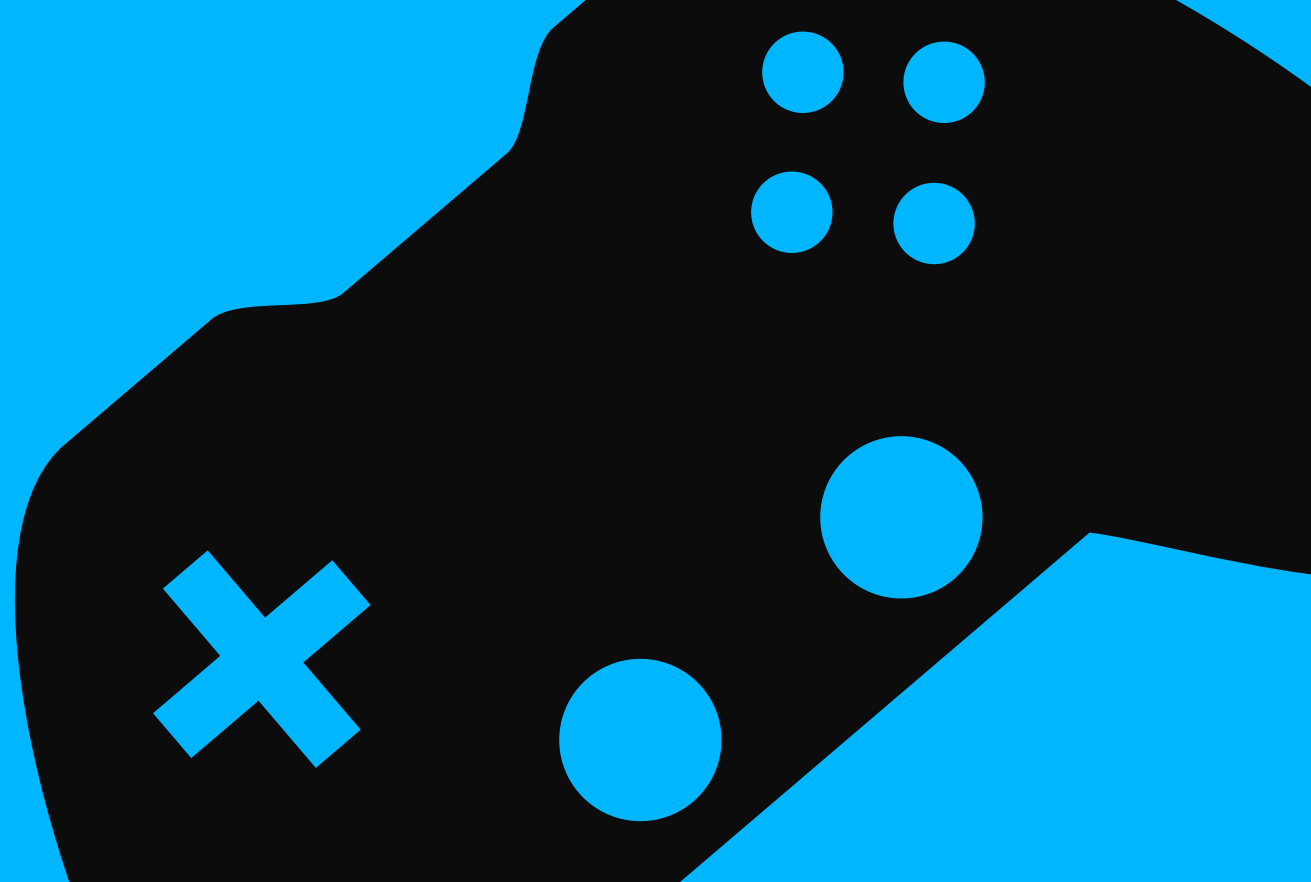
U S E R M A N U A L

B E T A V E R S I O N

INDEX

INDEX (2)
CONTROLS - EDITOR (3)
CONTROLS - PLAYMODE (7)
CONTROLS - MULTIPLAYER (9)
HOTBAR (10)
HIERARCHY - GROUPS (11)
HIERARCHY - LIST (7)
ELEMENT TAB (13)
TEXTURES LIBRARY (15)
COLOR CREATOR (16)
TRANSFORM TAB (17)
EDITABLE VALUES (18)
PROPERTIES TAB (19)
PROPERTIES BROWSER (20)
PROPERTIES CONTROLLERS (21)
PROPERTIES (22)
ANIMATION STUDIO (27)
SPECIAL OBJECTS (30)
TRIGGERS (37)
PLAYER EDITOR (38)
CAMERA EDITOR (41)
TRANSFORM GIZMO (44)
EDITOR SETTINGS (47)
LEVEL SETTINGS (50)
MULTIPLAYER (52)

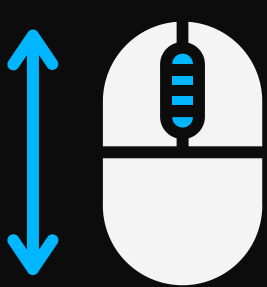
CONTROLS - EDITOR



MOVEMENT

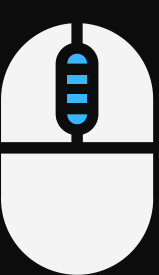
W A S D ----- **MOVE**

HOLD  ----- **LOOK**

 ----- **ZOOM**

E ----- **ASCEND**

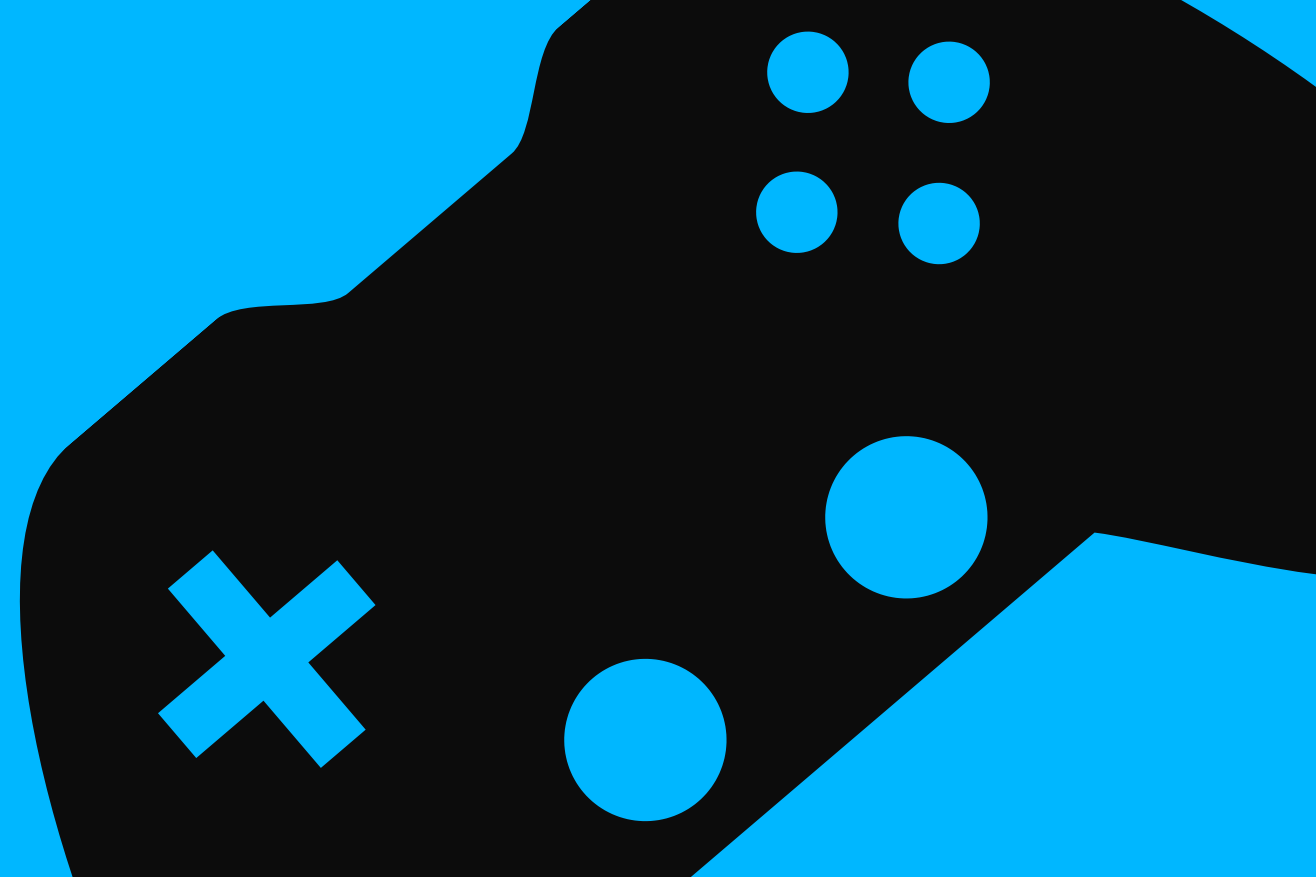
Q ----- **DESCEND**

HOLD  ----- **DRAW
VIEW**

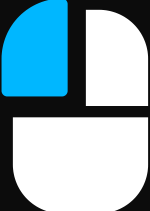
HOLD **SHIFT** ----- **INCREASED
MOVEMENT
SPEED**

HOLD **CTRL** ----- **DECREASED
MOVEMENT
SPEED**

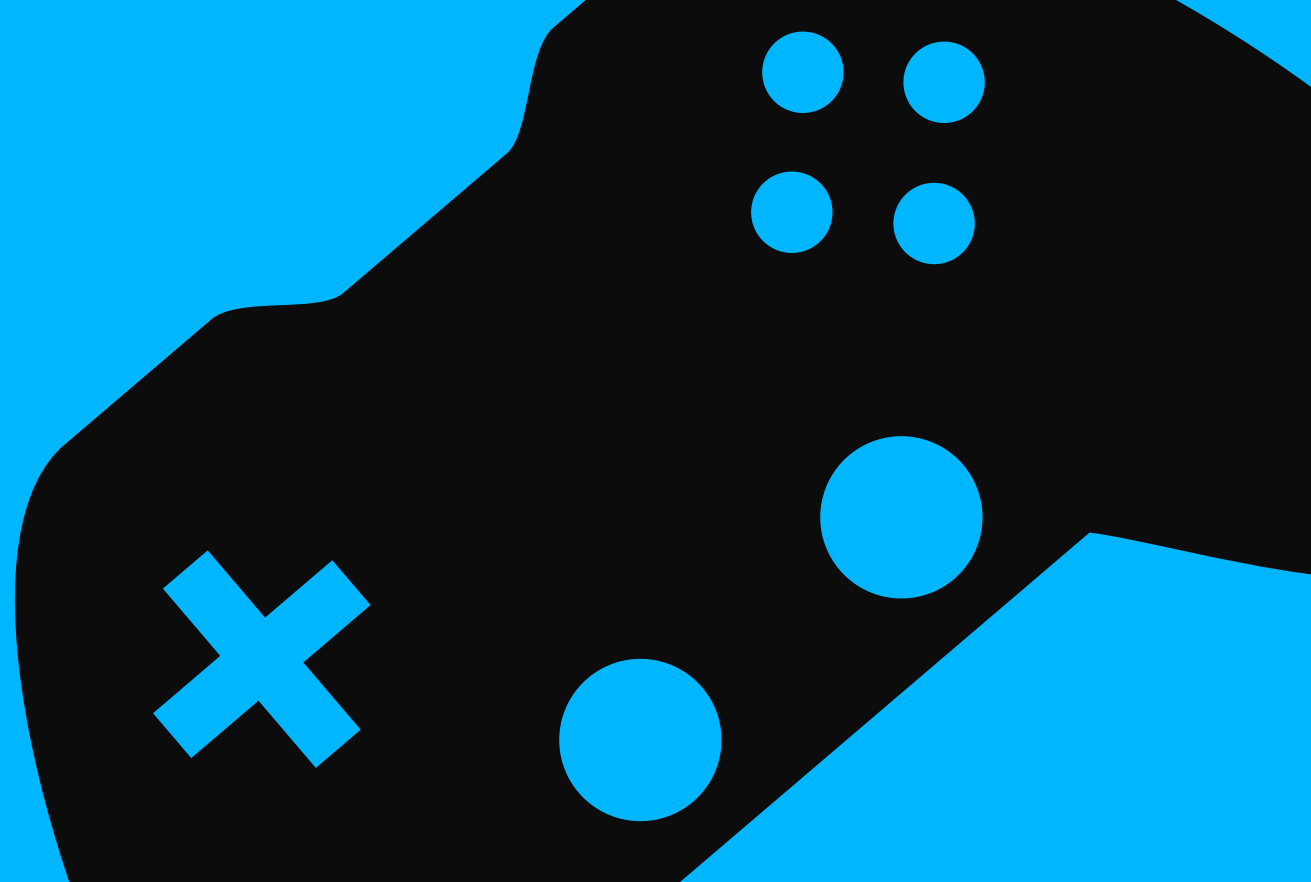
CONTROLS - EDITOR



OBJECT SPECIFIC

	ABOVE NON SELECTED OBJECT	-----	SELECT OBJECT
	ABOVE NON SELECTED OBJECT	SHIFT -----	ADD OBJECT TO SELECTION
	ABOVE SELECTED OBJECT	SHIFT -----	REMOVE OBJECT FROM SELECTION
DEL	-----		DELETE SELECTED OBJECT(S)
CTRL	C	-----	COPY SELECTED OBJECT(S)
CTRL	V	-----	PASTE COPIED OBJECT(S)
CTRL	X	-----	CUT SELECTED OBJECT(S)
CTRL	D	-----	DUPLICATE SELECTED OBJECT(S)
CTRL	Z	-----	UNDO LAST TRANSFORM
HOLD		-----	DRAW SELECTION BOX

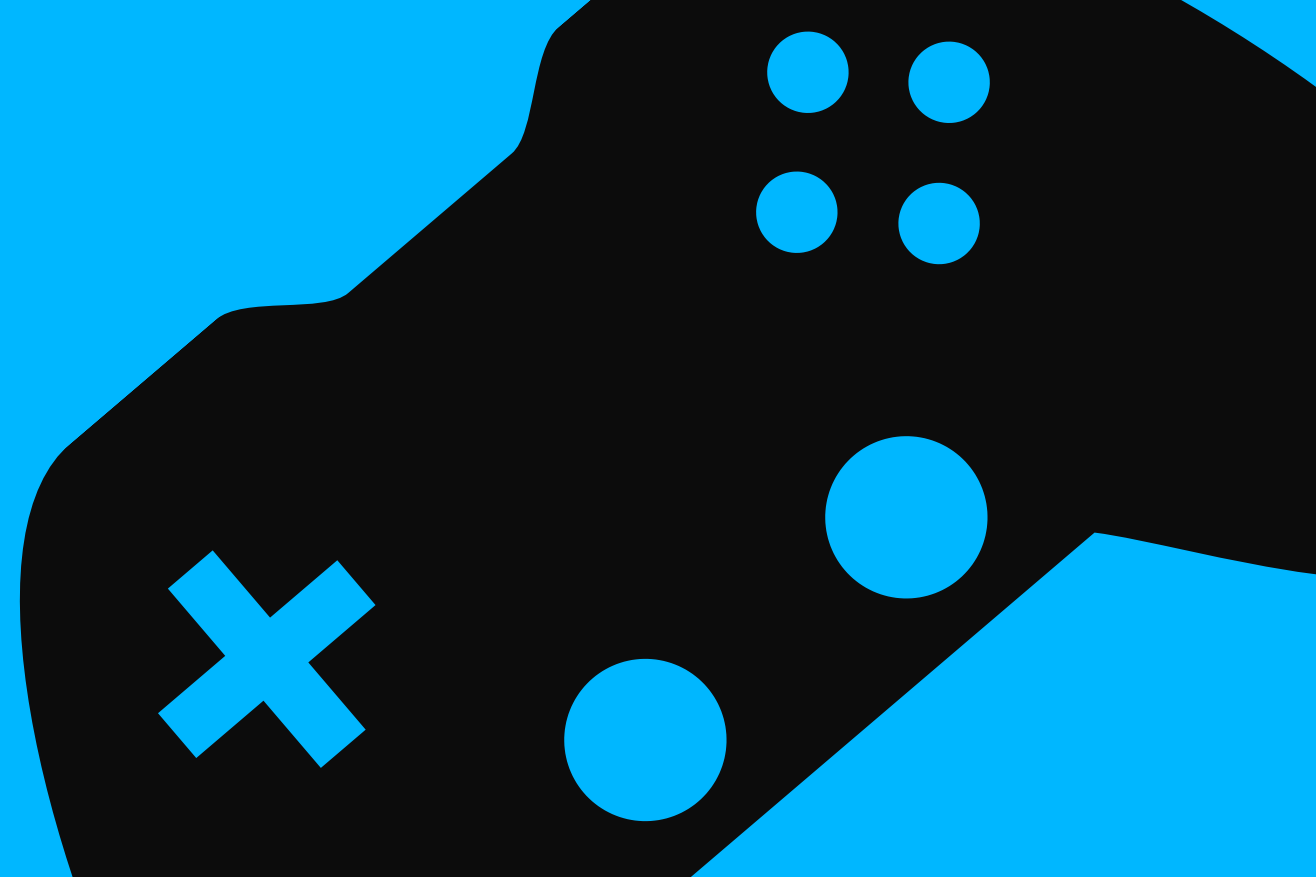
CONTROLS - EDITOR



GENERAL SHORTCUTS

	-----	OPEN MENU
	-----	USE SNAPPING TRANSFORM
	-----	SNAP TO CAMERA VIEW
	-----	INCREASE/DECREASE VALUE BY 1.0
	-----	INCREASE/DECREASE VALUE BY 10
	-----	INCREASE/DECREASE VALUE BY 0.1
	-----	NEXT/PREVIOUS HIERARCHY PAGE
	-----	FIRST/LAST HIERARCHY PAGE
	-----	FOCUS ON SELECTED OBJECT

CONTROLS - EDITOR



GENERAL SHORTCUTS

1	-----
2	-----
3	-----
4	-----
5	-----
6	-----
7	-----
8	-----

TRANSFORM
POSITION MODE

TRANSFORM
ROTATION MODE

TRANSFORM
SCALE MODE

TRANSFORM ALL
MODE

TRANSFORM
WORLD SPACE

TRANSFORM
LOCAL SPACE

TRANSFORM
PIVOT

TRANSFORM
CENTER

CAMERA UTILITIES

F1	-----
F2	-----

TOGGLE CAMERA
PREVIEW MODE

SNAP TO CAMERA
VIEW

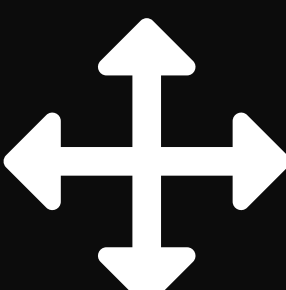
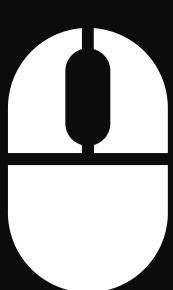
CONTROLS - PLAYMODE

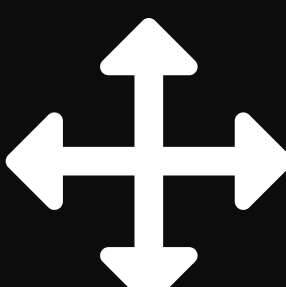
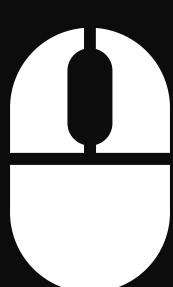


MOVEMENT

W A S D ----- **MOVE**

SPACE ----- **JUMP**

  ----- **LOOK (FIRST PERSON)**

  ----- **TURN (THIRD PERSON)**

SHIFT ----- **SPRINT**

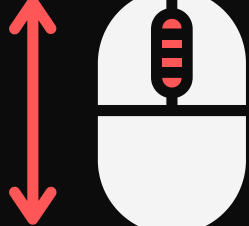
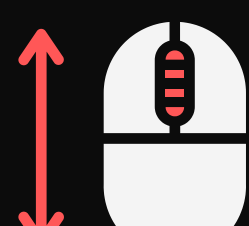
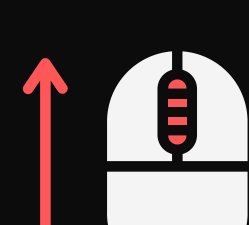
CTRL ----- **CROUCH**

HOLD **TAB** ----- **EDIT MODE**

CONTROLS - PLAYMODE



OBJECT SPECIFIC - WHILE IN EDIT MODE (HOLDING TAB)

	ABOVE NON SELECTED OBJECT		SELECT OBJECT
	ABOVE NON SELECTED OBJECT	SHIFT	ADD OBJECT TO SELECTION
	ABOVE SELECTED OBJECT	SHIFT	REMOVE OBJECT FROM SELECTION
	ABOVE ANY NUMERICAL VALUE		INCREASE/DECREASE VALUE BY 1.0
	ABOVE ANY NUMERICAL VALUE	SHIFT	INCREASE/DECREASE VALUE BY 10
	ABOVE ANY NUMERICAL VALUE	CTRL	INCREASE/DECREASE VALUE BY 0.1

CONTROLS - MULTIPLAYER



COMMUNICATION

T

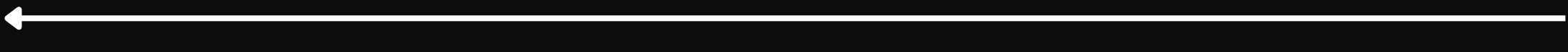
OR

ENTER

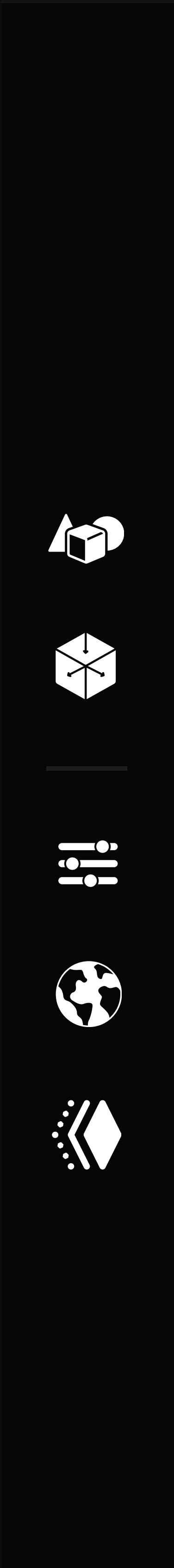
OPEN TEXT CHAT

V

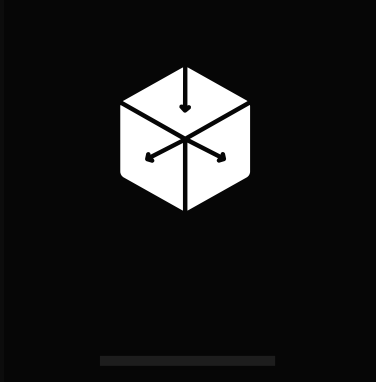
VOICE CHAT
(PUSH TO TALK)



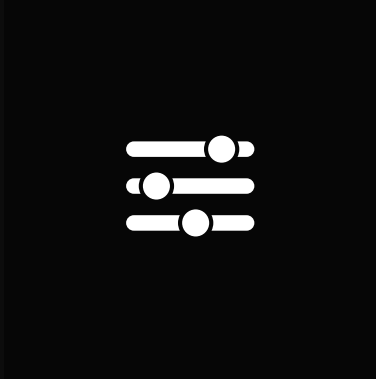
MENU



OBJECTS BROWSER



SPECIAL OBJECTS
BROWSER



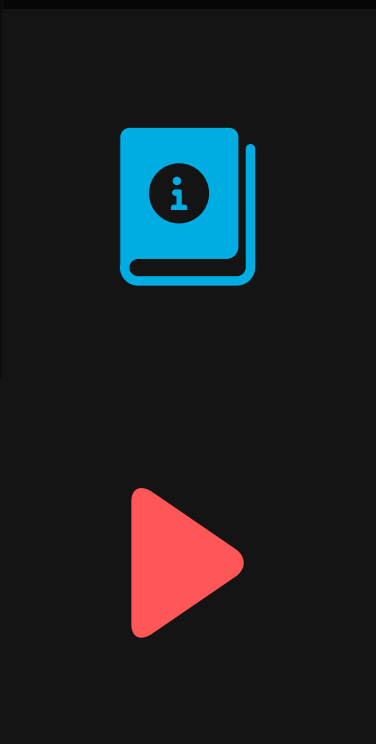
EDITOR SETTINGS



LEVEL SETTINGS



ANIMATION LAB



OPEN MANUAL



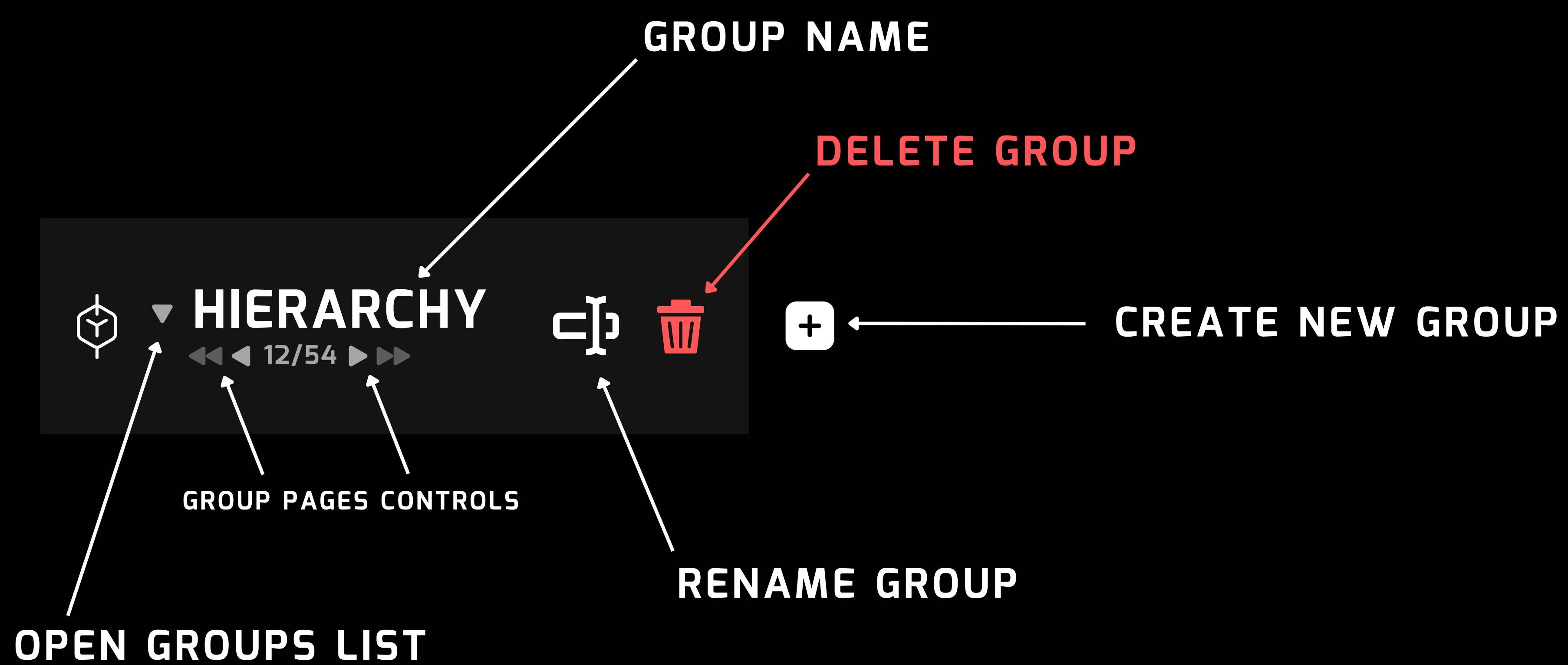
SAVE & START LEVEL

HIERARCHY - GROUPS



GROUPS PANEL

- **GROUPS** HELP KEEP YOUR LEVEL ORGANIZED.
- EVERY PROJECT STARTS WITH A DEFAULT GROUP CALLED **HIERARCHY**.
- SELECTING A GROUP FILTERS THE HIERARCHY, SHOWING ONLY THE OBJECTS **ASSIGNED TO THAT GROUP**.
- EACH PROJECT CAN HAVE **UP TO 99 GROUPS**, AND EACH GROUP CAN CONTAIN **ANY NUMBER OF OBJECTS**.
- THE DEFAULT GROUP HIERARCHY **CANNOT BE DELETED OR RENAMED**, AND ALL NEW OBJECTS ARE ASSIGNED TO IT BY DEFAULT.
- IF A GROUP IS **DELETED**, ALL OF ITS OBJECTS ARE AUTOMATICALLY MOVED BACK TO THE DEFAULT HIERARCHY GROUP.
- ASSIGNING OBJECTS TO GROUPS IS EXPLAINED IN **ELEMENT TAB** ON **PAGE 13**.

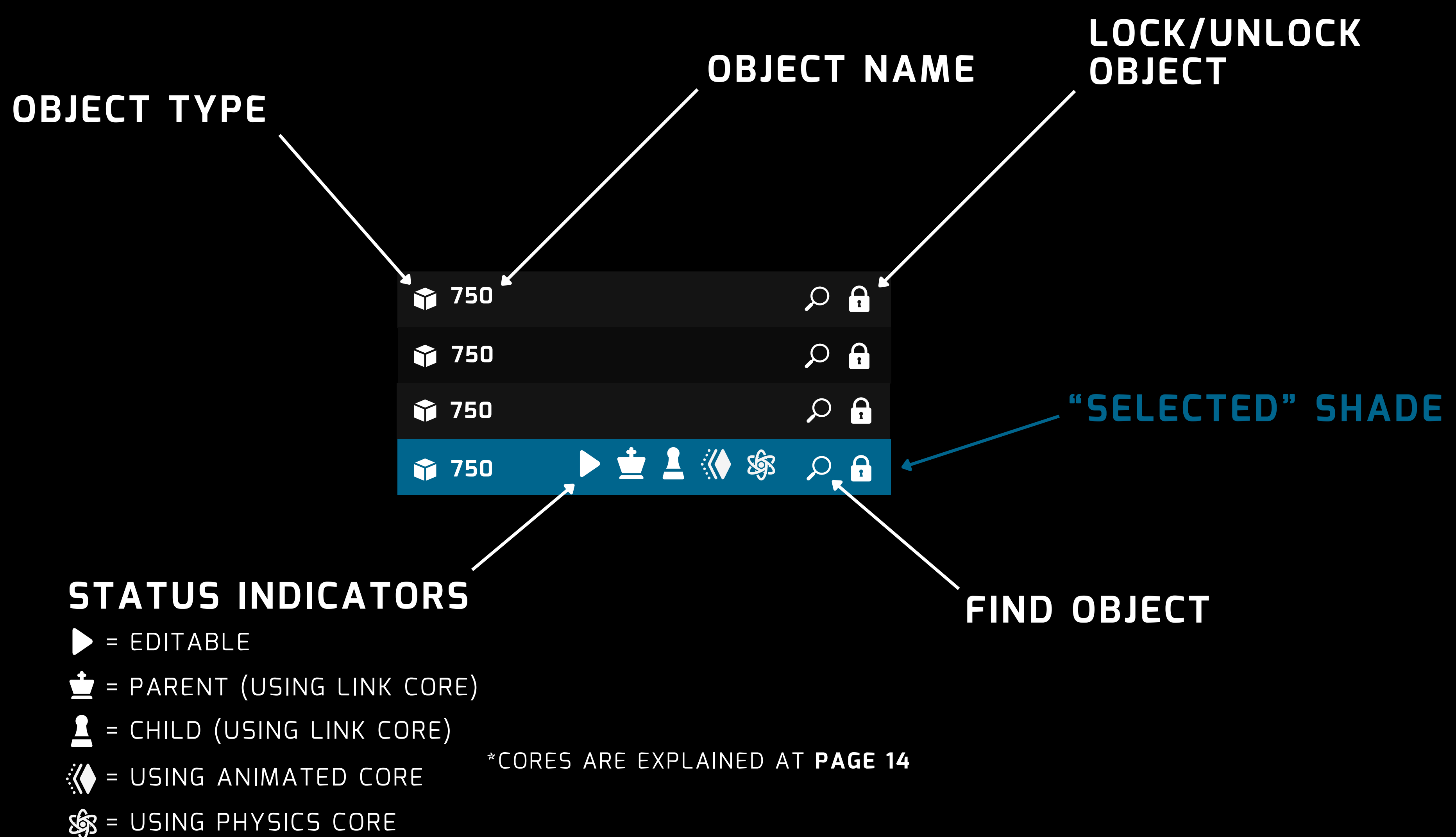


HIERARCHY - LIST



HIERARCHY LIST

- THE **HIERARCHY LIST** SHOWS ALL OBJECTS INSIDE THE **SELECTED GROUP**.
- IF A GROUP CONTAINS MORE THAN **35 OBJECTS**, THE LIST IS **SPLIT INTO PAGES**.
- USE THE **PAGE CONTROLS** IN THE **GROUPS PANEL** OR THE **ARROW KEYS** TO NAVIGATE THEM.
- SELECTING AN OBJECT **HIGHLIGHTS** ITS TAB WITH **YOUR PERSONAL COLOR**.
- IF THE SELECTED OBJECT IS ON ANOTHER PAGE, THE HIERARCHY LIST **AUTOMATICALLY JUMPS** TO THE CORRECT PAGE.
- HOLD **SHIFT** AND **CLICK** OBJECT TABS TO SELECT OR DESELECT MULTIPLE OBJECTS.
- **CLICKING** AN **ALREADY SELECTED** OBJECT TAB **HIDES** THE **TRANSFORM GIZMO** (**PAGE 44**) FOR THAT OBJECT.
- CLICK AN **OBJECT'S NAME** TO **RENAME** IT.



ELEMENT TAB



ELEMENT TAB

- AFTER AT LEAST ONE OBJECT HAS BEEN SELECTED, THREE TABS WILL APPEAR ON THE RIGHT SIDE OF YOUR SCREEN. THE FIRST ONE FROM THE TOP IS THE **ELEMENT TAB**.
- THE **ELEMENT TAB** IS DIVIDED INTO THREE SECTIONS, EACH ONE CONTROLLING A DIFFERENT ASPECT OF THE SELECTED OBJECTS.
- THE FIRST SECTION ALLOWS YOU TO CONTROL THE “**TRANSFORM MODIFIERS**” YOU’RE APPLYING TO THE OBJECT. **WITH “TRANSFORM MODIFIES” WE REFER TO ANYTHING THAT IN SOME WAY CHANGES THE VISIBLE CHARACTERISTICS OF AN OBJECT, LIKE IT’S SIZE, ROTATION AND POSITION IN THE WORLD.**
- THE SECOND SECTION ALLOWS YOU TO ASSIGN THE SELECTED OBJECT(S) TO A GROUP, OR REMOVE THEM FROM A GROUP THEY ALREADY BELONG TO. AS EXPLAINED ON **PAGE 11**, ALL OBJECTS ARE PART OF THE HIERARCHY GROUP BY DEFAULT.
- TO ASSIGN OBJECT(S) TO A GROUP, CREATE A GROUP FROM THE GROUPS HEADER (ALSO EXPLAINED ON **PAGE 11**), THEN CLICK THE SMALL ARROW (▼) IN THE **ELEMENT TAB** AND SELECT THE GROUP YOU WANT TO MOVE THE OBJECT(S) TO.
- TO REMOVE OBJECT(S) FROM A GROUP, SIMPLY CLICK THE BUTTON WITH A BROKEN CHAIN IN THIS SECTION AND ALL THE TARGETS WILL BE MOVED BACK TO THE DEFAULT “HIERARCHY” GROUP.

EDIT

DELETE OBJECT(S) (points to trash icon)

CHANGE TRANSFORM MODES:

- ↕ = CHANGE POSITION
- ↻ = CHANGE ROTATION
- ⊞ = CHANGE SCALE
- * = SHOW ALL MODES
- 🌐 = APPLY TRANSFORMS WHILE IGNORING THE OBJECT'S OWN ORIENTATION
- 📦 = APPLY TRANSFORMS RELATIVELY TO OBJECT'S OWN ORIENTATION
- ⦿ = WHEN MULTIPLE OBJECTS ARE SELECTED, APPLY TRANSFORMS STARTING FROM A MEDIAN POINT BETWEEN ALL OF THEM
- 📏 = WHEN MULTIPLE OBJECTS ARE SELECTED, APPLY TRANSFORMS STARTING FROM EVERY SINGLE OBJECT'S CENTER

SHOW GROUPS LIST (points to GROUP HIERARCHY dropdown)

REMOVE OBJECT(S) FROM GROUP (points to broken chain icon)

TRANSFORM SPACE MODE

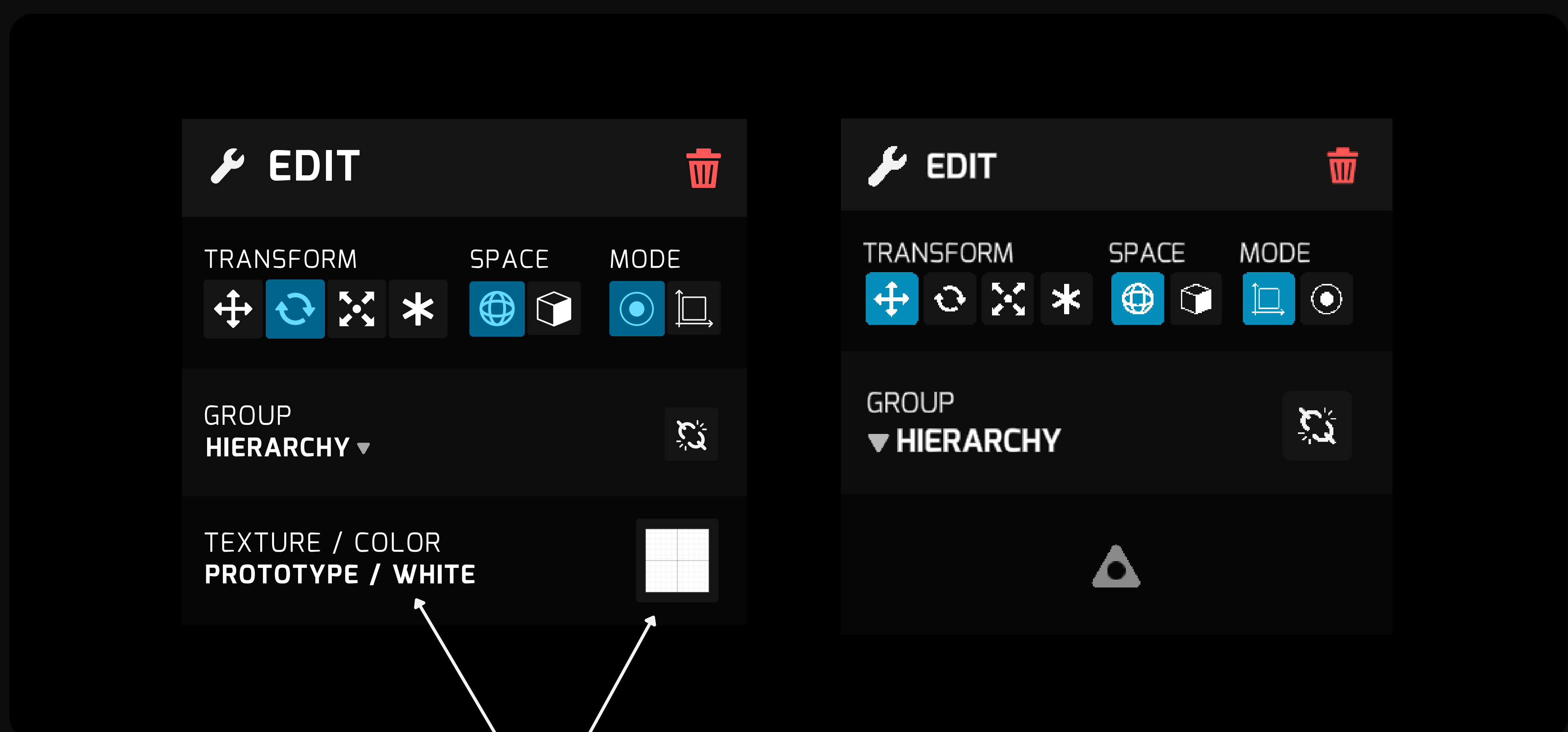
GROUP HIERARCHY ▼

TEXTURE / COLOR
PROTOTYPE / WHITE

ELEMENT TAB



- THE LAST SECTION OF THE **ELEMENT TAB** ALLOWS YOU TO CONTROL THE TEXTURE AND COLOR OF AN OBJECT, TO SUM IT UP, ITS VISUAL APPEARANCE.
- WITHIN PROJECT LEVELS, OBJECTS WHICH SUPPORT VISUAL MODIFICATIONS ALLOW **A COMBINATION OF A SOLID COLOR WITH A TEXTURE APPLIED ON TOP**. IF AN OBJECT DOES NOT ALLOW VISUAL MODIFICATIONS AN “ERROR” ICON WILL BLOCK THIS WHOLE SECTION.
- KEEP IN MIND THAT THE ELEMENT TAB WORKS ON **ALL** THE OBJECTS YOU HAVE SELECTED, THEREFORE, IF EVEN A SINGLE OBJECT AMONG ALL THE SELECTED ONES DOES NOT SUPPORT VISUAL MODIFICATIONS, THIS SECTION WILL BE BLOCKED AS PER BASIC MULTI-EDITING PRINCIPLES.



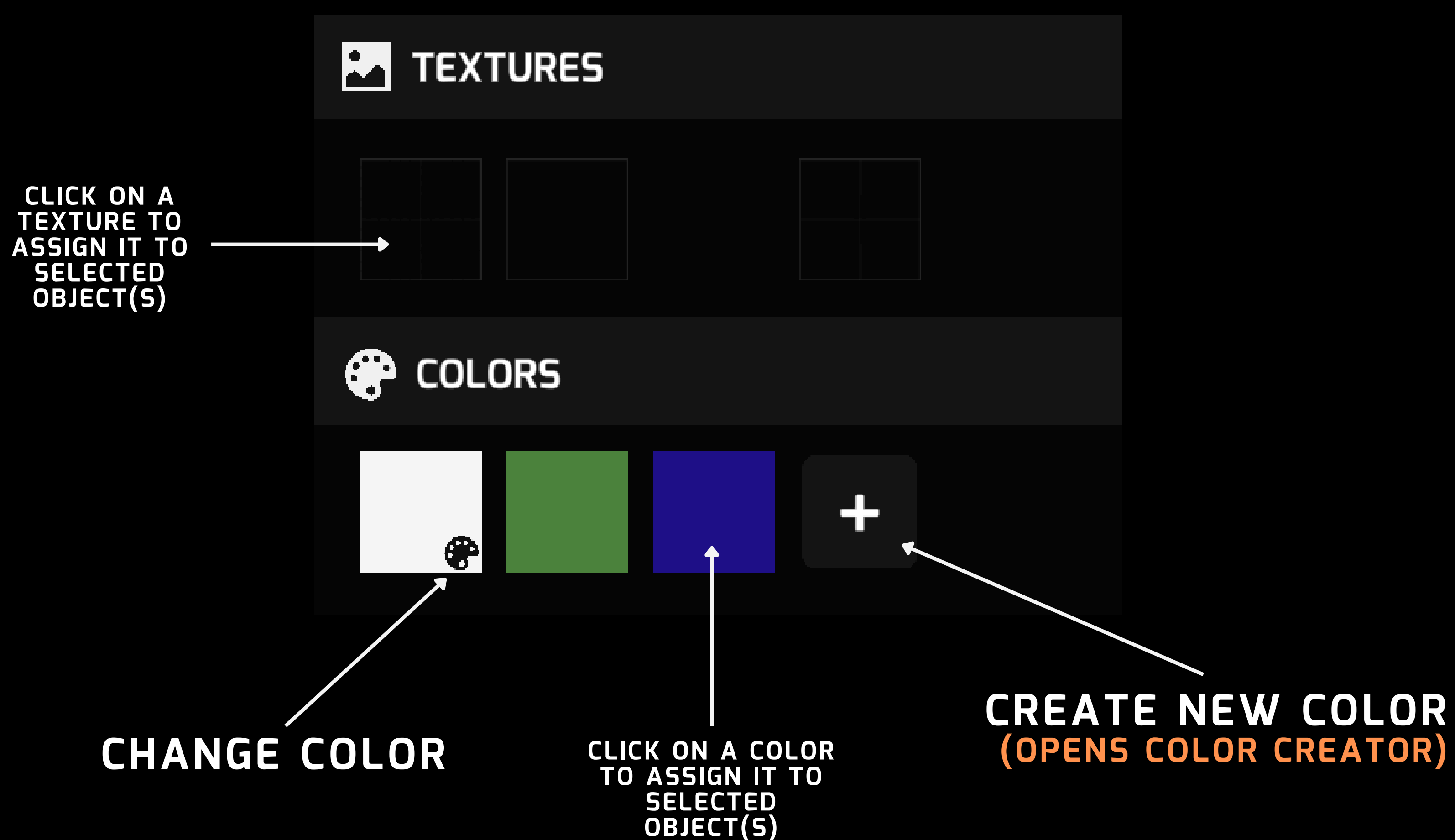
- THE LITTLE ICON ALL TO THE RIGHT IN THIS SECTION WILL DISPLAY THE COLOR AND TEXTURE OF THE OBJECT(S) THAT YOU HAVE CURRENTLY SELECTED, ALONG WITH THEIR NAMES.
- IF MULTIPLE OBJECTS ARE SELECTED AND THEY DO NOT ALL SHARE THE SAME TEXTURE AND COLOR, AN “UNDEFINED TEXTURE” PLACEHOLDER WILL BE DISPLAYED INSTEAD. THIS DOES NOT PREVENT YOU FROM CLICKING ON IT AND APPLYING A NEW COLOR AND OR TEXTURE TO ALL THE SELECTED OBJECTS AS EXPLAINED IN **PAGE 19**.

TEXTURES LIBRARY

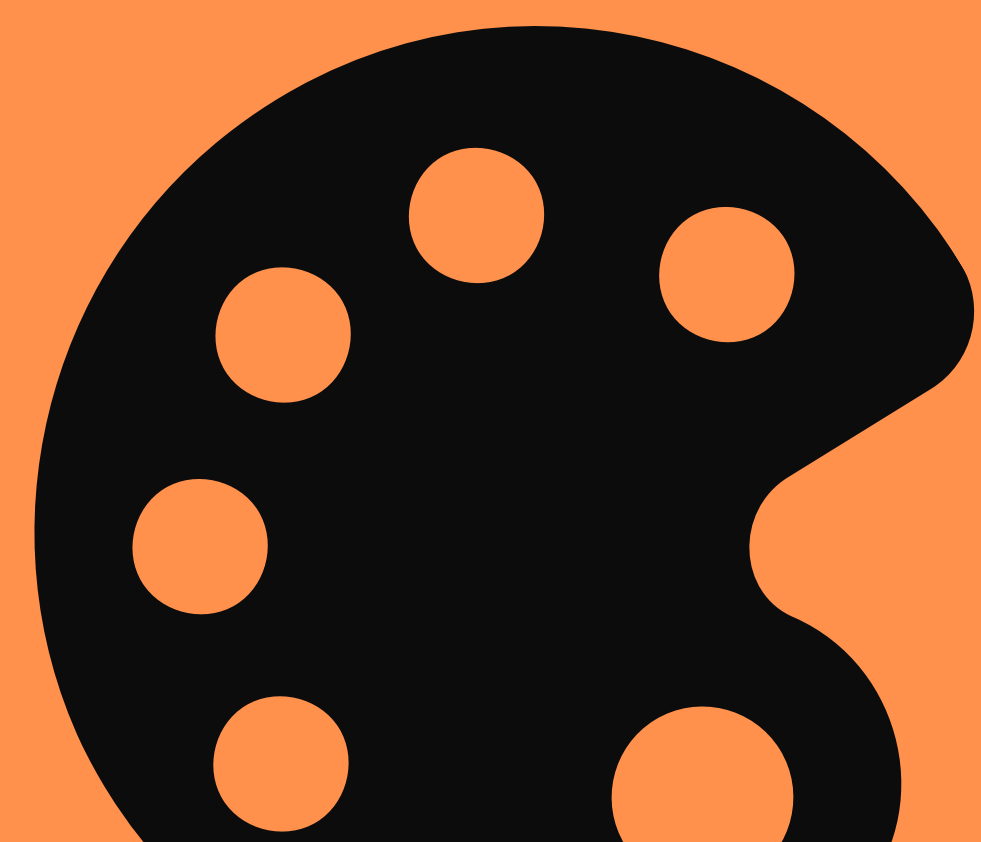


TEXTURES LIBRARY

- IF THE LAST SECTION OF THE ELEMENT TAB IS NOT BLOCKED, YOU CAN CLICK ON IT AND OPEN THE **TEXTURES LIBRARY**.
- HERE YOU CAN SEE A COLLECTION OF ALL THE AVAILABLE VISUAL MODIFIES THAT YOU CAN APPLY TO YOUR OBJECTS SPLIT INTO TWO SECTIONS; **TEXTURES AND COLORS**.
- BY SIMPLY **CLICKING ON A TEXTURE OR A COLOR** WITHIN THE LIBRARY PANEL YOU WILL APPLY SAID TEXTURE/COLOR TO ALL THE CURRENTLY SELECTED OBJECTS
- WHILE UPLOADING CUSTOM TEXTURES IS LIMITED TO DEVELOPER BUILDS OF PROJECT LEVELS DUE TO SECURITY REASONS, EVERYONE IS ALLOWED TO CREATE CUSTOM COLORS BY CLICKING ON THE (+) BUTTON AT THE END OF THE COLORS LIBRARY BY USING THE **COLOR CREATOR** WHICH WILL POP UP ON THE LEFT. ALL COLOR CREATORS WITHIN THE EDITOR ARE THE SAME AND THEY ARE FULLY EXPLAINED IN **PAGE 16**. YOU CAN HAVE UP TO 99 DIFFERENT COLORS IN A PROJECT.
- WHILE ABOVE A COLOR'S ICON, A LITTLE BUTTON WITH A COLOR PALETTE WILL APPEAR IN ITS BOTTOM RIGHT CORNER. BY CLICKING ON IT, THE COLOR CREATOR WILL OPEN UP AGAIN AND ALLOW YOU TO CHANGE THAT COLOR. ALL THE OBJECTS IN THE WORLD WHICH ARE USING THAT COLOR WILL BE CHANGED AUTOMATICALLY.

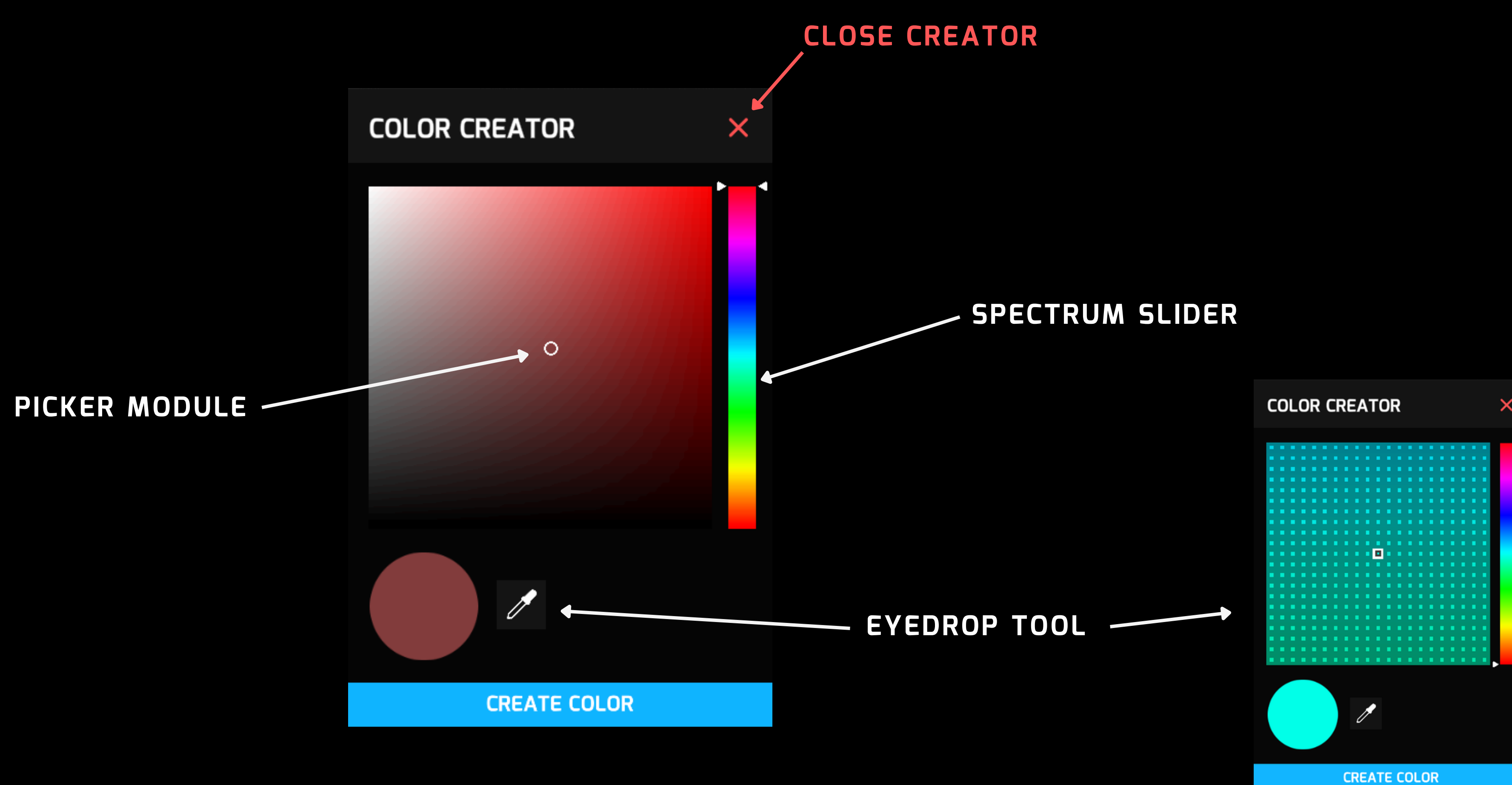


COLOR CREATOR



COLOR CREATOR

- THE **COLOR CREATOR IS** A TOOL THAT ALLOWS YOU TO PICK THE RIGHT COLORS FOR YOUR PROJECTS WITH HIGH PRECISION. IT WILL AUTOMATICALLY OPEN EVERY TIME YOU CLICK ABOVE A VALUE REPRESENTED BY A COLOR.
- CREATORS CAN NAVIGATE BOTH THE MAIN PANEL IN THE CENTER AND THE VERTICAL SLIDE ON THE RIGHT WITH THE MOUSE CURSOR TO IDENTIFY A SPECIFIC COLOR.
- THE COLOR YOU'RE CURRENTLY SELECTING IS HIGHLIGHTED IN THE BIG CIRCLE IN THE BOTTOM LEFT SIDE OF THE PANEL.
- THE **EYEDROP TOOL ICON** ALLOWS YOU TO PICK AN ALREADY EXISTING COLOR ON YOUR SCREEN WITH ABSOLUTE PRECISION, JUST CLICK ON THE SMALL ICON AND THEN ABOVE THE COLOR YOU WANT TO SELECT.
- ONCE YOU'RE SATISFIED WITH THE COLOR YOU'VE CHOSEN, HITTING **"CREATE" OR "UPDATE" COLOR** ON THE BOTTOM OF THE CREATOR WILL AUTOMATICALLY ASSIGN THE CHOSEN COLOR TO THE FIELD YOU WERE WORKING ON.

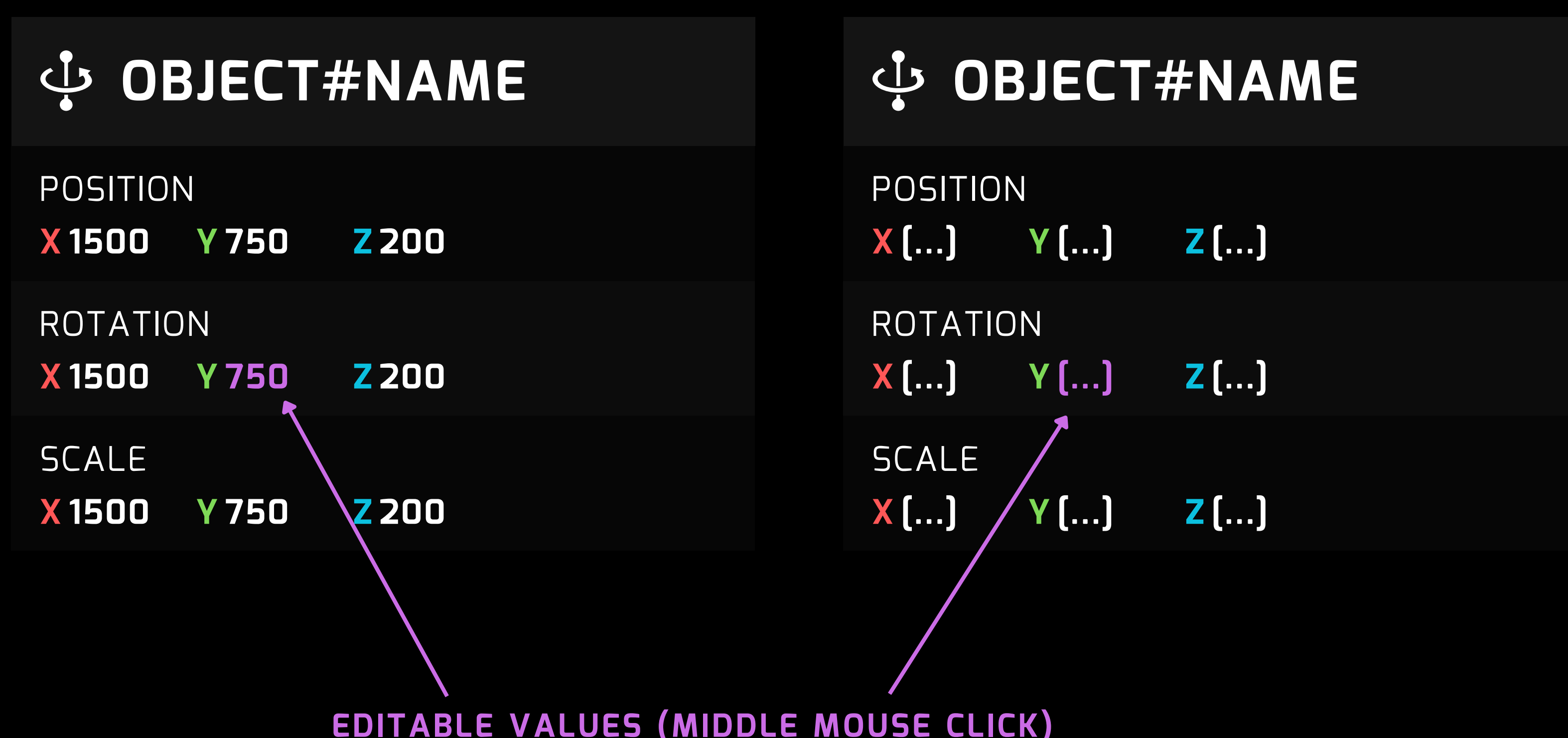


TRANSFORM TAB



TRANSFORM TAB

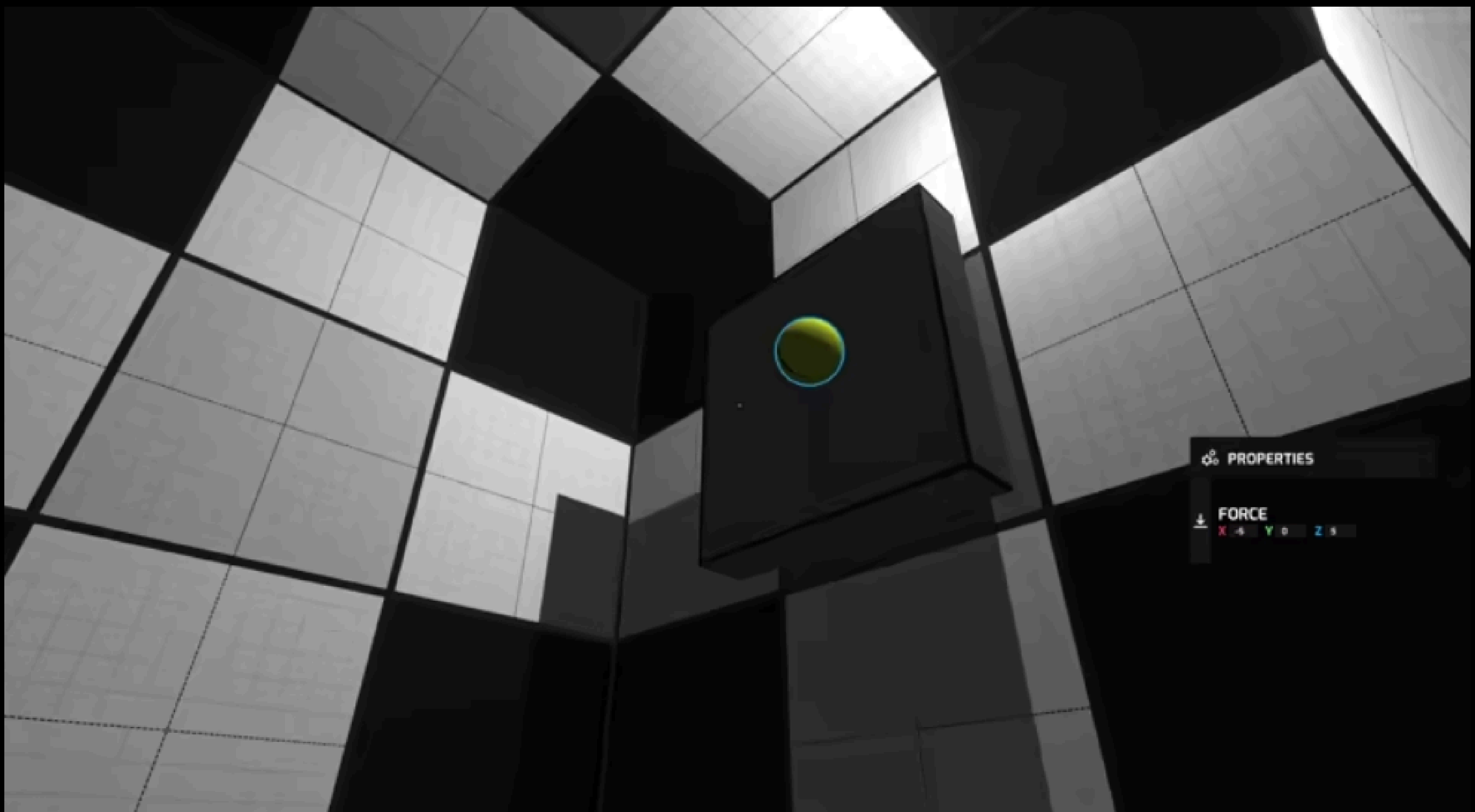
- AFTER AT LEAST ONE OBJECT HAS BEEN SELECTED, THREE TABS WILL APPEAR ON THE RIGHT SIDE OF YOUR SCREEN. THE ONE IN THE MIDDLE IS THE **TRANSFORM TAB**.
- THE TRANSFORM TAB IS ALSO DIVIDED INTO THREE SECTIONS, ONE FOR EACH **TRANSFORM MODIFICATION** APPLICABLE TO THE SELECTED OBJECT: **POSITION**, **ROTATION**, AND **SCALE**.
- EVERY SECTION HAS THREE VALUES, ONE FOR EACH DIMENSION, INDICATED BY THREE LETTERS AND COLORS, **X**, **Y** AND **Z**.
- EVERY MODIFY APPLIED TO AN OBJECT THROUGH THE **TRANSFORM GIZMO** WILL BE IMMEDIATELY REFLECTED IN THE TRANSFORM TAB.
- OF COURSE, THE REVERSE IS ALSO TRUE. MODIFYING VALUES BY CLICKING ON THEM OR USING THE MOUSE WHEEL SHORTCUTS (**PAGE 5**) IN THE TRANSFORM TAB WILL CAUSE THE OBJECT TO CHANGE IN THE LEVEL.
- IF MORE THAN ONE OBJECT IS SELECTED, THE TRANSFORM TAB, AS PER **MULTI-EDITING BASIC PRINCIPLES**, WILL ONLY SHOW MATCHING VALUES ACROSS ALL OBJECTS. IF EVEN A SINGLE OBJECT CAUSES A MISMATCH, THE TRANSFORM TAB WILL SHOW AN **INDEFINITE VALUE INDICATOR (...)** WHICH CAN HOWEVER STILL BE EDITED TO GIVE ALL THE OBJECTS A COMMON VALUE.
- WHILE ABOVE A VALUE IN THE TRANSFORM TAB, **CLICKING THE MIDDLE MOUSE BUTTON (OR MOUSE WHEEL)** WILL CAUSE THAT VALUE TO BECOME **EDITABLE** AND TURN PURPLE.



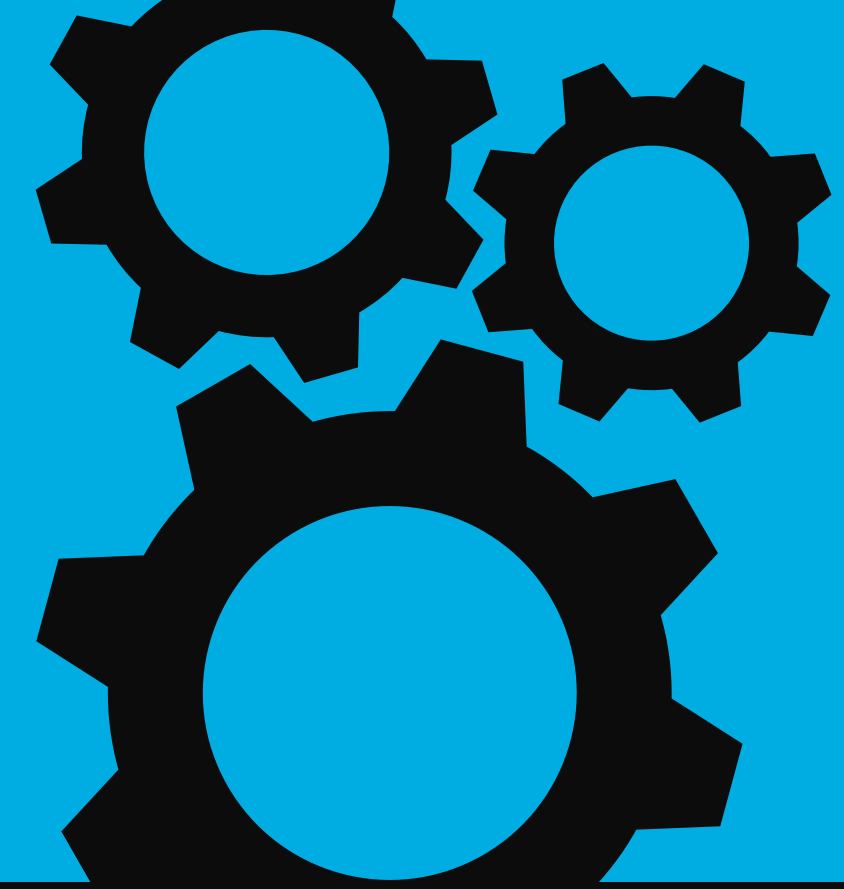
EDITABLE VALUES

EDITABLE VALUES

- AS YOU CAN SEE IN THE PREVIOUS PAGE'S PICTURE, THE TWO MIDDLE VALUES HAVE BEEN MADE **EDITABLE**, BUT WHAT DOES THAT MEAN?
- PROJECT LEVELS HAS A UNIQUE GAMEPLAY MECHANIC CALLED "**REALTIME EDITABLE VALUES**"
- WHEN A VALUE IS MARKED AS **EDITABLE**, PEOPLE PLAYING YOUR LEVEL **CAN MODIFY IT IN ANY WAY THEY WANT** AND **SEE THE RESULTS IN REALTIME** BY USING **EDIT MODE**, WHICH IS ACTIVATED BY HOLDING DOWN THE **[TAB]** KEY WHILE PLAYING ANY LEVEL.
- WHILE IN EDIT MODE, **THE WHOLE WORLD WILL FADE TO BLACK AND WHITE EXCEPT FOR THE OBJECTS THAT HAVE AT LEAST SOMETHING MARKED AS EDITABLE.**
- PLAYERS CAN THEN JUST **CLICK ON THE OBJECT WITH THE MOUSE CURSOR** (WHICH WILL BECOME VISIBLE) AND **MODIFY THE AVAILABLE EDITABLE VALUES** BY USING THE UIs THAT WILL POP UP.
- EDITABLE VALUES CAN BE EITHER THE **TRANSFORM VALUES** OF AN OBJECT FROM THE TRANSFORM TAB OR **ANY OF THE VALUES OF ITS PROPERTIES**. ALL ABOUT PROPERTIES AND HOW THEY CAN BE MADE EDITABLE IS EXPLAINED IN THE FOLLOWING PAGES.



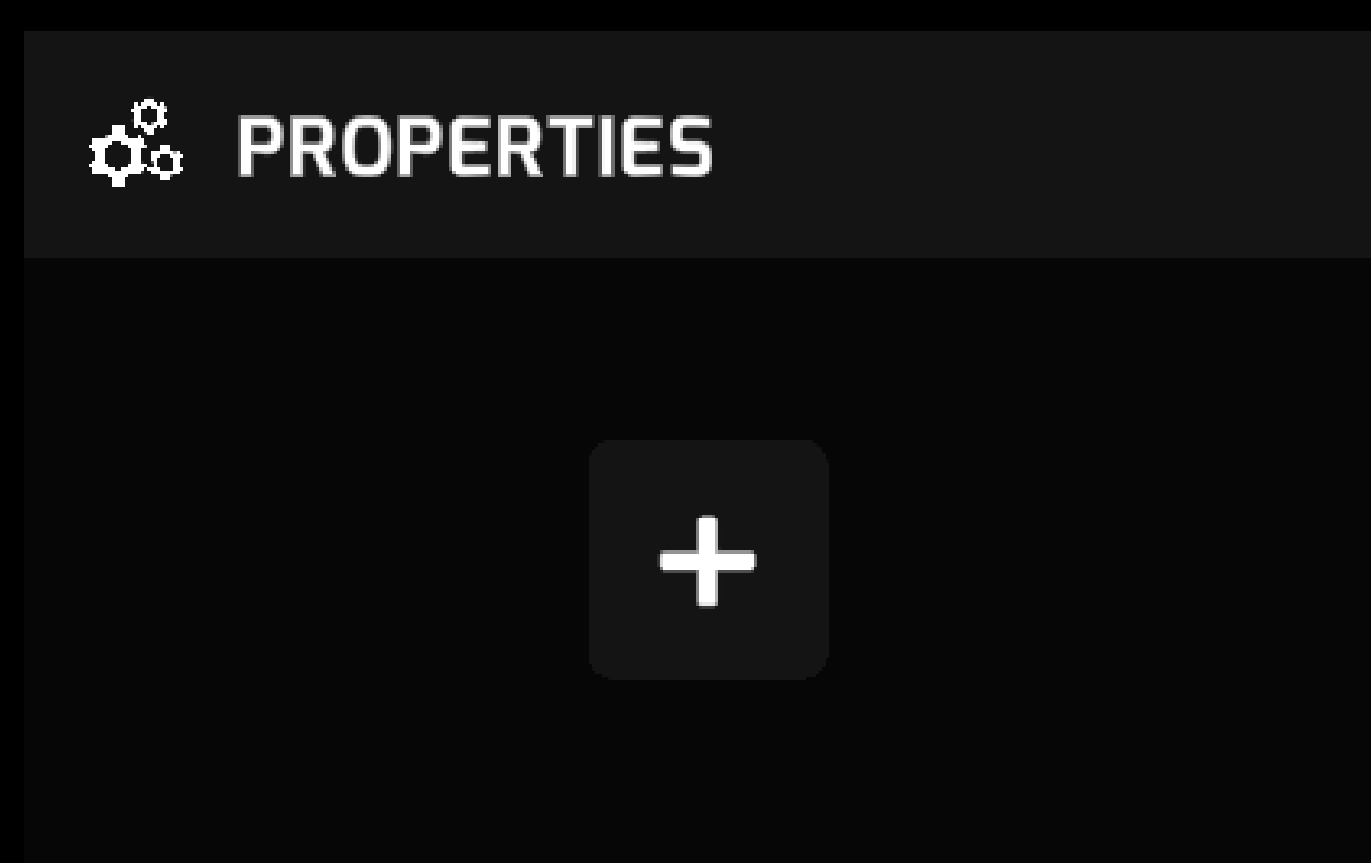
PROPERTIES TAB



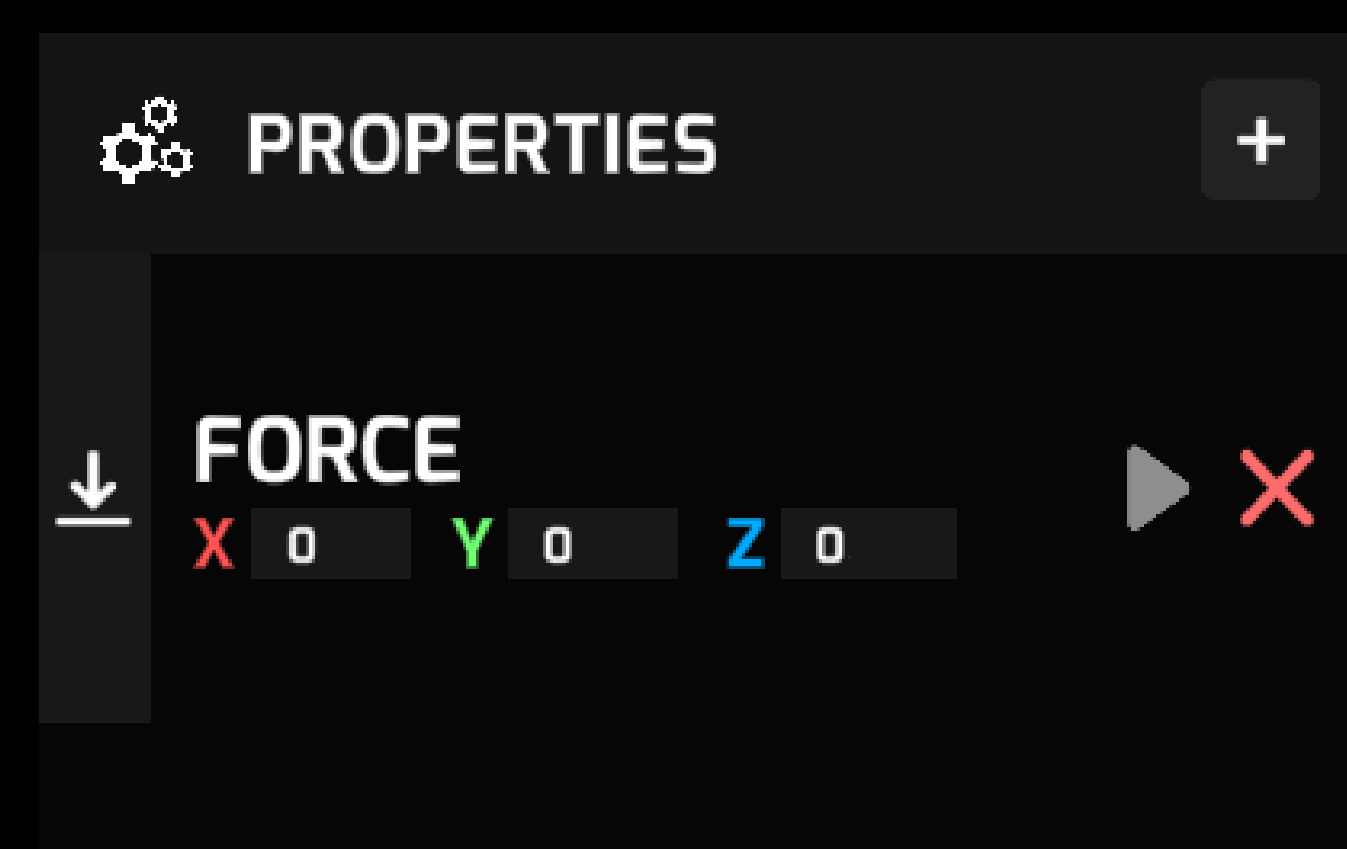
PROPERTIES TAB

- AFTER AT LEAST ONE OBJECT HAS BEEN SELECTED, THREE TABS WILL APPEAR ON THE RIGHT SIDE OF YOUR SCREEN. THE ONE IN THE BOTTOM IS THE **PROPERTIES TAB**. **(PICTURE 1)**
- THE PROPERTIES TAB WILL DISPLAY **ALL THE PROPERTIES CURRENTLY APPLIED TO ALL THE OBJECTS YOU HAVE CURRENTLY SELECTED**. **(PICTURE 2)**
- TO ADD A PROPERTY, SIMPLY CLICK ON THE (+) BUTTON AND CHOOSE THE DESIRED PROPERTY FROM THE **PROPERTIES BROWSER** THAT WILL POP UP IMMEDIATELY TO THE LEFT. ALL PROPERTIES ARE EXPLAINED ONE BY ONE IN THE NEXT PAGES.
- I SAID “PROPERTIES” A LOT OF TIMES, DIDN’T I?
- IF YOU HAVE MULTIPLE OBJECTS SELECTED WHICH DON’T SHARE THE SAME PROPERTIES, A WARNING WILL BE DISPLAYED ON THE PROPERTIES TAB. THIS DOES NOT PREVENT YOU FROM ADDING A NEW PROPERTY THAT WILL BE APPLIED TO ALL THE SELECTED OBJECTS ANYWAY. **(PICTURE 3)**
- IF, ON THE OTHER HAND, ALL OBJECTS SHARE THE SAME EXACT PROPERTY(IES), YOU CAN **MULTI-EDIT** THEM. AND IF THEIR VALUES DON’T MATCH EXACTLY, AN “**INDEFINITE VALUE**” INDICATOR (-) WILL BE SHOWN. REMEMBER THAT EVEN WITH THE INDEFINITE VALUE INDICATOR IN PLACE YOU CAN STILL MODIFY THE VALUE FOR ALL THE SELECTED OBJECTS. **(PICTURE 4)**

PICTURE 1



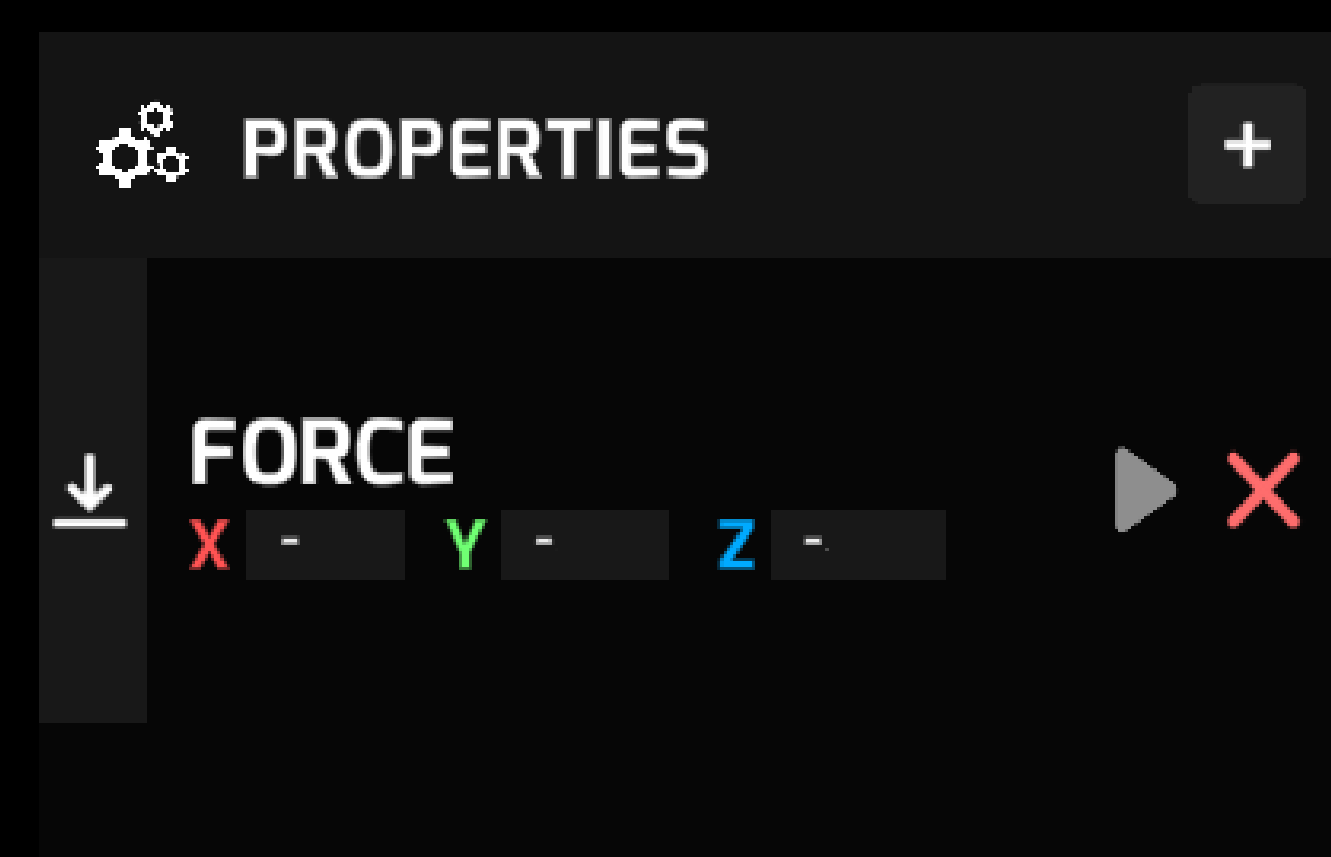
PICTURE 2



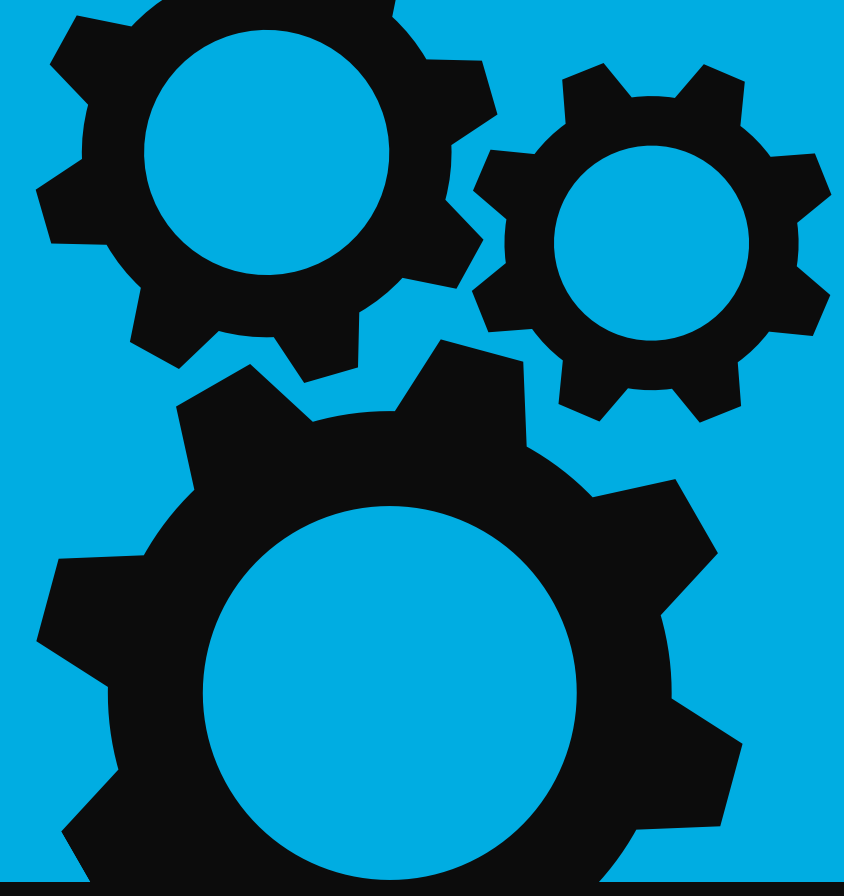
PICTURE 3



PICTURE 4

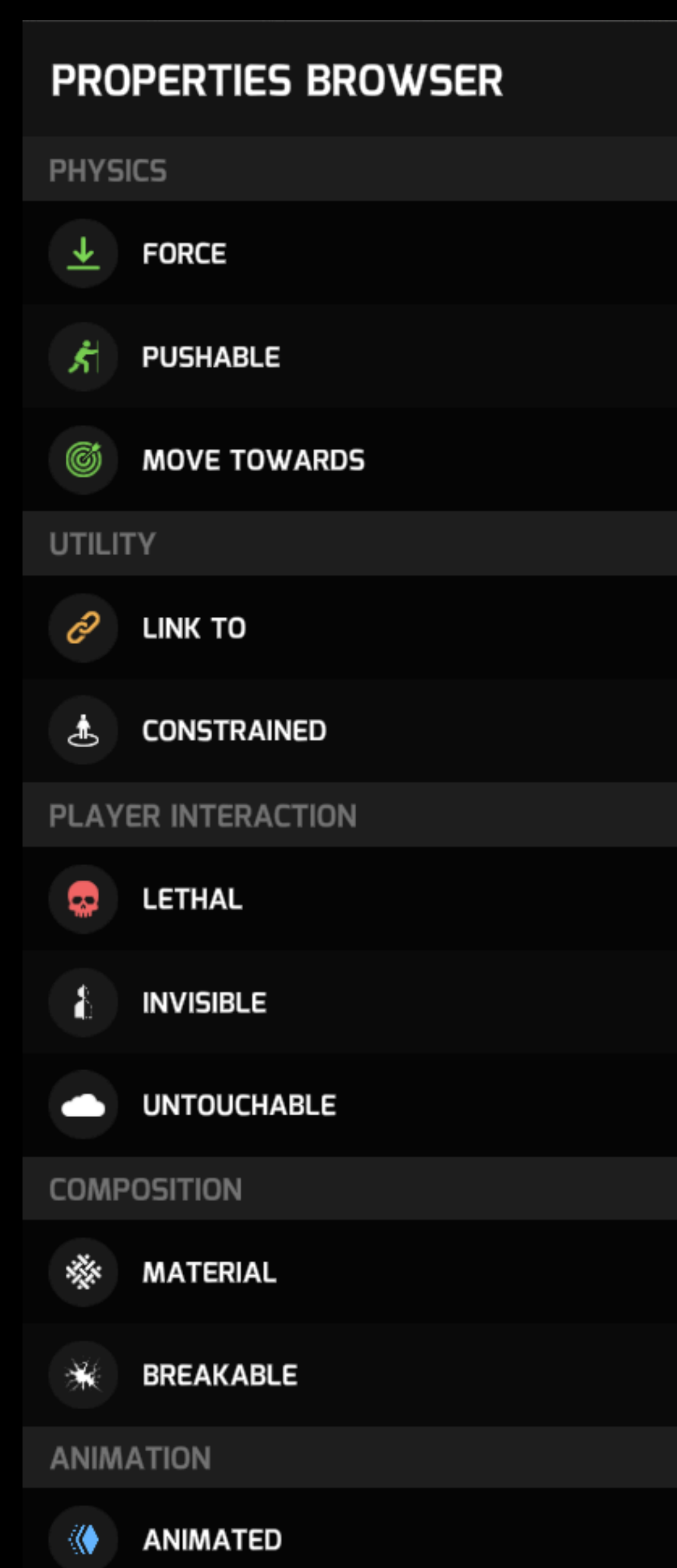


PROPERTIES BROWSER

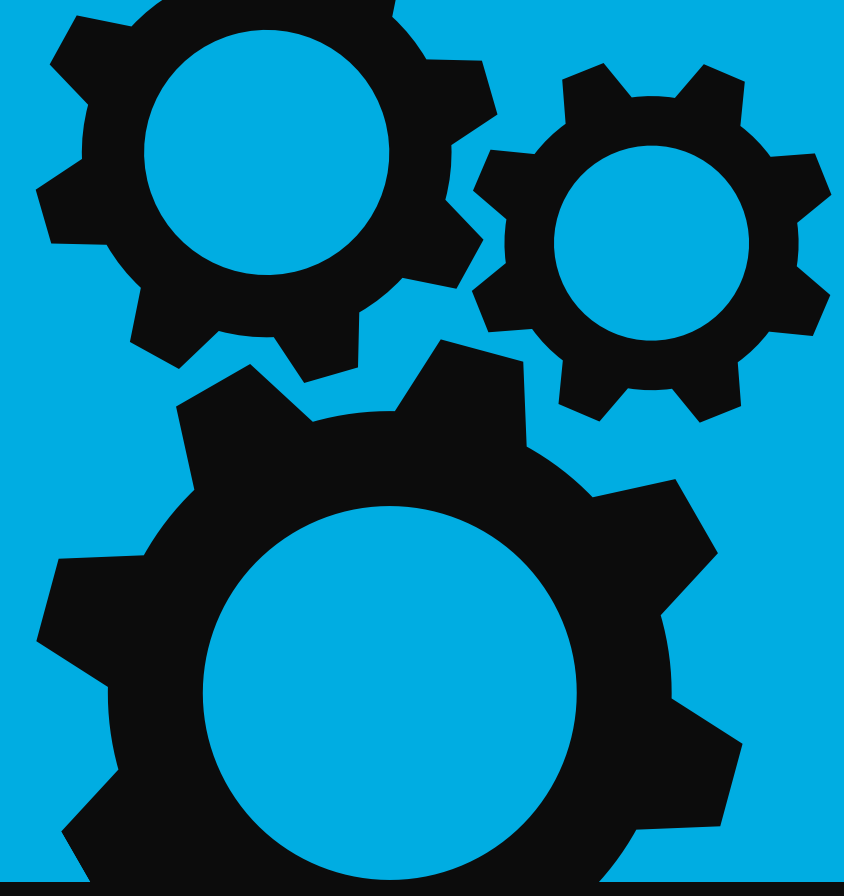


PROPERTIES BROWSER

- AFTER CLICKING ON THE (+) BUTTON IN THE PROPERTIES TAB, THE **PROPERTIES BROWSER** WILL POP UP ON YOUR SCREEN.
- THE PROPERTIES BROWSER DISPLAYS ALL THE AVAILABLE PROPERTIES CONVENIENTLY ORGANIZED IN GROUPS.
- SIMPLY CLICKING ON A PROPERTY'S NAME, ICON OR RETICLE WILL ADD SAID PROPERTY TO ALL THE CURRENTLY SELECTED OBJECTS.
- YOU MAY HAVE NOTICED SOME PROPERTIES HAVE COLORED ICONS, THIS IS TO HIGHLIGHT THAT THEY ARE **CORE PROPERTIES**.
- **CORE PROPERTIES** DIFFER FROM REGULAR ONES BECAUSE THEY SUBSTANTIALLY CHANGE THE WAY AN OBJECT BEHAVES.
- **GREEN PROPERTIES** USE THE **PHYSICS CORE**, **CYAN PROPERTIES** USE THE **ANIMATION CORE**, **RED PROPERTIES** USE THE **LETHAL CORE** AND **YELLOW PROPERTIES** USE THE **LINK CORE**.
- YOU ARE ALLOWED TO USE MULTIPLE CORE PROPERTIES ON A SINGLE OBJECT **AS LONG AS YOU DON'T MIX PHYSICS AND ANIMATION PROPERTIES**, WHICH THE EDITOR **WON'T ALLOW** TO AVOID SIMULATION PROBLEMS.

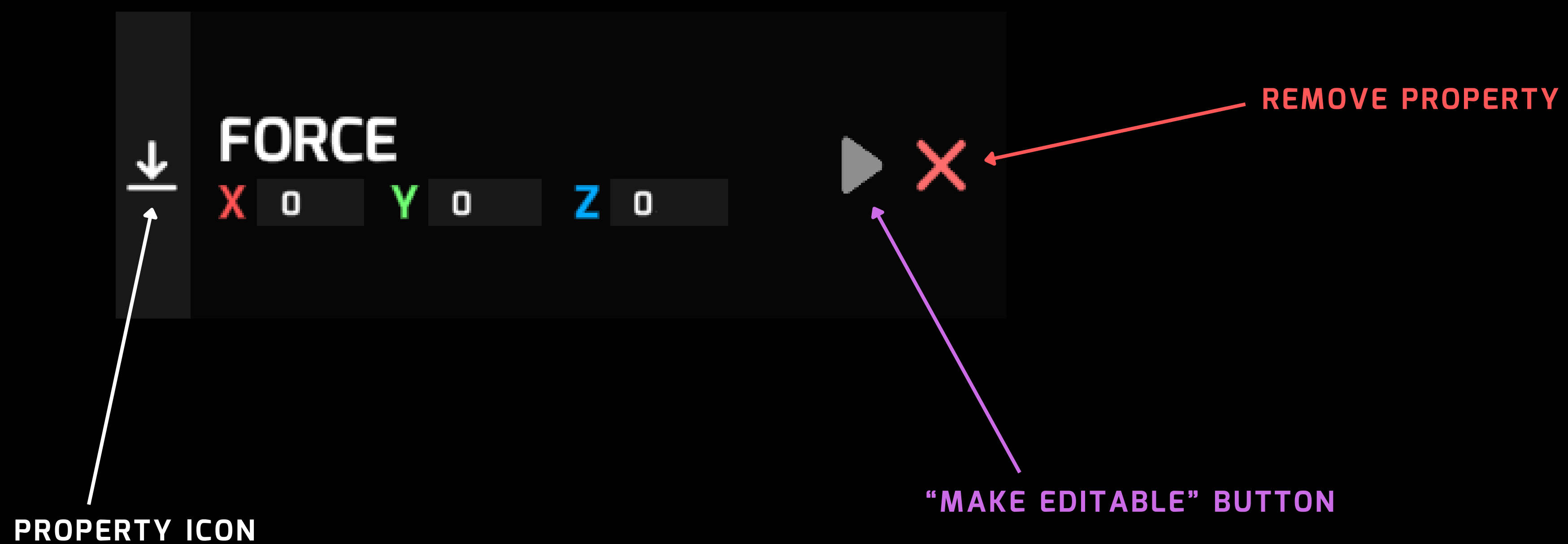


PROPERTIES CONTROLLERS

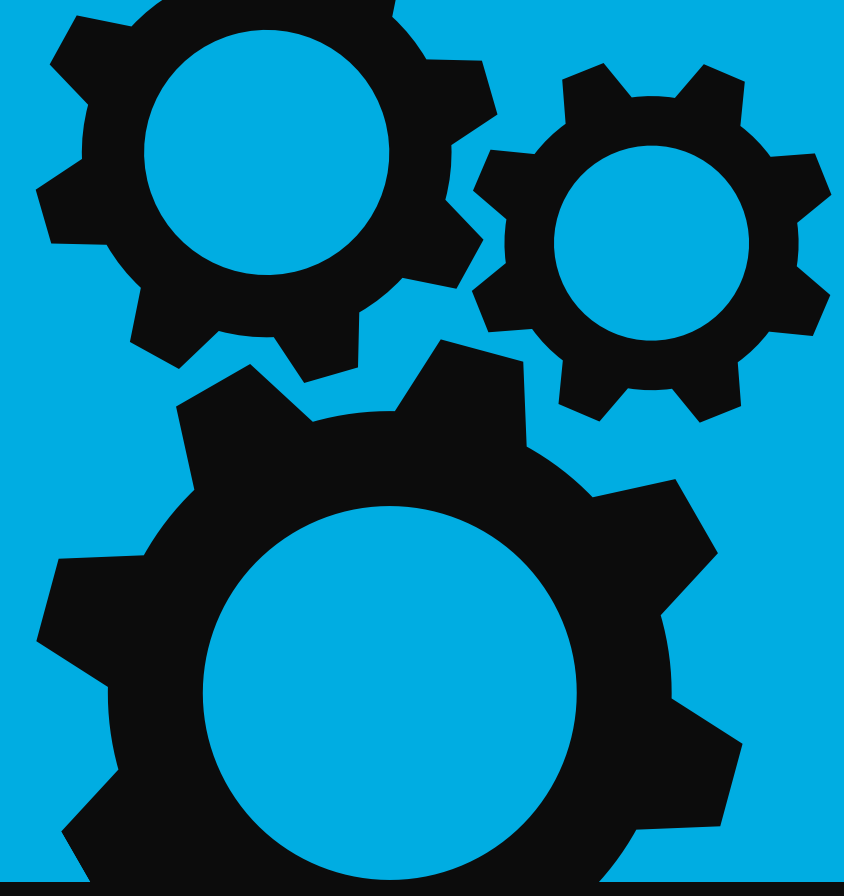


PROPERTIES CONTROLLERS

- YOU CAN THINK OF PROPERTIES IN PROJECT LEVELS LIKE YOU THINK OF ADJECTIVES IN REAL LIFE GRAMMAR. THEY ARE A WAY TO ALLOW AN OBJECT TO DO “**SOMETHING MORE**” THAN IT NORMALLY COULD.
- ADDING PROPERTIES TO AN OBJECT IS EXPLAINED IN THE PREVIOUS PAGES.
- ONCE A PROPERTY HAS BEEN ADDED TO AN OBJECT, YOU WILL BE ABLE TO MODIFY IT’S VALUES BY USING THE RELATIVE **PROPERTY CONTROLLER** THAT WILL DISPLAY ON THE **PROPERTIES TAB**.
- EVERY CONTROLLER IS DIFFERENT IN ORDER TO ACCOMODATE ITS SPECIFIC PROPERTY’S QUIRKS, BUT THEY ALL SHARE TWO COMMON ELEMENTS, THE **EDITABLE BUTTON** AND THE **REMOVE BUTTON**.
- WHILE **THE LATTER** IS INTUITIVE, **THE FORMER** ALLOWS YOU TO MARK THAT SPECIFIC PROPERTY ON THAT SPECIFIC OBJECT(S) AS **EDITABLE** SO THAT PLAYERS CAN MODIFY ITS VALUES IN EDIT MODE WHILE PLAYING. THIS IS EXPLAINED IN DEPTH ON **PAGE 18**.



PROPERTIES

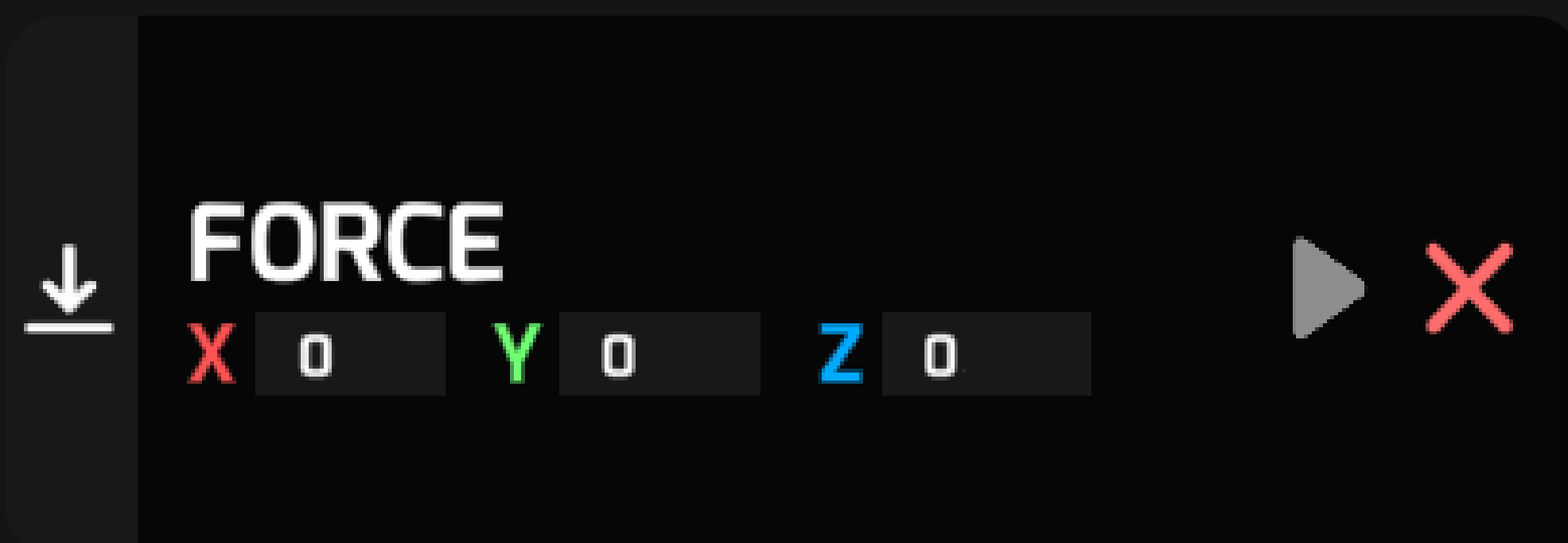


PROPERTIES

- I WILL NOW FOLLOW WITH A QUICK RUNDOWN OF ALL THE AVAILABLE PROPERTIES AND THEIR RESPECTIVE CONTROLLERS.

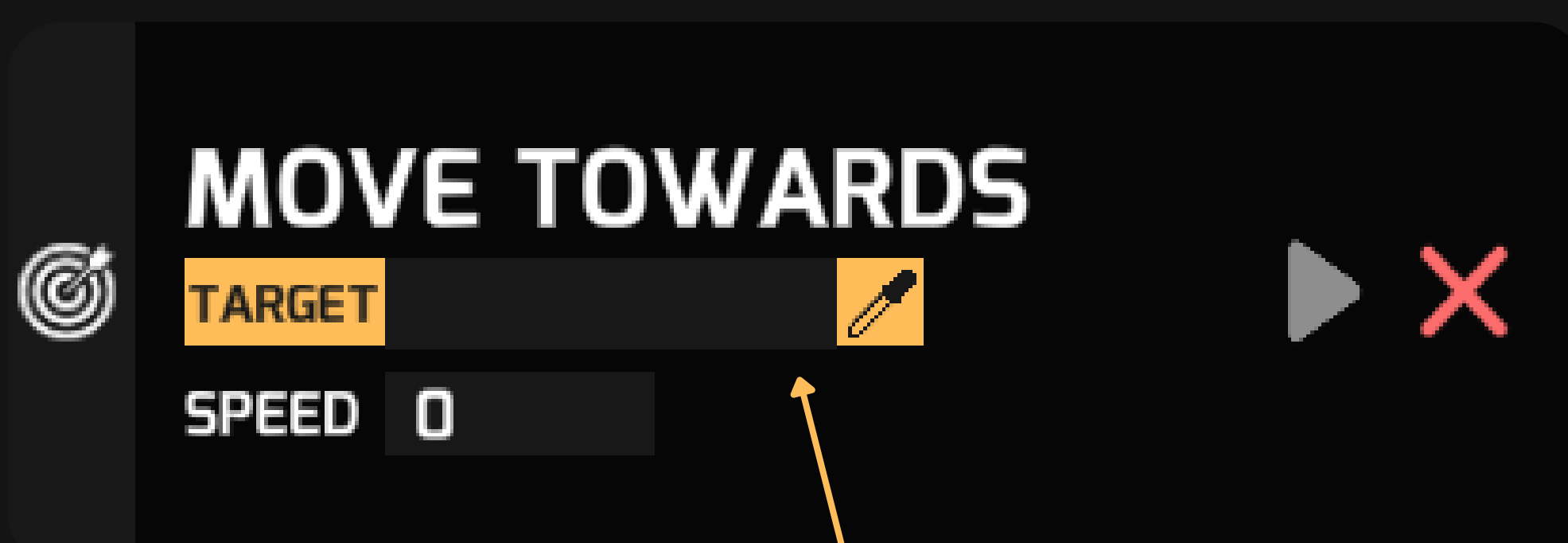
FORCE

CONSTANTLY PUSHES THE OBJECT IN THE SPECIFIED DIRECTION.



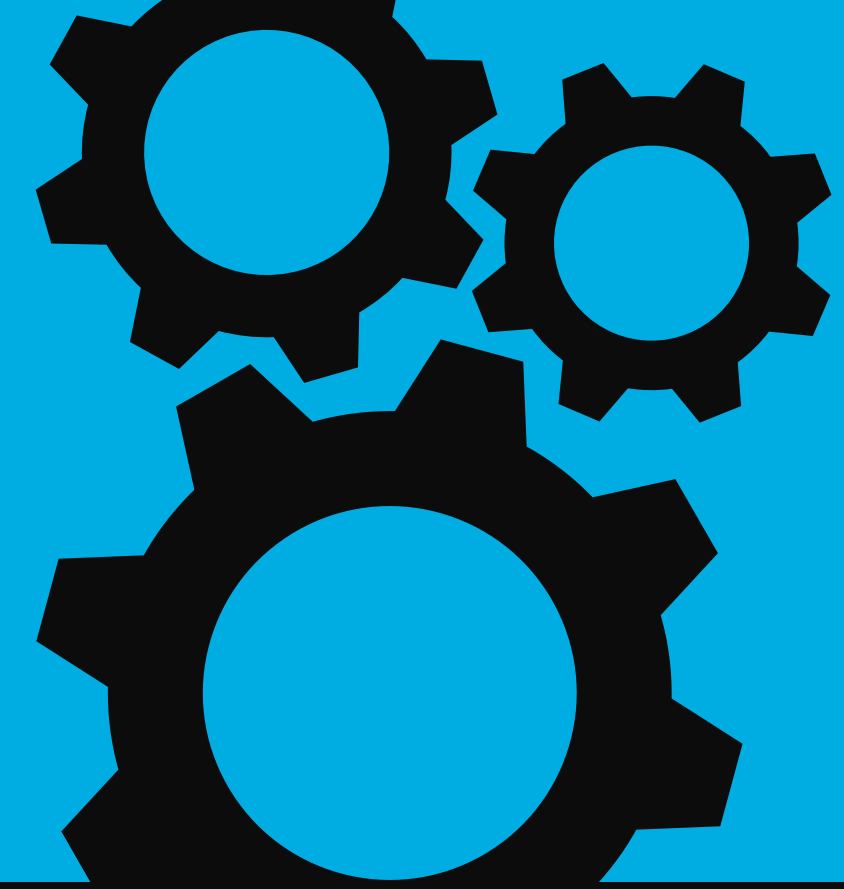
MOVE TOWARDS

MAKE AN OBJECT CONSTANTLY MOVE TOWARDS A SPECIFIED TARGET AT A SPECIFIED SPEED.



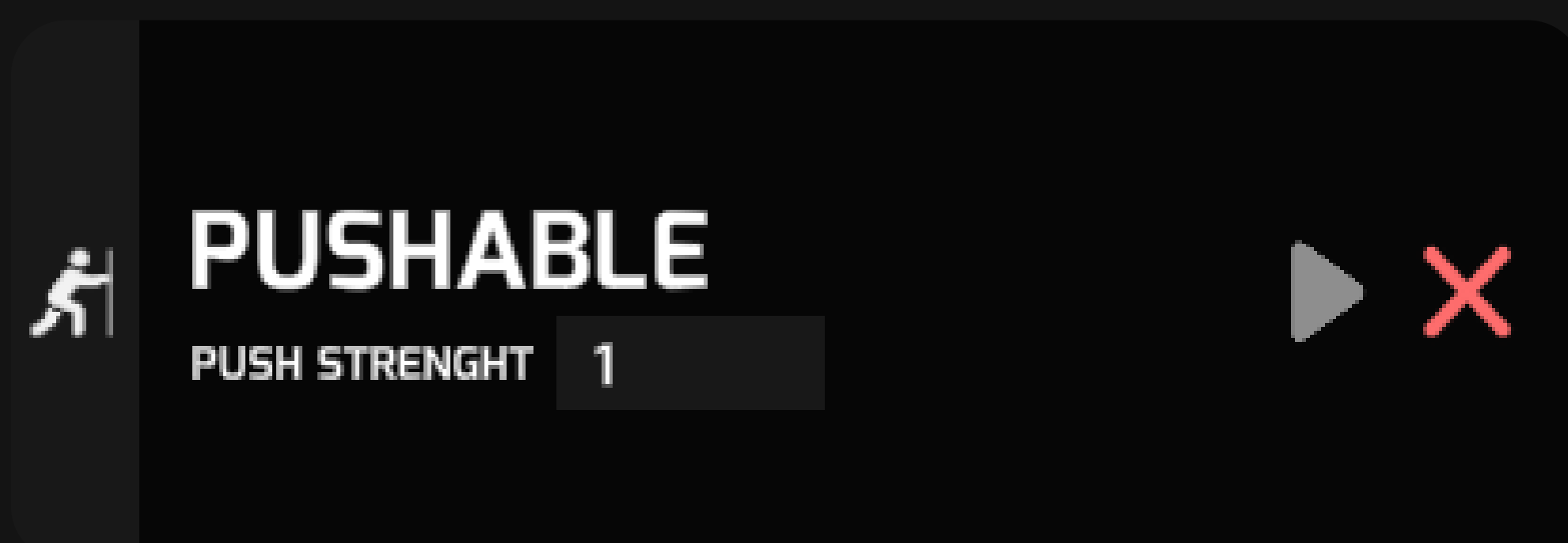
PS: THIS IS A "PICKER", CLICKING IT WILL MAKE THE INTERFACE DISAPPEAR TO ALLOW YOU TO SELECT THE OBJECT THAT YOU WANT TO PICK AS A TARGET. JUST CLICK ON THE DESIRED OBJECT AND THE EDITOR WILL DO EVERYTHING FOR YOU!

PROPERTIES



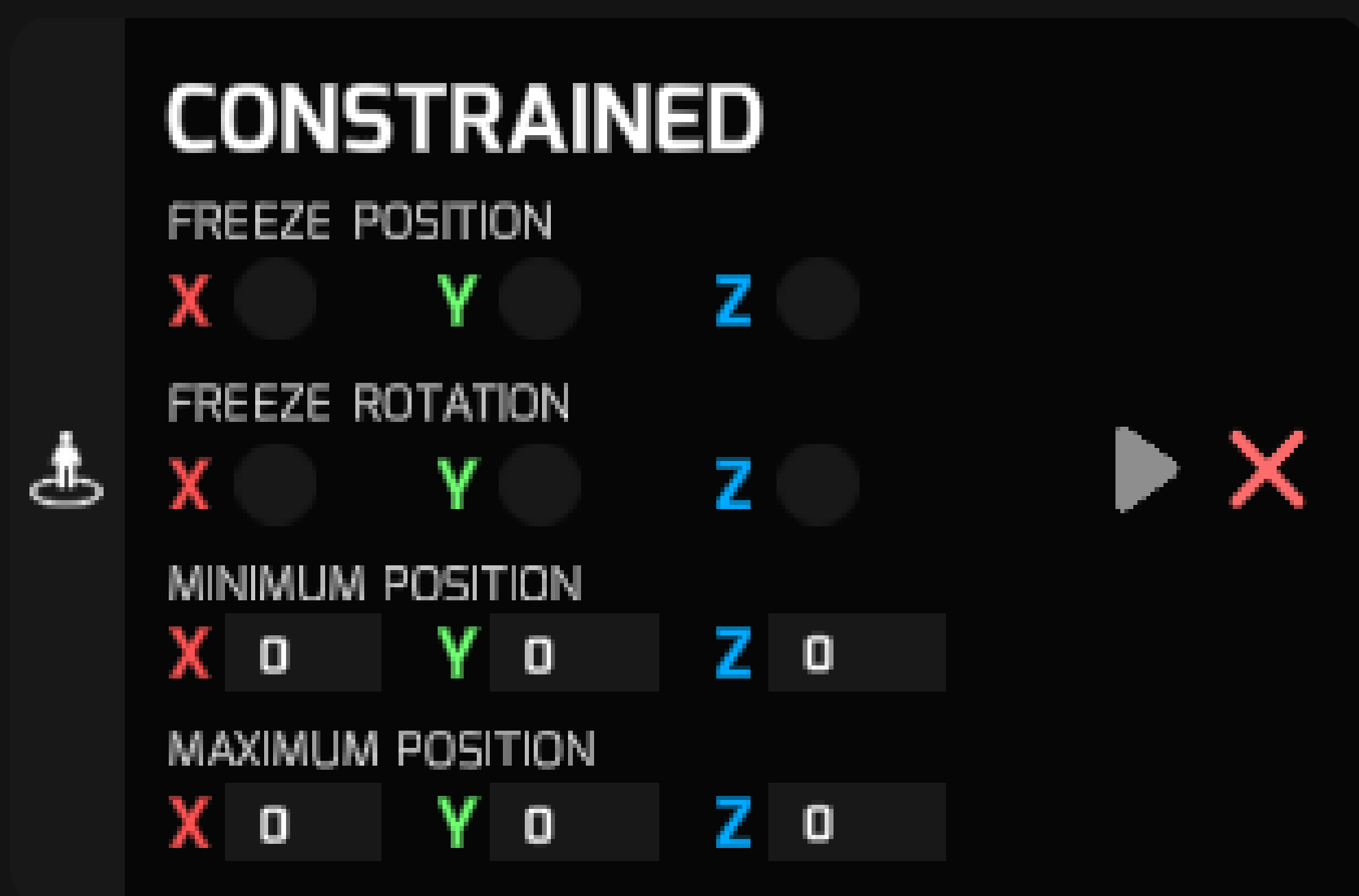
PUSHABLE

APPLIES A SERIES OF INTERNAL MODIFIES TO ALLOW AN OBJECT TO BE PUSHED EASILY. "PUSH STRENGTH" WORKS AS A MULTIPLIER FOR THE INTENSITY OF THE FORCES APPLIED TO MAKE IT EASIER OR HARDER TO PUSH.

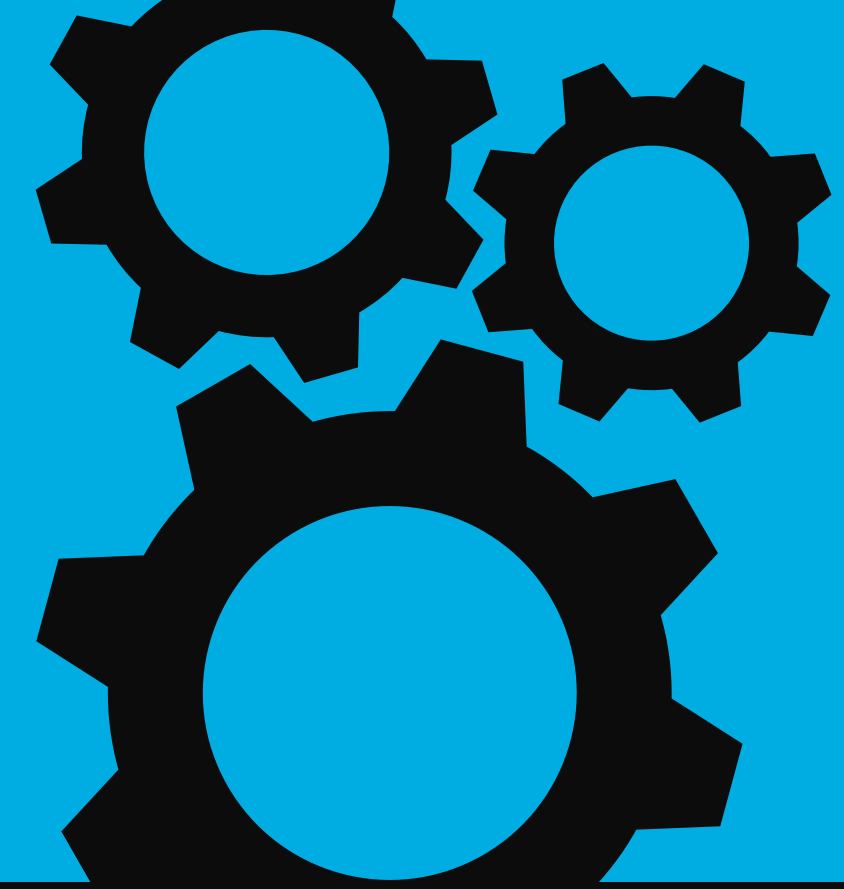


CONSTRAINED

ALLOWS TO APPLY RESTRICTIONS TO HOW FREELY AN OBJECT CAN MOVE. YOU CAN LOCK IT'S POSITION AND ROTATION INDIVIDUALLY ON EACH AXIS OR DEFINE A MINIMUM AND A MAXIMUM POSITION IN WHICH IT CAN MOVE



PROPERTIES



LETHAL

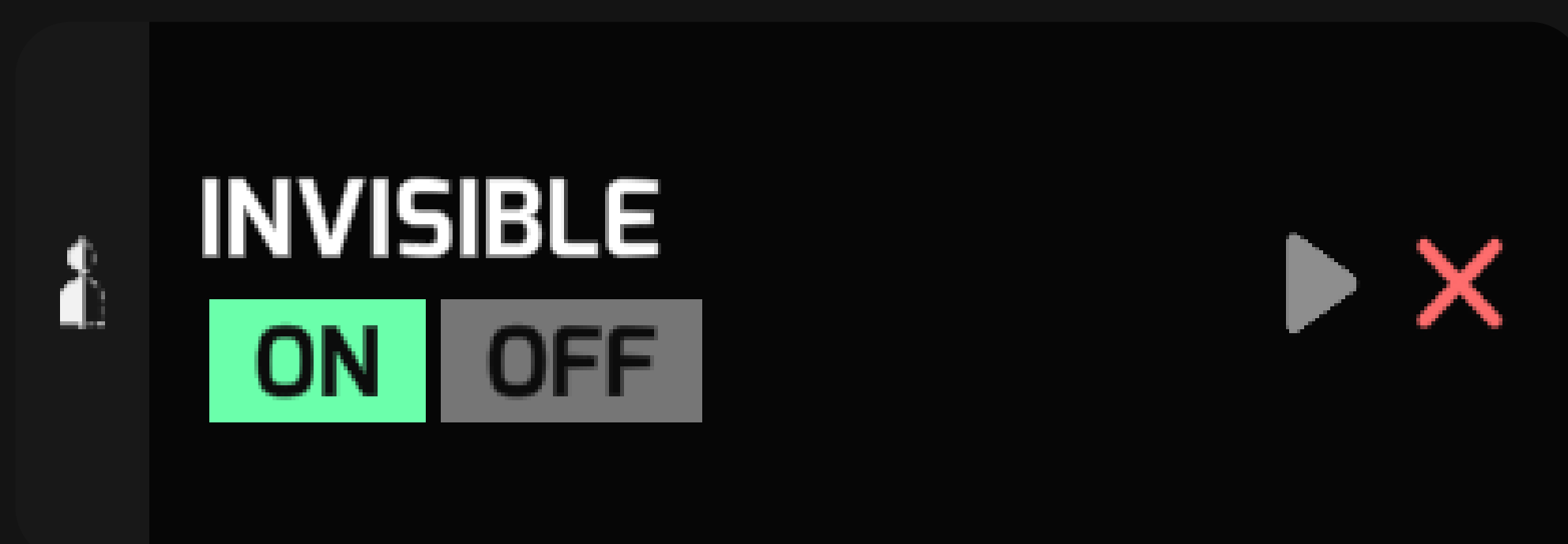
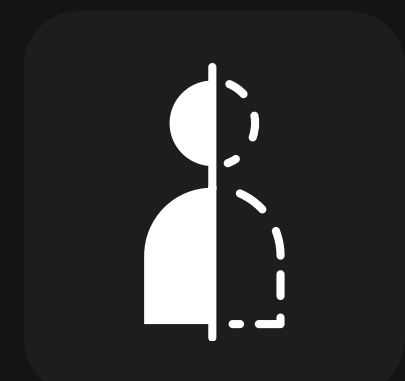
MARKS AN OBJECT AS **LETHAL**. IF THE PLAYER COMES INTO CONTACT WITH A LETHAL OBJECT, IT WILL CAUSE A **GAME OVER**.

FAST MODE ALLOWS YOU TO DECIDE WHETHER THE WHOLE LEVEL SHOULD RELOAD FROM ZERO UPON THE PLAYER'S DEATH OR IF HE SHOULD JUST BE TELEPORTED BACK TO THE STARTING POSITION / LAST ACTIVATED CHECKPOINT.



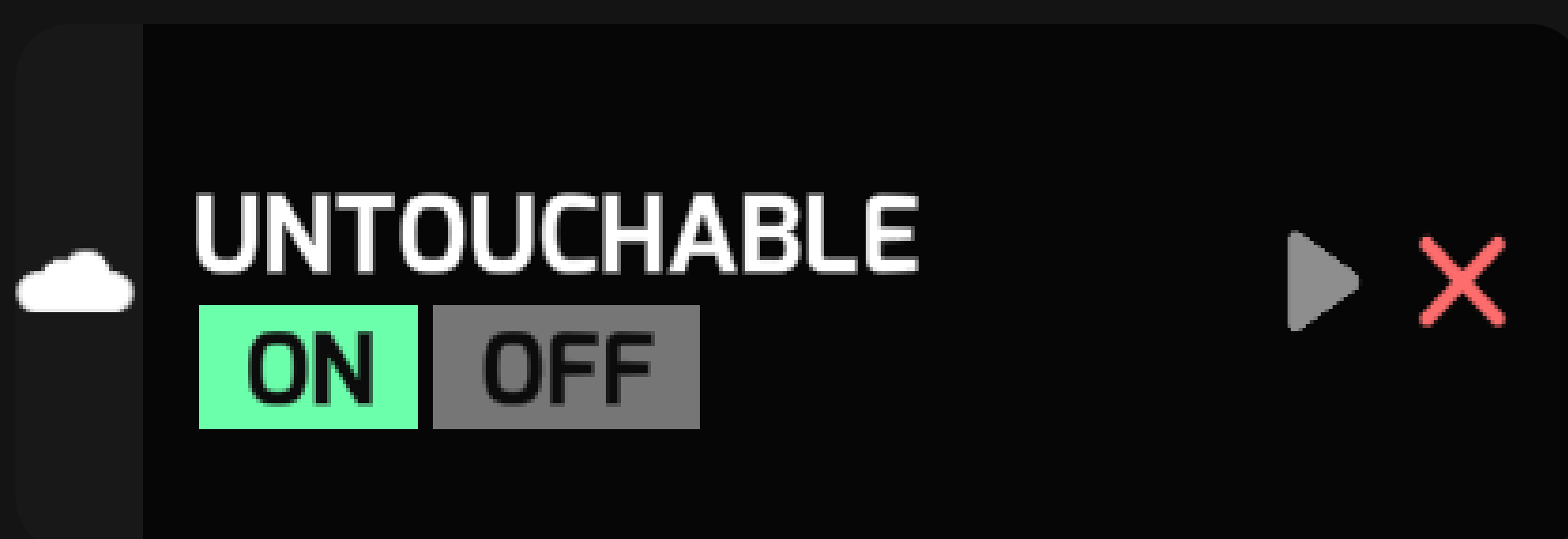
INVISIBLE

MAKES AN OBJECT INVISIBLE. CAN BE TURNED ON AND OFF.

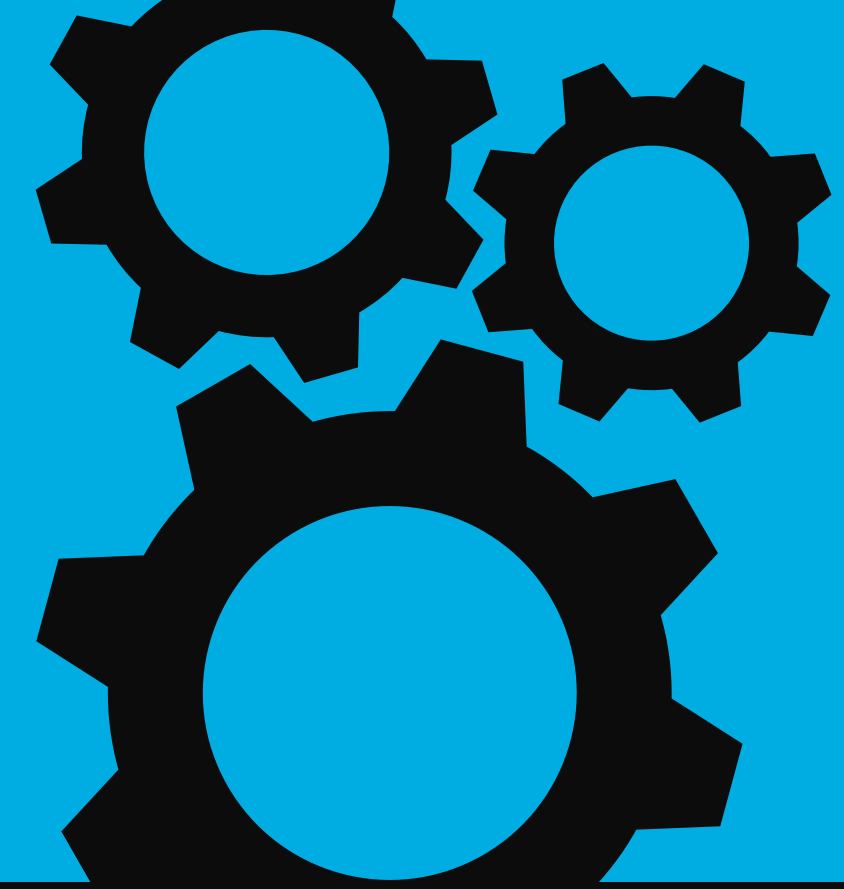


UNTOUCHABLE

MAKES AN OBJECT UNTOUCHABLE. CAN BE TURNED ON AND OFF.



PROPERTIES



MATERIAL

ALLOWS YOU TO SPECIFY HOW OBJECTS THAT COME INTO CONTACT WITH THIS OBJECT WILL BEHAVE. ALSO ALLOWS TO DECIDE THE MASS OF THE OBJECT.



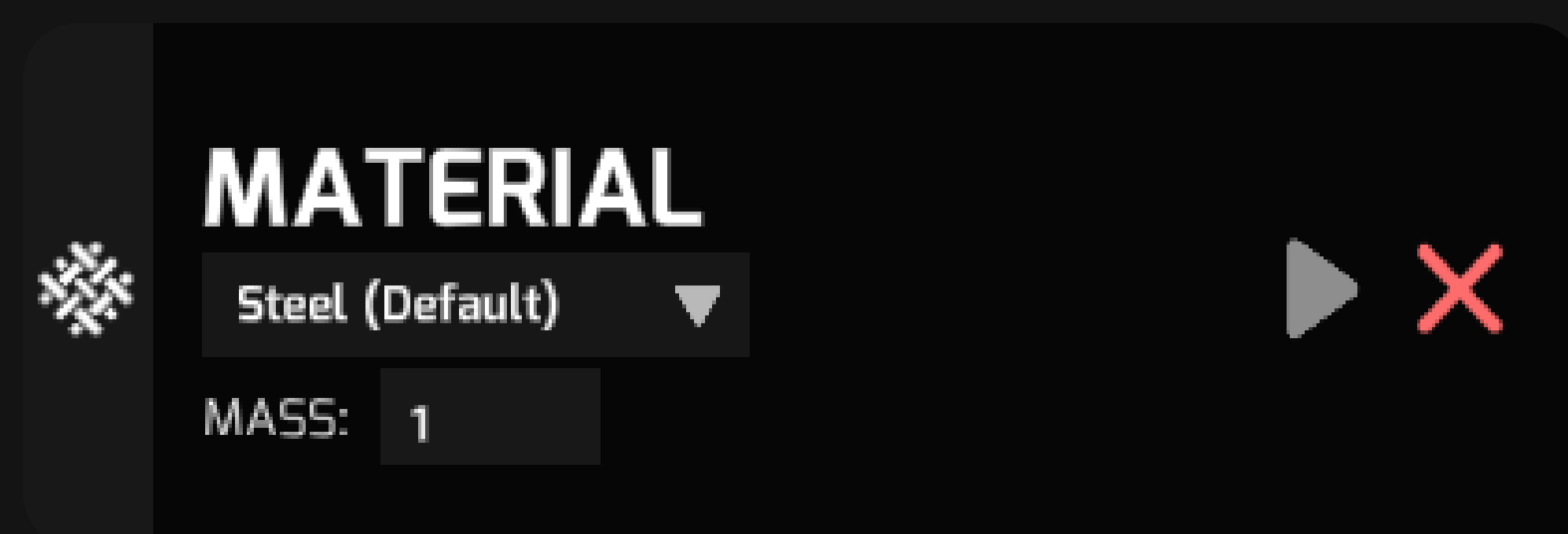
STEEL (DEFAULT) = AVERAGE FRICTION, NO ELASTICITY

ICE = ZERO FRICTION, NO ELASTICITY

ASPHALT = HIGH FRICTION, NO ELASTICITY

GUM = AVERAGE FRICTION, MAX ELASTICITY

MUD = MAX FRICTION, NO ELASTICITY



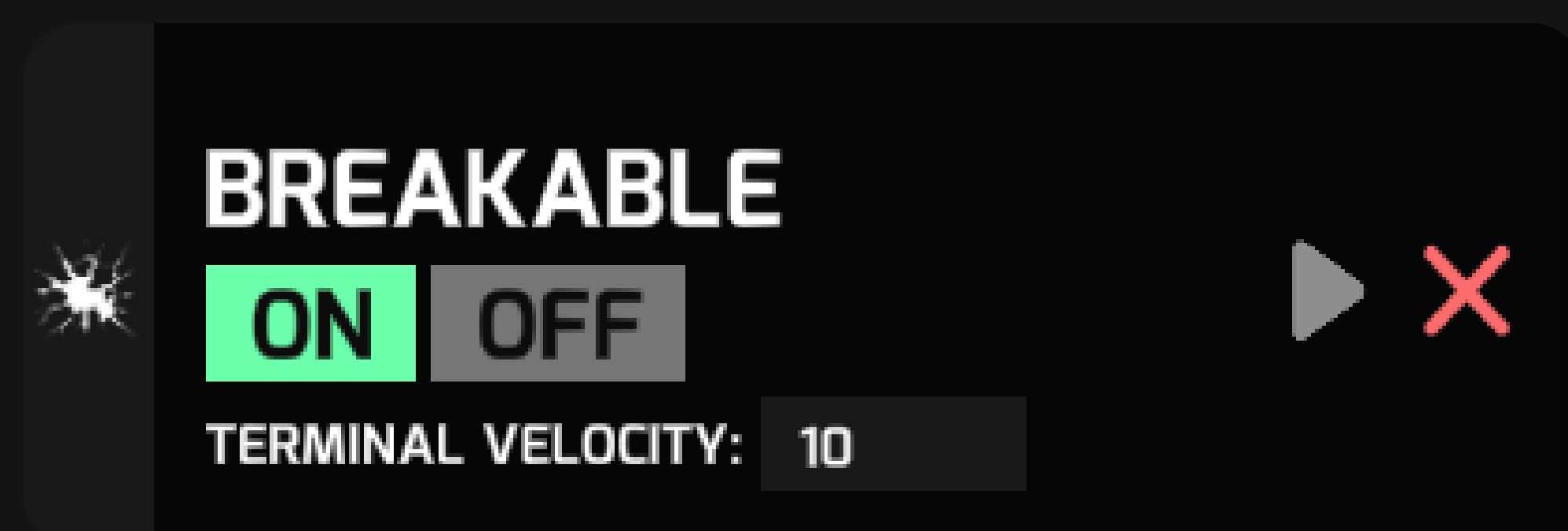
BREAKABLE

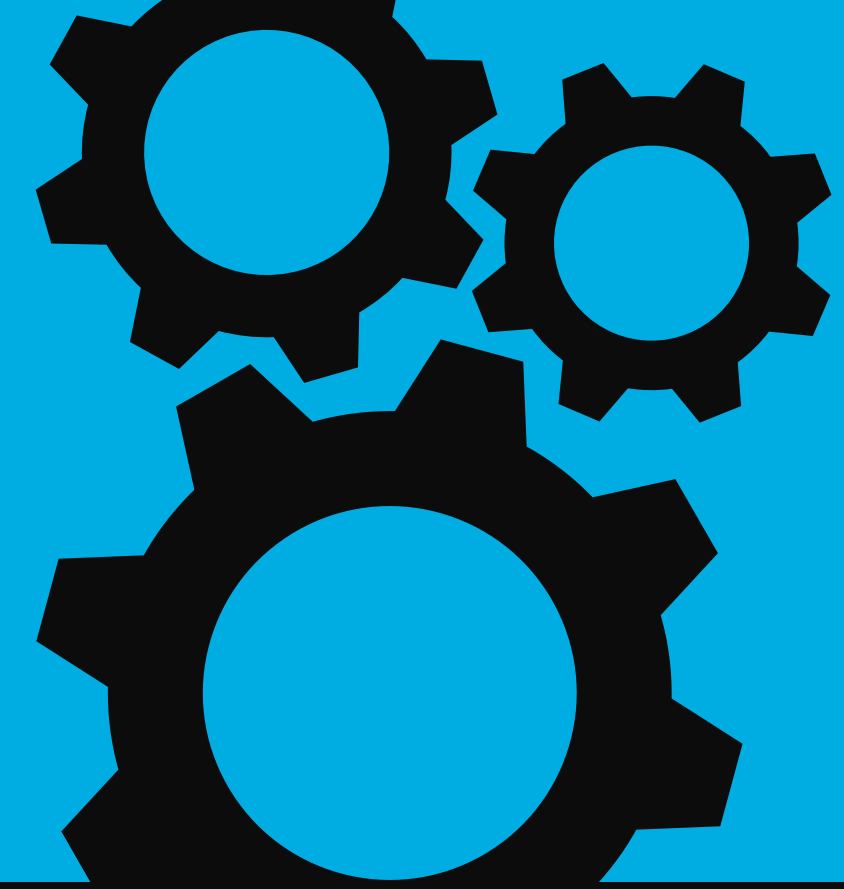
MARKS AN OBJECT AS BREAKABLE.

BREAKABLE OBJECTS WILL COLLAPSE INTO PIECES IF SHOT BY A DRONE, HIT BY A LASER OR IF THEY SUSTAIN AN IMPACT AT AN HIGHER SPEED THAN THE SPECIFIED TERMINAL SPEED.

ONCE BROKEN, THE PIECES WILL STILL FOLLOW THE GRAVITY THAT WAS APPLIED TO THE OBJECT WHEN IT WAS INTACT.

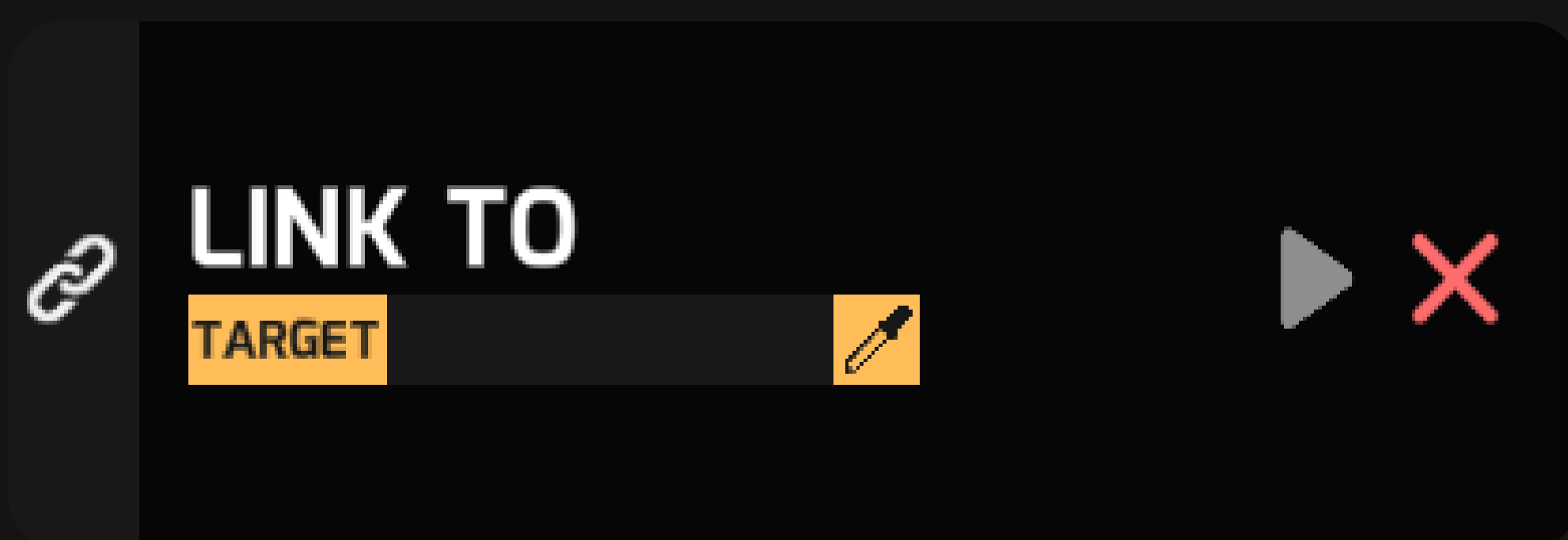
PIECES WILL DISAPPEAR ONCE THE **CLEAR DEBRIS** TIMER SPECIFIED IN THE **EDITOR SETTINGS (PAGE 47)** RUNS OUT.





LINK TO

LINKS THE OBJECT TO A TARGET. THE LINKED OBJECT (THE ONE WITH THE PROPERTY ON) WILL THEN MIMIC EVERY TRANSFORM CHANGE THAT THE TARGETED OBJECT DOES. THIS MEANS MOVING, ROTATING, AND SCALING ALONG WITH IT. MULTIPLE OBJECTS CAN BE LINKED TO THE SAME TARGET.



ANIMATED

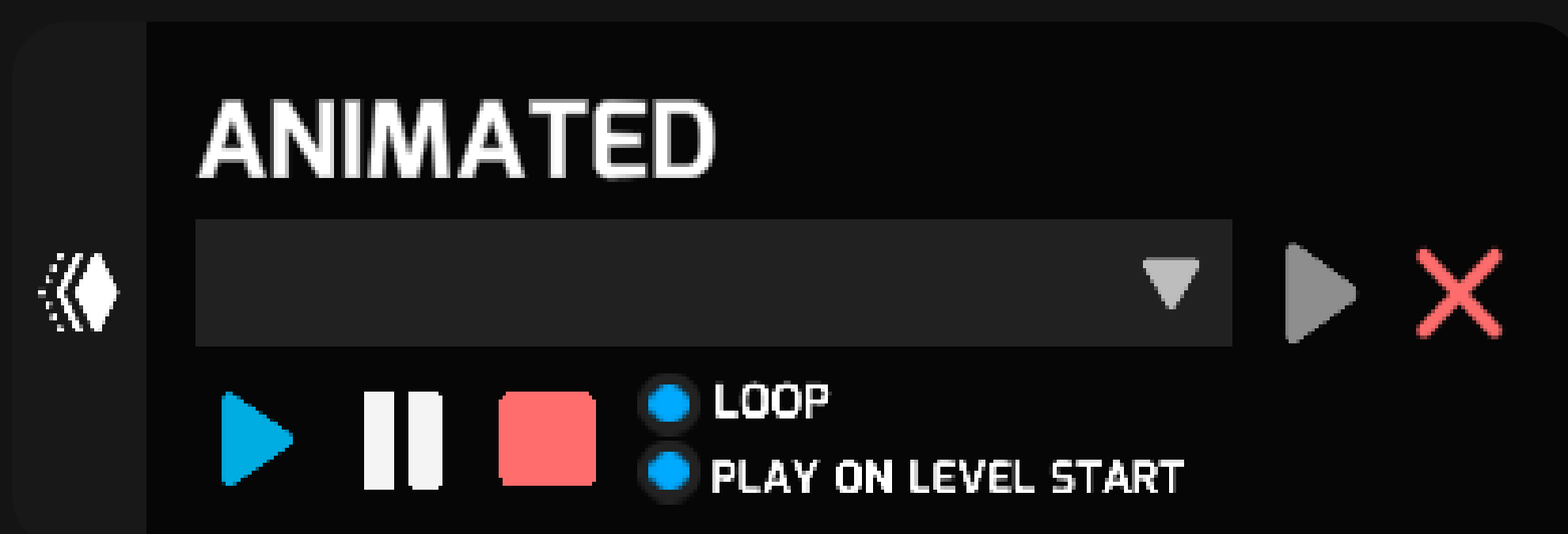
ALLOWS YOU TO BIND AN OBJECT TO AN **ANIMATION**.

YOU MUST FIRST CREATE AN **ANIMATION** IN THE **ANIMATION STUDIO (PAGE 27)** ONCE YOU HAVE AT LEAST A VALID ANIMATION, USE THE DROPDOWN MENU TO ASSIGN IT TO THE OBJECT.

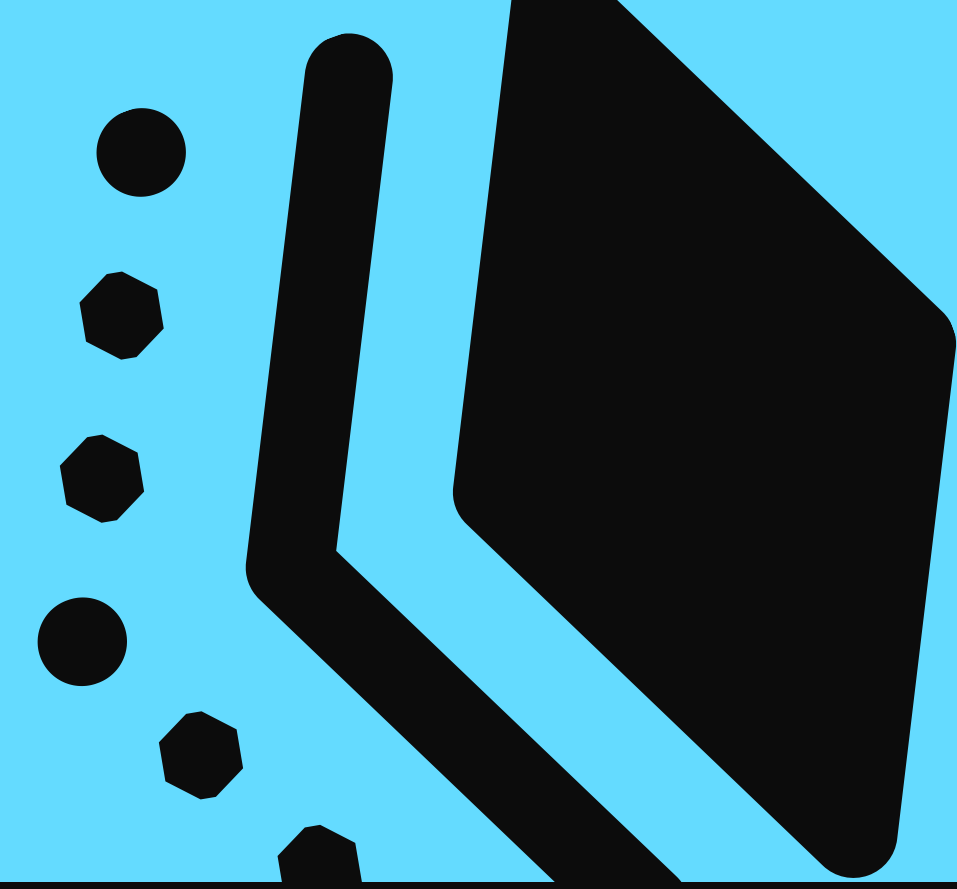
YOU CAN THEN PREVIEW THE ANIMATION IN THE EDITOR BY USING THE **PLAY**, **PAUSE** AND **STOP** BUTTONS.

"PLAY ON LEVEL START" ALLOWS YOU TO DECIDE IF YOU WANT THE ANIMATION TO START AS SOON AS THE LEVEL DOES OR IF YOU WANT IT TO WAIT UNTIL SOMETHING (A TRIGGER) CALLS IT.

YOU CAN ALSO DECIDE IF YOU WANT THE ANIMATION TO LOOP FOREVER ONCE IT ENDS OR PLAY ONLY ONCE.

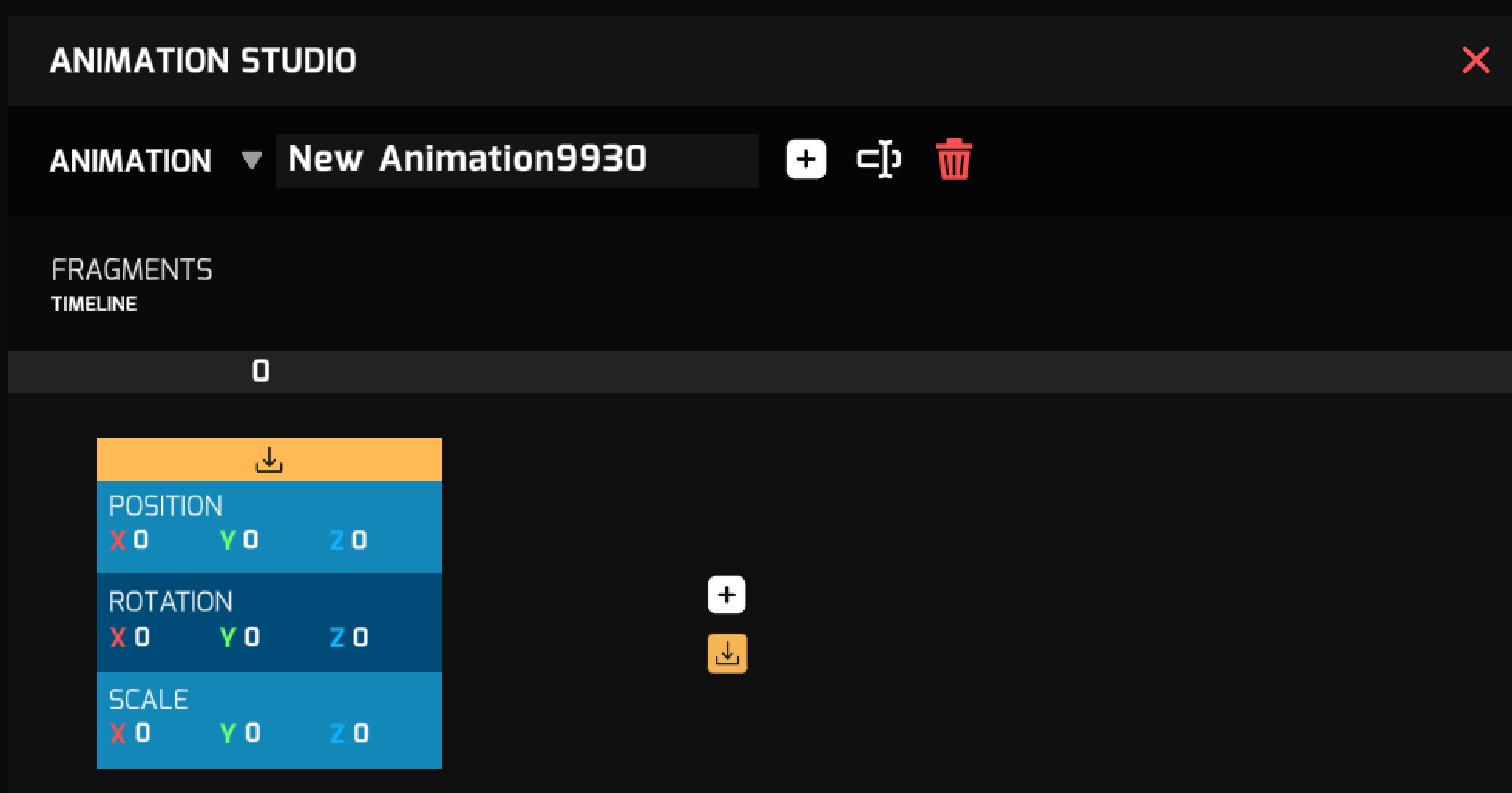


ANIMATION STUDIO



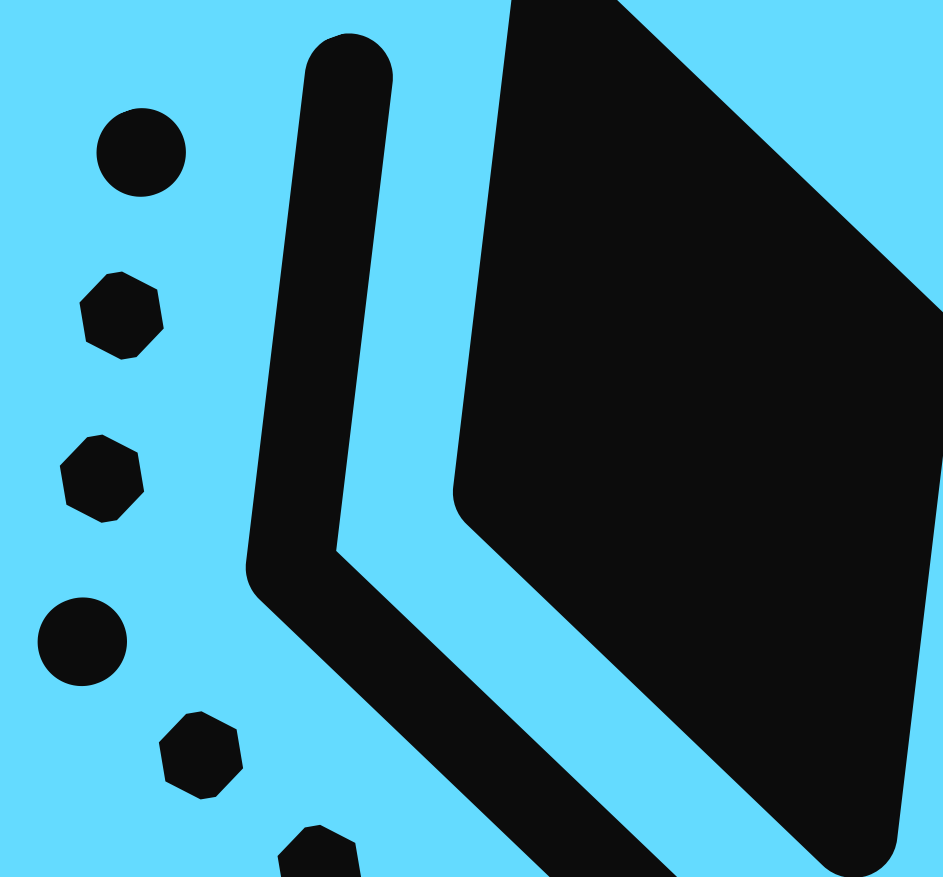
ANIMATION STUDIO

- THE ANIMATION STUDIO IS ARGUABLY THE MOST COMPLICATED TOOL WITHIN PROJECT LEVELS, SO PLEASE BEAR WITH ME IF I FAIL TO PROVIDE A CLEAR EXPLANATION. I'LL TRY MY BEST.
- FIRST OF ALL, **OPEN THE ANIMATION STUDIO** FROM **THE HOTBAR** AND **CREATE AN ANIMATION** BY CLICKING THE **+** BUTTON, YOU SHOULD NOW SEE SOMETHING SIMILAR TO THIS:



- ANIMATING AN OBJECT ESSENTIALLY MEANS TELLING IT HOW IT SHOULD “CHANGE” ACROSS A PERIOD OF TIME. THIS IS EXACTLY WHAT THE ANIMATION STUDIO ALLOWS YOU TO DO.
- THE FIRST **FRAGMENT** OF YOUR ANIMATION IS THE **BLUE SQUARE** YOU SEE ON THE LEFT. THIS REPRESENTS THE STATE OF YOUR OBJECT WHEN THE ANIMATION BEGINS.
- YOU CAN EITHER INPUT ALL OF ITS VALUES ONE BY ONE (POSITION XYZ, ROTATION XYZ, ETC...) OR YOU CAN TELL THE EDITOR TO DO IT FOR YOU BY SIMPLY HAVING THE OBJECT SELECTED AND CLICKING ON THE YELLOW “DOWNLOAD LIKE” ICON ON THE TOP OF THE FRAGMENT, CALLED THE **INHERIT** BUTTON.
- NOW THAT YOU’VE SET UP THE INITIAL STATE OF YOUR ANIMATION, YOU NEED TO ADD MORE FRAGMENTS TO TELL YOUR OBJECT HOW TO CHANGE OVER TIME AND ACHIEVE A FULL ANIMATION. TO DO SO, YOU HAVE TWO OPTIONS, REPRESENTED BY THE TWO LITTLE BUTTONS IMMEDIATELY NEXT TO YOUR (SO FAR) ONLY **BLUE FRAGMENT**.

ANIMATION STUDIO



- THE WHITE + BUTTON WILL CREATE **AN EMPTY NEW ANIMATION FRAGMENT**, WHILE THE THE YELLOW ONE WILL CREATE **A NEW FRAGMENT BY COPYING THE VALUES** OF THE OBJECT YOU CURRENTLY HAVE SELECTED. EITHER WAY, YOU WILL END UP WITH SOMETHING LIKE THIS:

ANIMATION STUDIO

ANIMATION ▾ New Animation9930

+

⌘

🗑

FRAGMENTS
TIMELINE

0

3

↓

POSITION

X 0 Y 0 Z 0

ROTATION

X 0 Y 0 Z 0

SCALE

X 1 Y 1 Z 1

3

↓

POSITION

X 5 Y 0 Z -5

ROTATION

X 0 Y 0 Z 0

SCALE

X 1 Y 1 Z 1

+

↓

- WHAT YOU’RE DOING HERE IS ESSENTIALLY TELLING YOUR OBJECT: “HEY. YOU ARE INITIALLY LIKE **THIS** AND AFTER AN X AMOUNT OF SECONDS I WANT YOU TO BE **LIKE THIS**”.
- **TO DECIDE THE X AMOUNT OF SECONDS THAT THE OBJECT WILL TAKE TO CHANGE BETWEEN THE TWO STATES** YOU SIMPLY HAVE TO **MODIFY THE SMALL NUMBER PLACED BETWEEN TWO FRAGMENTS**.
- LET’S ADD ONE MORE FRAGMENT AND END UP WITH SOMETHING LIKE THIS:

ANIMATION STUDIO

ANIMATION ▾ New Animation9930

+

⌘

🗑

FRAGMENTS
TIMELINE

0

3

8

↓

POSITION

X 0 Y 0 Z 0

ROTATION

X 0 Y 0 Z 0

SCALE

X 1 Y 1 Z 1

3

↓

POSITION

X 5 Y 0 Z -5

ROTATION

X 0 Y 0 Z 0

SCALE

X 1 Y 1 Z 1

5

↓

POSITION

X 10 Y 0 Z 10

ROTATION

X 90 Y 0 Z 0

SCALE

X 1 Y 1 Z 1

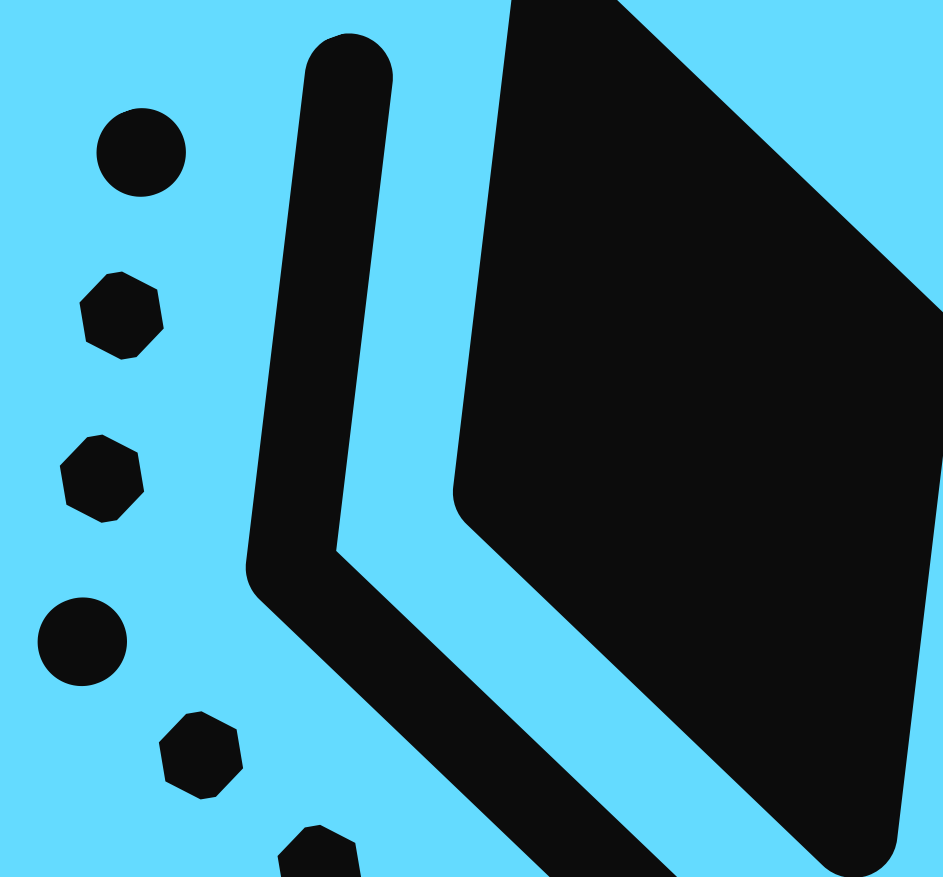
+

↓

- AND WHAT WE ESSENTIALLY HAVE HERE IS AN OBJECT THAT WILL START FROM THE CENTER OF THE WORLD (0,0,0), THAT AFTER 3 SECONDS WILL BE AT (5, 0, -5) AND THAT AFTER 5 MORE SECONDS WILL BE IN (10,0,10) ROTATED BY 90 DEGREES ON THE X AXIS. TAKING A TOTAL OF 8 SECONDS. THAT’S IT, YOU HAVE CREATED AN ANIMATION!

28

ANIMATION STUDIO



- ALL THAT’S LEFT TO DO NOW IS TO SELECT YOUR OBJECT, ADD AN **ANIMATED** PROPERTY TO IT, AND SELECT YOUR ANIMATION FROM THE DROPDOWN MENU. YOU’RE GOOD TO GO!

ANIMATION STUDIO

ANIMATION ▾ New Animation9930

FRAGMENTS
TIMELINE

0	3	8
POSITION X 0 Y 0 Z 0 ROTATION X 0 Y 0 Z 0 SCALE X 1 Y 1 Z 1	POSITION X 5 Y 0 Z -5 ROTATION X 0 Y 0 Z 0 SCALE X 1 Y 1 Z 1	POSITION X 10 Y 0 Z 10 ROTATION X 90 Y 0 Z 0 SCALE X 1 Y 1 Z 1

PROPERTIES

ANIMATED

New Animation9930

▶

⏸

■

● LOOP

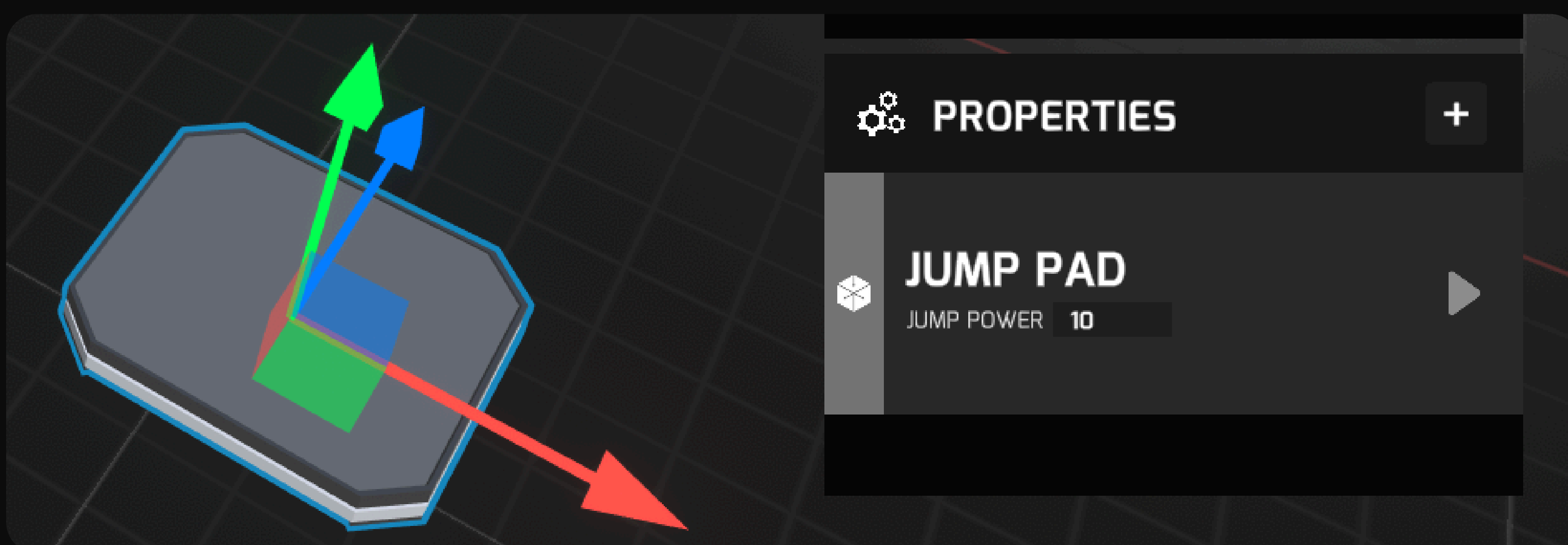
● PLAY ON LEVEL START

SPECIAL OBJECTS



SPECIAL OBJECTS

- UNLIKE STANDARD OBJECTS WHICH ARE ONLY PHYSICAL BODIES WITH NO ACTUAL USE, ALL SPECIAL OBJECTS ARE ABLE TO PERFORM A “SPECIAL ACTION”, THANKS TO THEIR **ATTRIBUTE**.
- AN **ATTRIBUTE** IS A SPECIAL QUIRK THAT IS UNIQUE TO A CERTAIN OBJECT AND IT’S DISPLAYED LIKE A PROPERTY CONTROLLER ON THE PROPERTIES TAB ONCE YOU’VE SELECTED THE CORRESPONDING OBJECT, HERE’S AN EXAMPLE FROM A **JUMP PAD**:



- AS YOU CAN SEE, ATTRIBUTES CAN BE MADE **EDITABLE** BUT THEY CAN’T BE **REMOVED** OR **ADDED**.
- I WILL NOW FOLLOW WITH A LIST AND A BRIEF EXPLANATION FOR ALL THE CURRENTLY AVAILABLE SPECIAL OBJECTS, ALONG WITH THEIR **ATTRIBUTE CONTROLLERS**.

SPECIAL OBJECTS



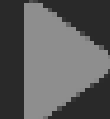
JUMP PAD

PROJECTS AN OBJECT OR A PLAYER WHICH HAS A **PHYSICS CORE PROPERTY** IN THE DIRECTION CURRENTLY PERPENDICULAR TO ITS ORIENTATION WITH A SPECIFIED **JUMP POWER**.



JUMP PAD

JUMP POWER **10**



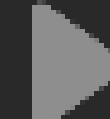
GRAVITY PAD

CHANGES A PLAYER'S OR OBJECT'S GRAVITY TO THE ONE PERPENDICULAR TO ITS OWN ORIENTATION. A NEW GRAVITY FORCE INTENSITY CAN ALSO BE SPECIFIED, BOTH POSITIVE AND NEGATIVE.



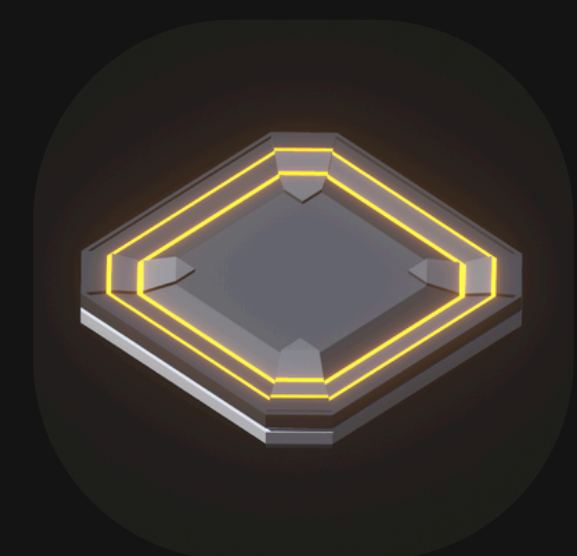
GRAVITY PAD

NEW GRAVITY PULL **10**



RESIZE PAD

RESIZES AN OBJECT BY MULTIPLYING HIS SCALE BY THE SPECIFIED NUMBER.
(0.1x = SMALLER, 10x = LARGER)



RESIZE PAD

0.1x 10x
1

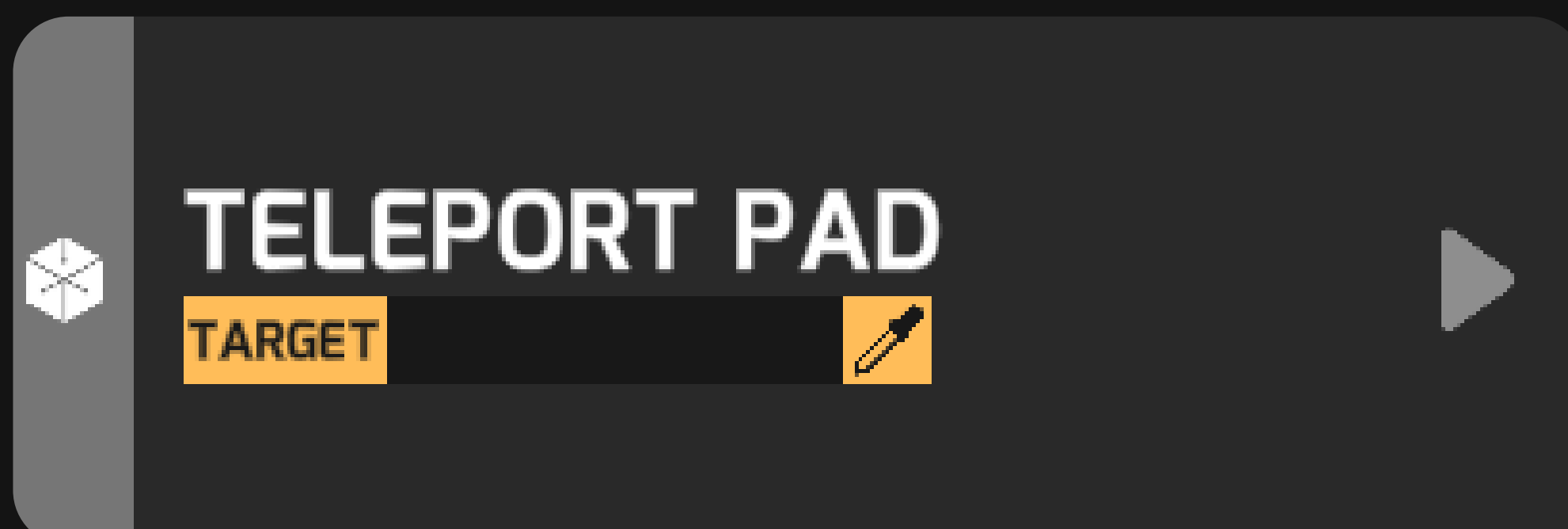


SPECIAL OBJECTS



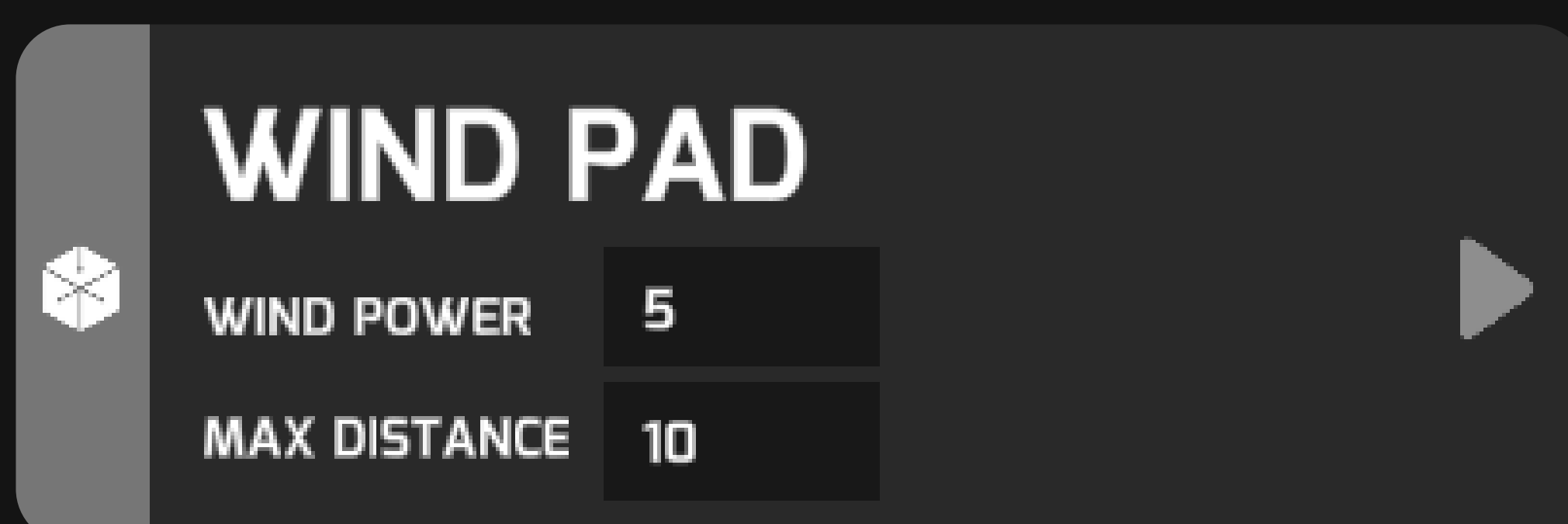
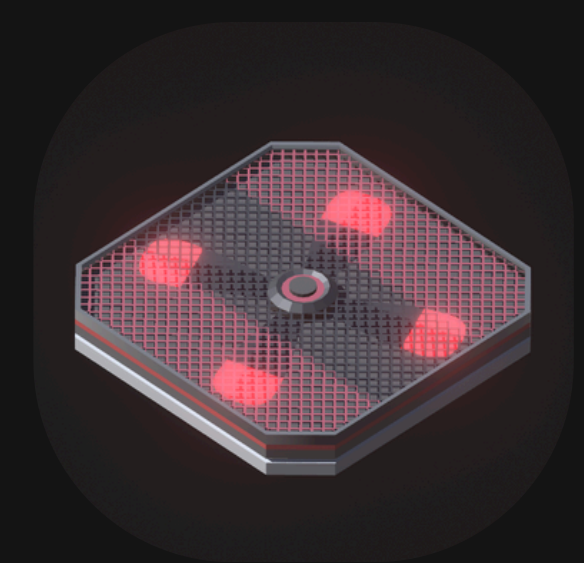
TELEPORT PAD

TELEPORTS ANYTHING THAT TOUCHES IT TO A TARGET OBJECT. YOU CAN ALSO PICK ANOTHER PORTAL AS A TARGET TO CREATE A LOOP. OBJECTS WILL MAINTAIN SPEED WHEN CROSSING PORTALS.



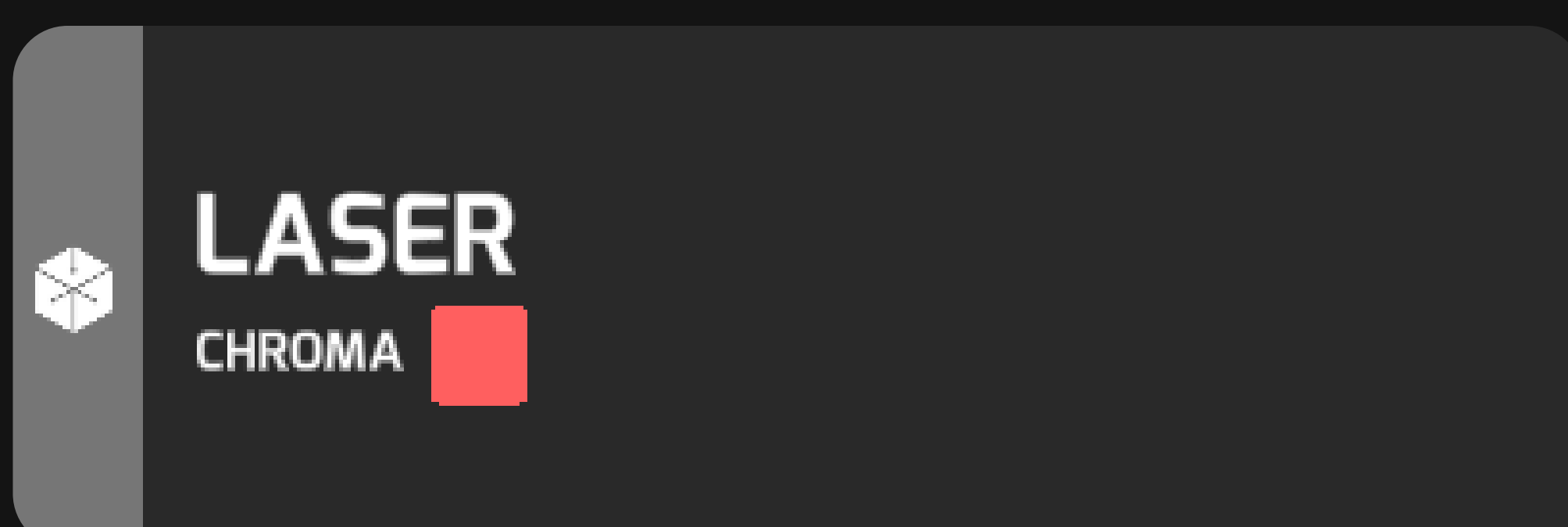
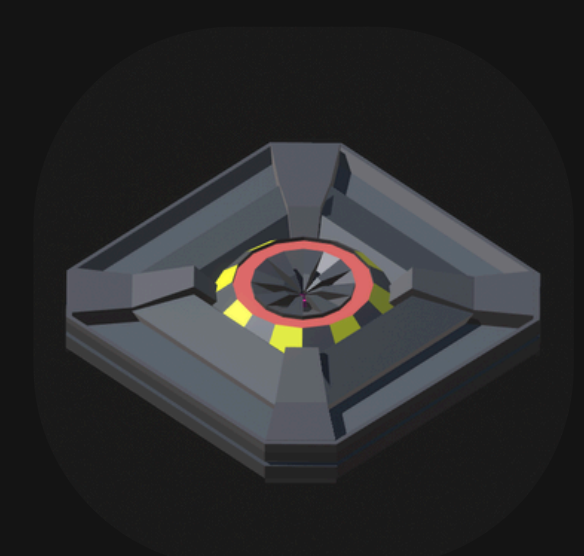
WIND PAD

BLOWS OBJECTS OR PLAYERS AWAY IF SET TO POSITIVE POWER, SUCKS THEM IN IF SET TO NEGATIVE POWER. A MAX DISTANCE OF EFFECT CAN BE SPECIFIED.



LASER PAD

SHOOTS A LASER BEAM OF THE SPECIFIED COLOR. LASERS WILL KILL PLAYERS, DESTROY BREAKABLE OBJECTS AND CROSS PORTALS.



SPECIAL OBJECTS



MINI CAMERA

IF CLICKED UPON, ALLOWS TO SEE THROUGH IT BY CLICKING THE **USE** BUTTON. YOU CAN ALSO SELECT ANOTHER MINICAMERA WHEN LOOKING THROUGH ONE.



MINI CAMERA

USE

DRONE

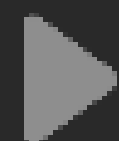
A DRONE WILL TARGET AND SHOOT ANY PLAYERS OR **PHYSICS ENABLED OBJECTS** WHEN THEY WALK INTO ITS LIGHT. BREAKING LINE OF SIGHT ABORTS THE SHOT. TIME REQUIRED FOR SHOOTING A TARGET AFTER SEEING IT CAN BE MODIFIED WITH **AIM TIME**.



DRONE

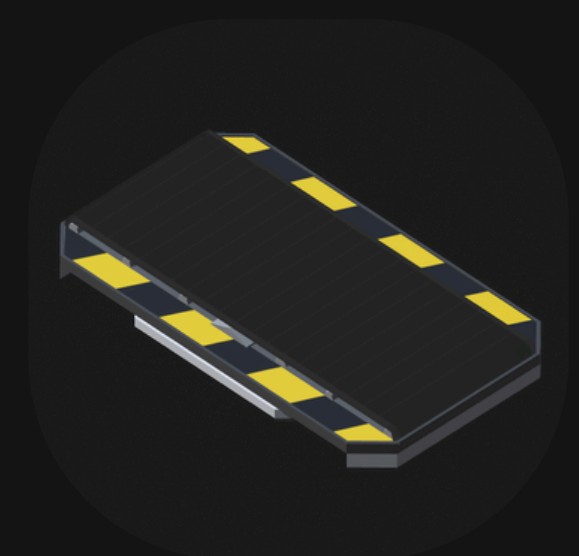
AIM TIME

1.5



CONVEYOR BELT

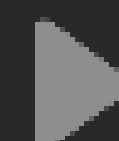
DRAGS OBJECTS THAT LAND UPON ITS SURFACE WITH THE SPECIFIED SPEED.



CONVEYOR BELT

BELT SPEED

1

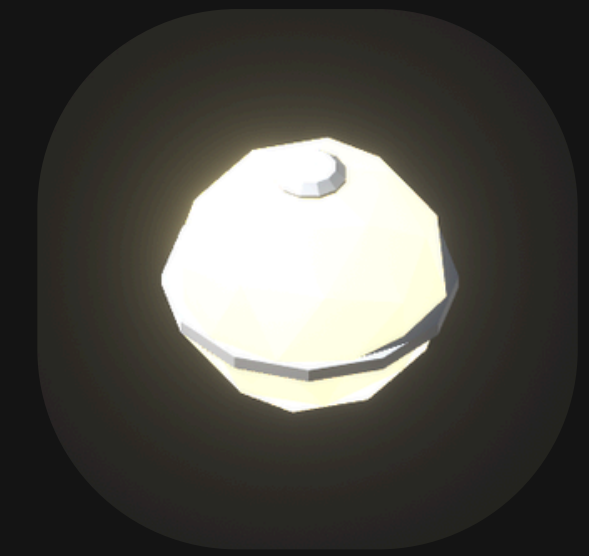


SPECIAL OBJECTS



LIGHT

A POINT LIGHT THAT ILLUMINATES ITS SURROUNDINGS. YOU CAN SPECIFY ITS POWER, THE MAX DISTANCE OF EFFECT AND THE COLOR.



**LIGHT**

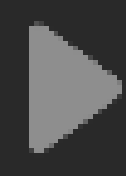
POWER

1

MAX DISTANCE

10

CHROMA



SPEED PAD

GIVES A SHORT SPEED BURST TO ANYTHING THAT TOUCHES IT. IF IT'S A PLAYER, YOU CAN CONFIGURE NEW VALUES FOR ITS MOVEMENT SPEEDS.



**SPEED PAD**

SPEED BURST

25

NEW WALK SPEED

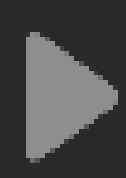
3.5

NEW RUN SPEED

7

NEW CROUCH SPEED

1.75



TIME PAD

ALLOWS YOU TO CHANGE THE FLOW OF TIME, EITHER GLOBALLY OR JUST FOR THE OBJECT THAT LANDS ON IT.



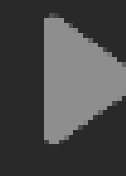
**TIME PAD**

NEW TIMEFLOW

1

GLOBAL

☐



SPECIAL OBJECTS



CHECKPOINT

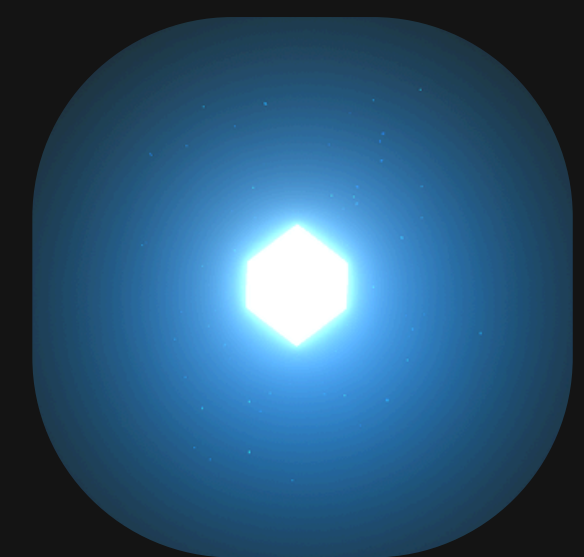
IF A PLAYER TOUCHES A CHECKPOINT HE WILL RESTART FROM THERE WHEN A GAME OVER IS TRIGGERED. CHECKPOINTS PERSIST ACROSS DIFFERENT SESSIONS.

GENERATE SAVEFILE ALLOWS YOU TO DECIDE WHETHER YOU WANT THIS CHECKPOINT TO SAVE THE WHOLE LEVEL'S STATE (WHICH IS EXPENSIVE AND POSSIBLY LONG, SO CHOOSE CAREFULLY) OR JUST THE PLAYER'S STATE.



CHECKPOINT

GENERATE SAVEFILE  

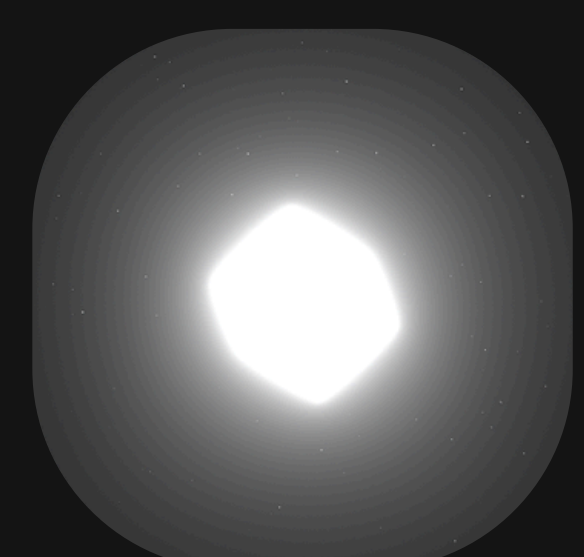


GOAL

THE END OF A LEVEL. IF THE PLAYER GETS CLOSE A CUTSCENE WILL PLAY AND THE LEVEL WILL END. ALL LEVELS MUST BE COMPLETED BEFORE THEY CAN BE UPLOADED.



GOAL

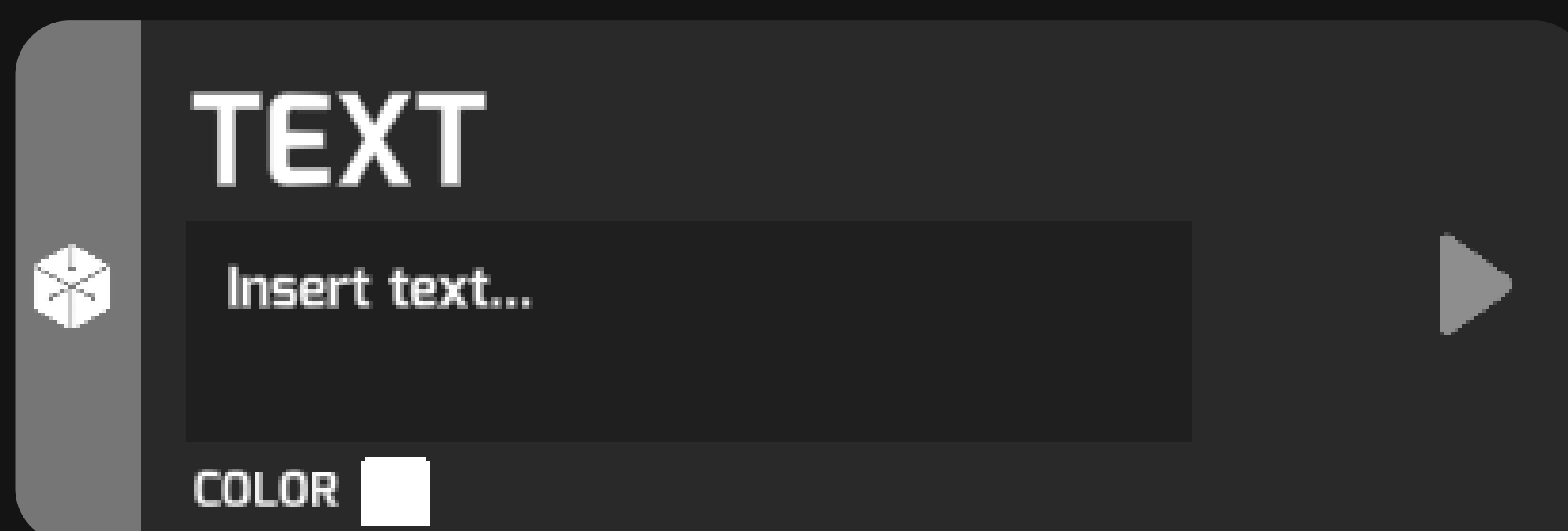


SPECIAL OBJECTS



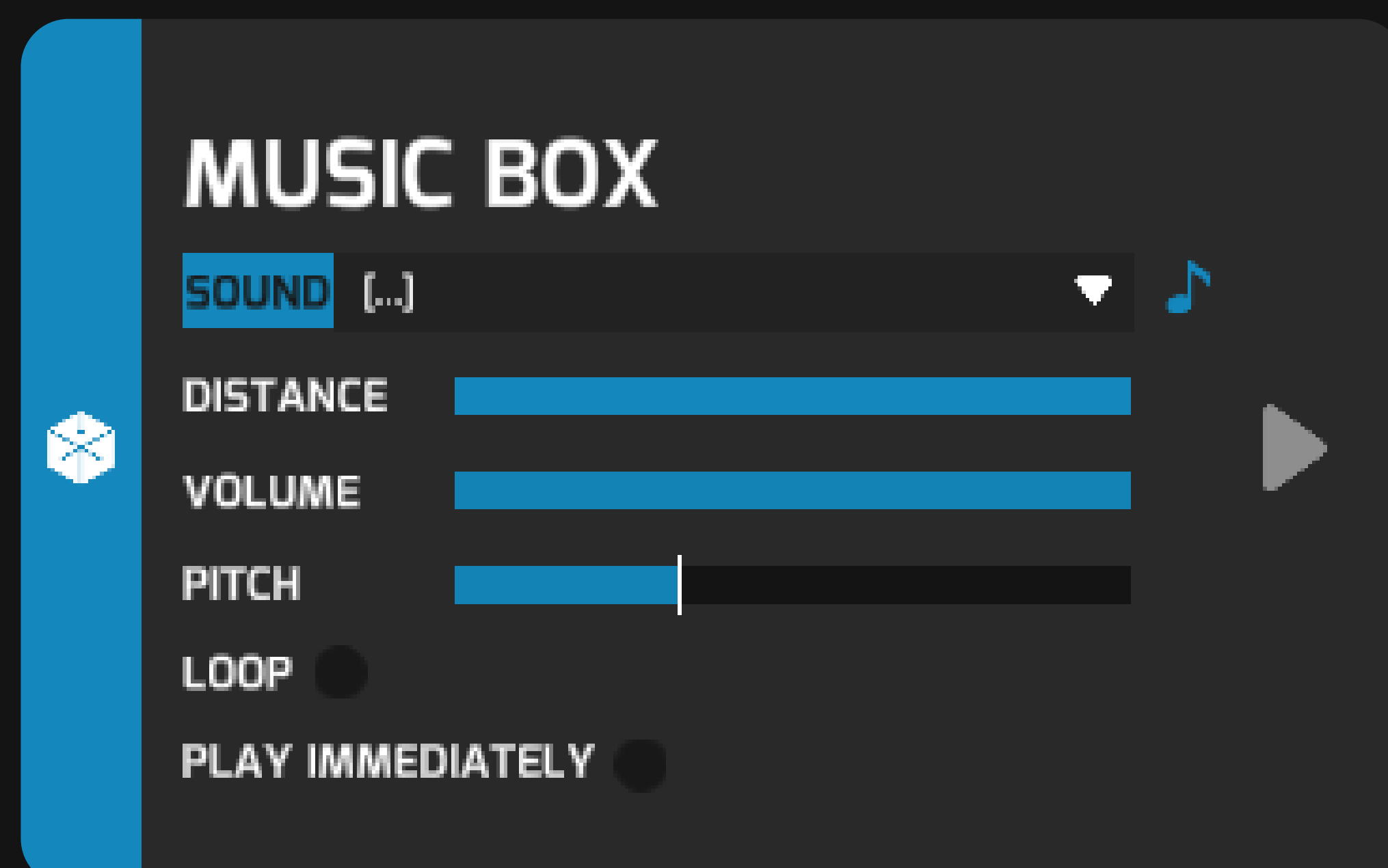
TEXT

A 3D TEXT WHOSE COLOR CAN BE SPECIFIED.
RICH FORMATTING ETC... IS ALSO SUPPORTED.

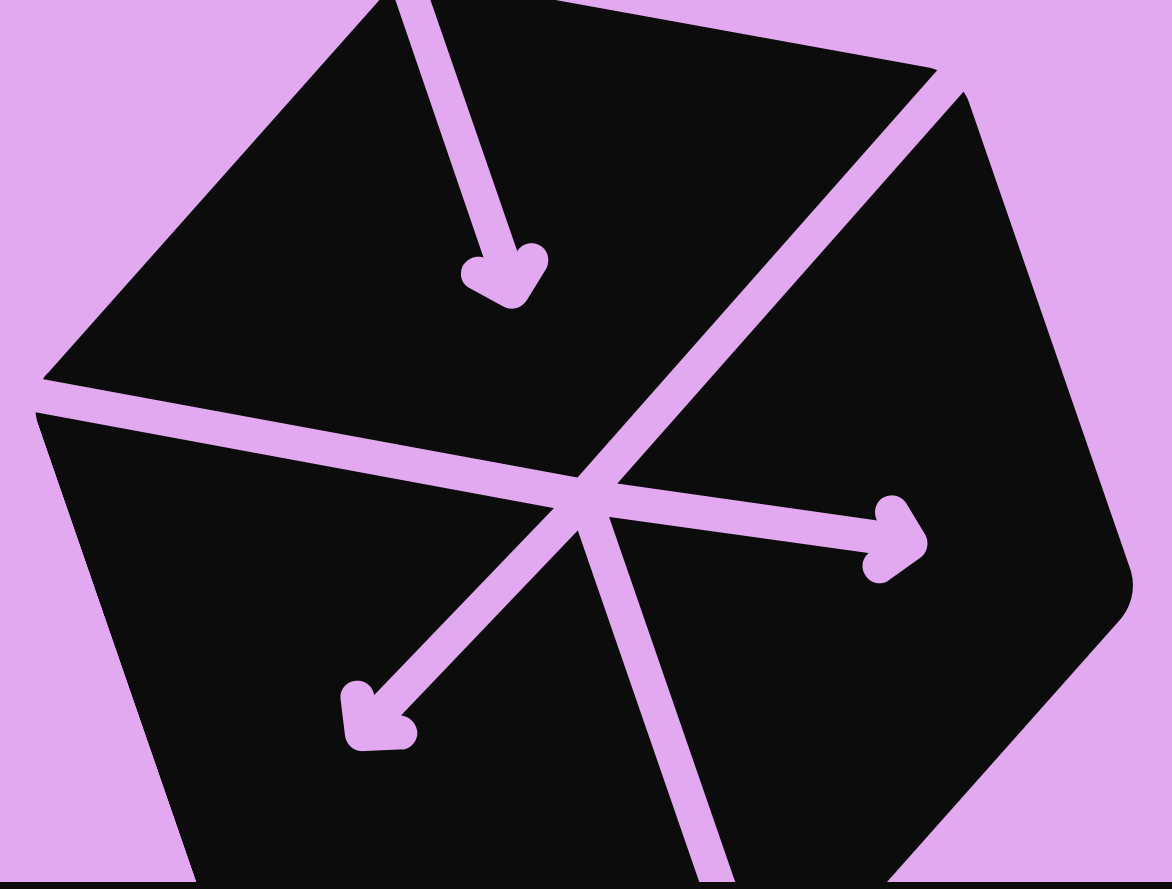


MUSIC BOX

A MUSIC BOX IS A MODULE THAT STORES AND PLAYS BACK A SPECIFIED SOUND UPON REQUEST. USE THE DROPDOWN MENU TO PICK A SOUND AND PRESS THE NOTE ICON NEXT TO IT TO LISTEN TO A PREVIEW. THE SLIDERS ALLOW YOU TO RESPECTIVELY CONTROL THE MAXIMUM DISTANCE AT WHICH THE SOUND CAN BE HEARD, ITS VOLUME AND ITS PITCH. THE EDITOR WILL HELP YOU VISUALIZE THE AREA IN WHICH THE SOUND WILL PLAY BY CREATING A BLUE BUBBLE AROUND THE MUSIC BOX AS YOU REGULATE THE PARAMETERS. IF THE “DISTANCE” SLIDER IS SET TO ZERO, THE SOUND WILL BE HEARD IN THE WHOLE LEVEL REGARDLESS OF DISTANCE. BY USING THE TOGGLES BELOW YOU CAN ALSO SPECIFY WHETHER THE SOUND SHOULD KEEP PLAYING INFINITELY ONCE IT ENDS OR PLAY ONLY ONCE. “PLAY IMMEDIATELY” SPECIFIED WHETHER THE SOUND SHOULD WAIT FOR AN EXTERNAL INPUT (EX: TRIGGER) OR JUST START PLAYING IMMEDIATELY ONCE THE LEVEL IS LOADED.



TRIGGERS



TRIGGERS

TRIGGERS ARE ONE OF THE MOST IMPORTANT COMPONENTS OF ALL MODERN VIDEOGAMES.

THEY ARE, ESSENTIALLY, DESIGNATED AREAS WHICH DETECT WHEN SOMETHING (A PLAYER, AN OBJECT, A BULLET, ETC...) ENTERS, EXITS OR STAYS WITHIN THEM. IN PROJECT LEVELS, TRIGGERS ARE SPECIAL OBJECTS SHAPED LIKE CUBES THAT YOU CAN SETUP TO EXECUTE VARIOUS TASKS WHEN ACTIVATED, OR, AS THEIR NAME SUGGESTS, “TRIGGERED”.



LET’S HAVE A LOOK AT THEIR CONTROLLER:

TRIGGER

WHEN **ENTERS** **THIS AREA:**

CREATE OBJECT

OBJECT: CUBE NAME:

WHERE

POSITION	0	5	0
ROTATION	0	0	0
SCALE	1	1	1

AS YOU CAN SEE, THE FIRST THINGS THAT A TRIGGER NEEDS TO KNOW IS WHO IS ITS TARGET AND WHICH ACTION THE TARGET HAS TO PERFORM TO ACTIVATE THE TRIGGER.

USE THE **PICKER** TO SELECT AN OBJECT OR A PLAYER WITHIN YOUR LEVEL, THEN USE THE DROPDOWN IMMEDIATELY TO THE RIGHT TO SPECIFY WHETHER THE TRIGGER SHOULD BE “TRIGGERED” WHEN THE TARGET ENTERS OR EXITS ITS VOLUME. ONCE THAT’S BEEN TAKEN CARE OF, YOU NEED TO SELECT AN **EVENT** FROM THE DROPDOWN MENU IMMEDIATELY BELOW THE PICKER:

TRIGGER

WHEN **ENTERS** **THIS AREA:**

STOP MUSIC BOX

CREATE OBJECT

DELETE OBJECT

Duplicate Object

TELEPORT OBJECT

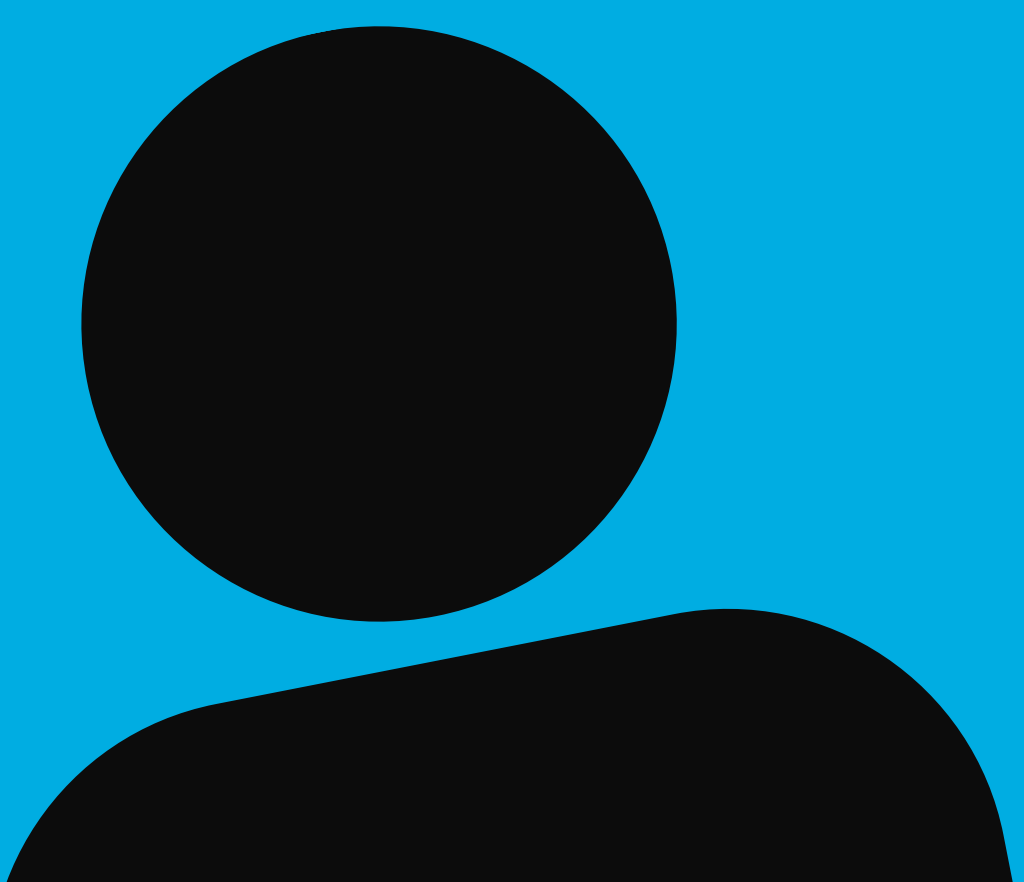
START MUSIC BOX

▶ STOP MUSIC BOX

START ANIMATION

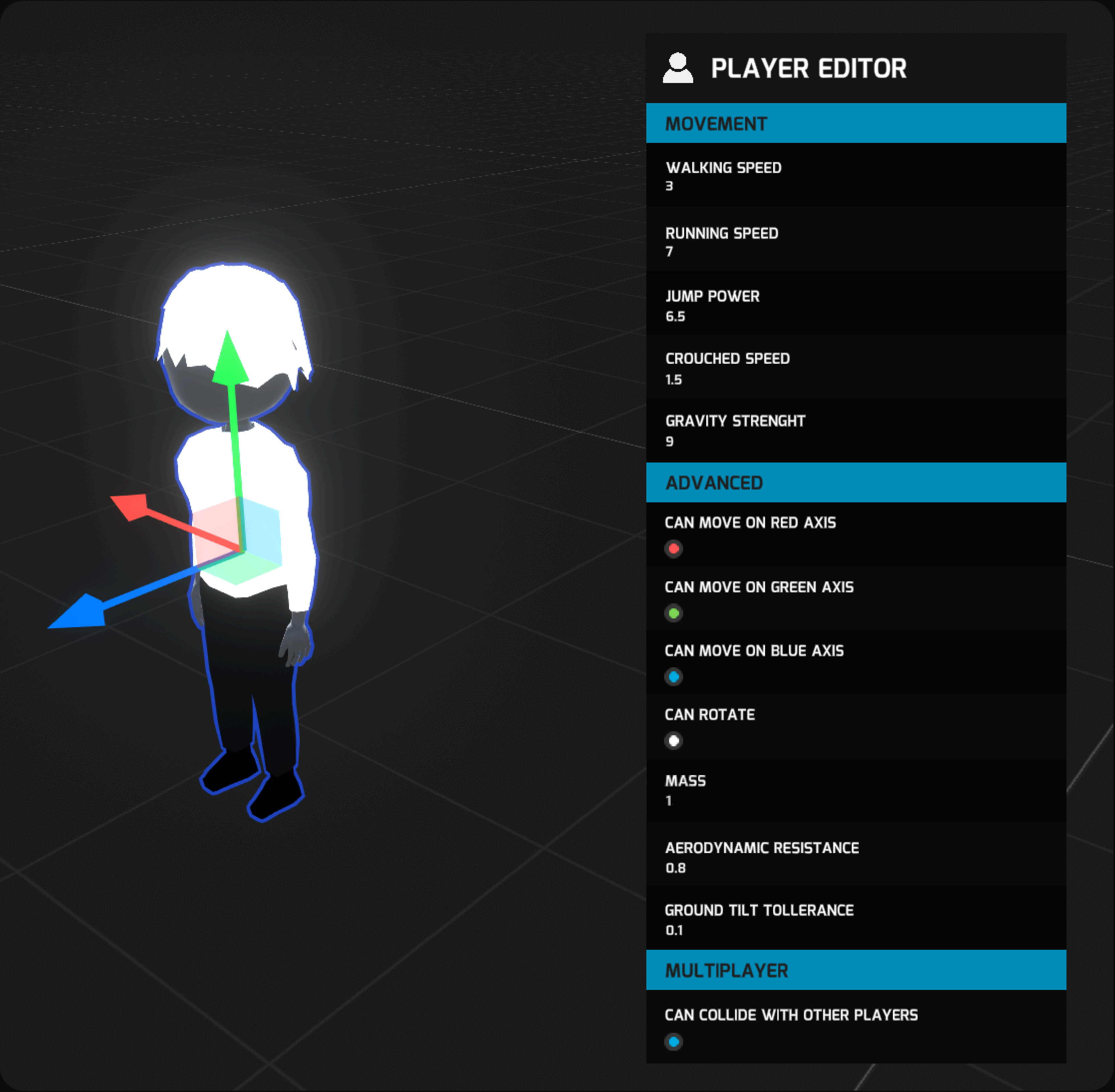
AS YOU’VE PROBABLY GUESSED, ONCE THE DESIGNATED TARGET ENTERS OR EXITS THE TRIGGER AREA, THE EVENT THAT YOU’VE CHOSEN (AND SETTED UP) WILL BE EXECUTED AUTOMATICALLY. TRIGGERS ARE TRICKY ENTITIES AND THEY MAY CAUSE SOME PRETTY SERIOUS PROBLEMS, SO PLEASE, USE THEM WITH REASON, ESPECIALLY IN MULTIPLAYER.

PLAYER EDITOR

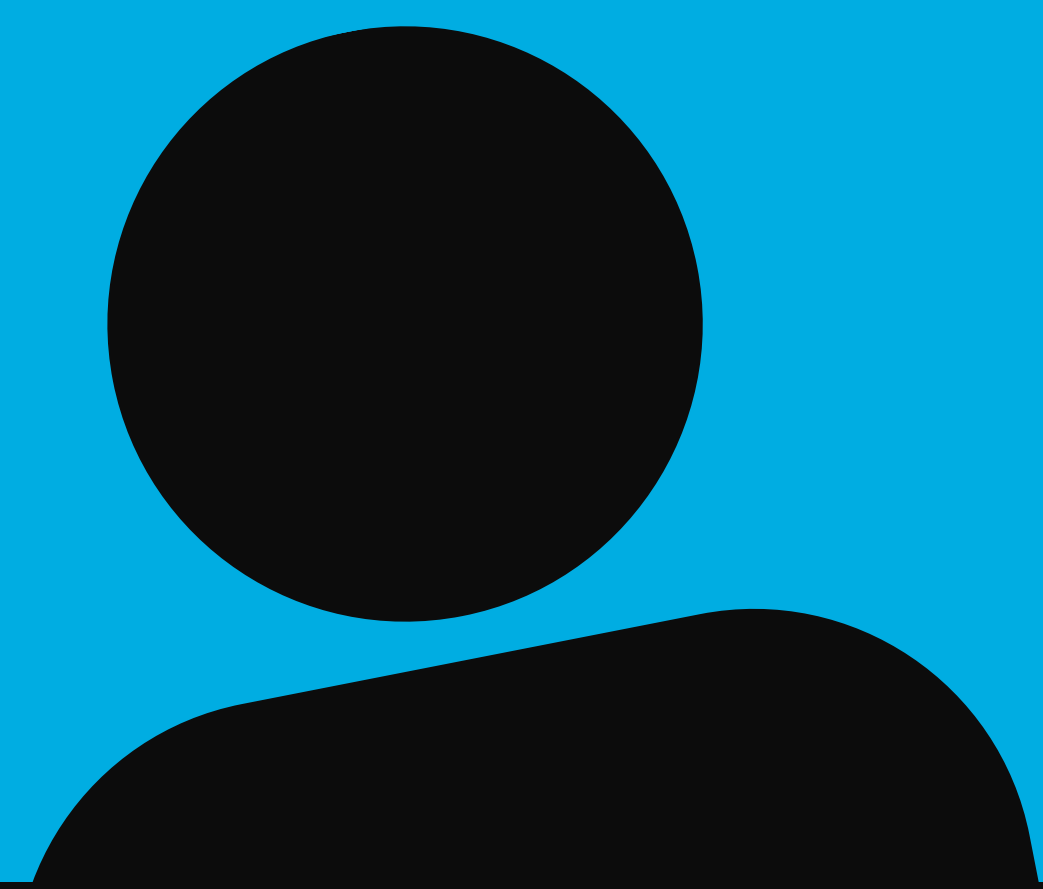


PLAYER EDITOR

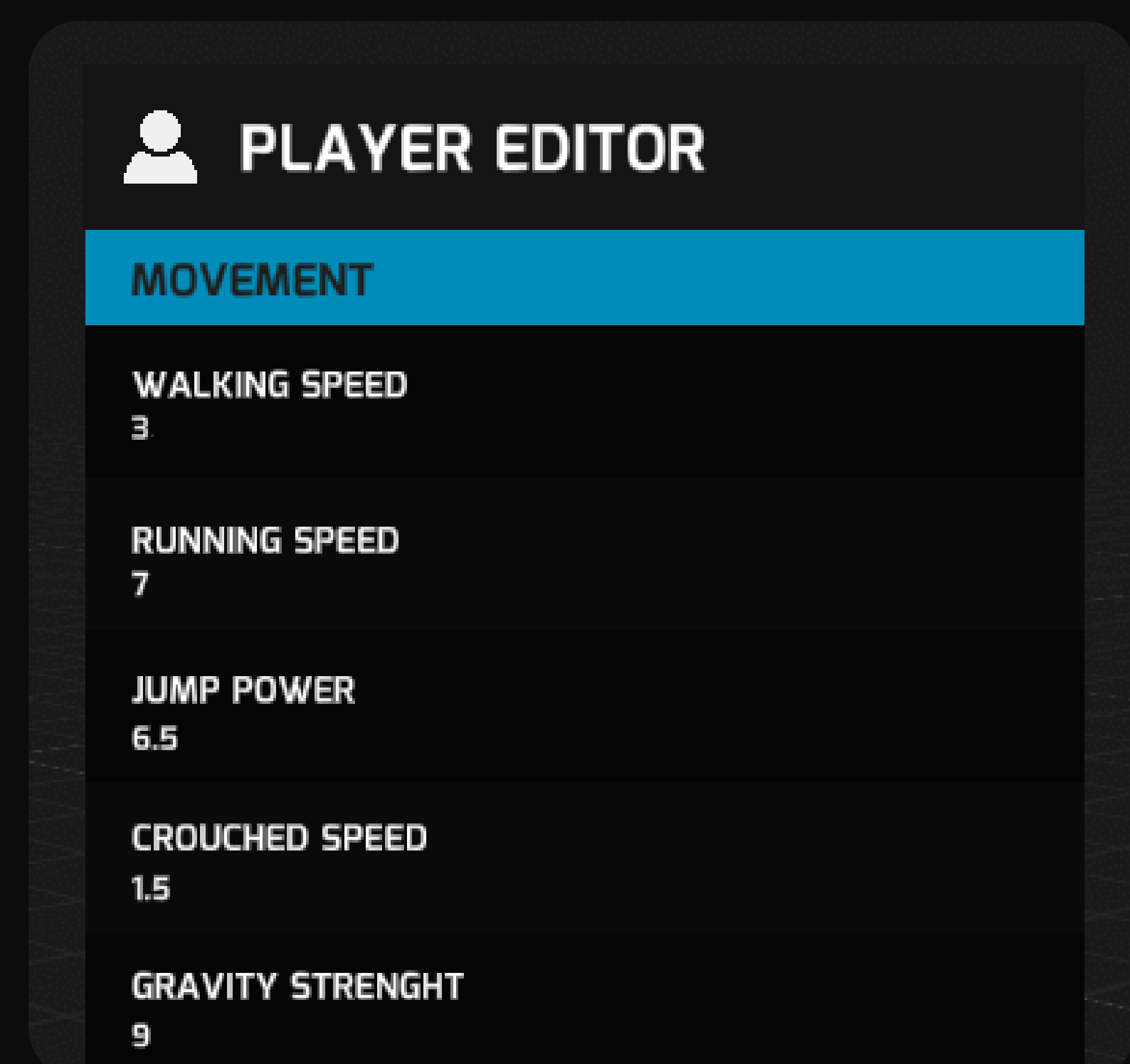
- THE PLAYER EDITOR IS A SPECIAL TAB WITHIN PROJECT LEVELS THAT ALLOWS YOU TO CONTROL HOW THE PLAYER BEHAVES AND INTERACTS WITH THE WORLD.
- IN ORDER TO MAKE IT SHOW UP, YOU JUST HAVE TO SELECT THE PLAYER:



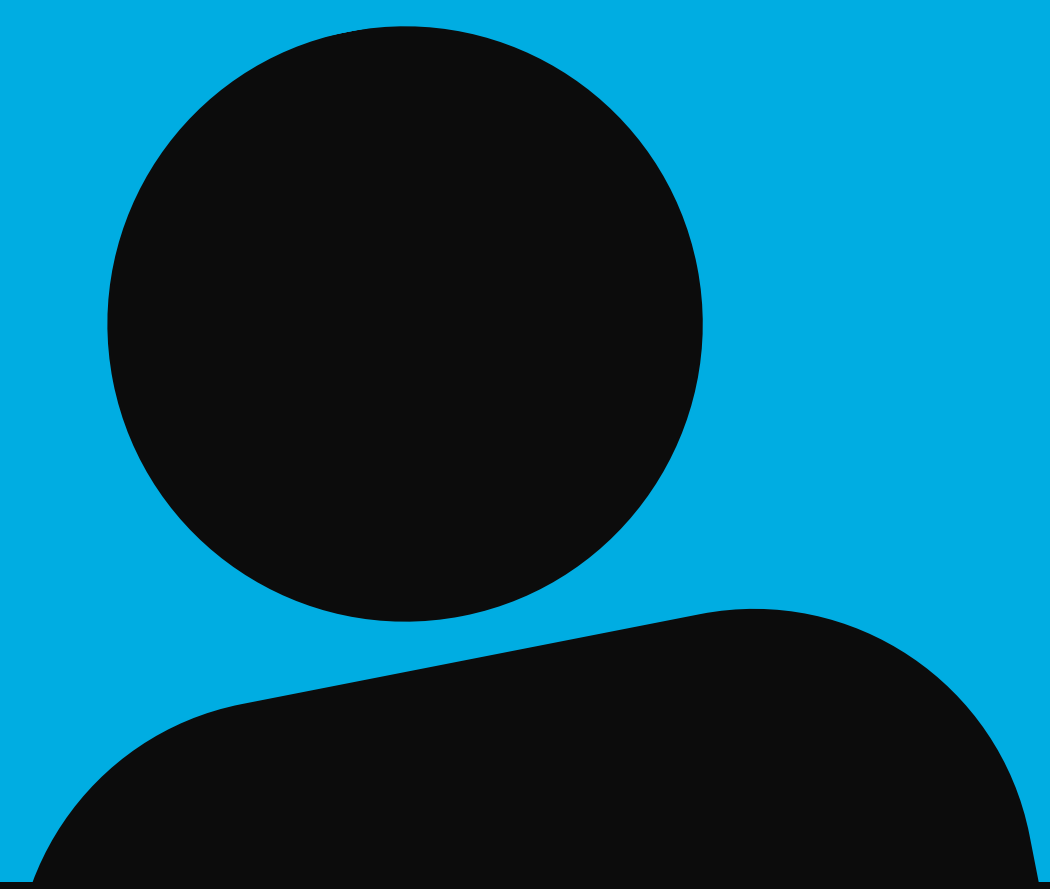
PLAYER EDITOR



- ALTHOUGH MANY SETTINGS ARE PRETTY INTUITIVE, I'LL GO OVER EVERY SINGLE ONE OF THEM FOR THE SAKE OF DOING THINGS RIGHT.
- THE **MOVEMENT** SECTION ALLOWS YOU TO CONTROL THE MOVEMENT SPEEDS OF THE PLAYER DURING VARIOUS SITUATIONS, FROM THE TOP YOU HAVE **WALKING SPEED**, **RUNNING SPEED** AND **CROUCHED SPEED**.
- **JUMP POWER** ALLOWS YOU TO CONTROL HOW HIGH THE PLAYER CAN JUMP.
- **GRAVITY STENGHT** INDICATES HOW STRONG THE PLAYER WILL BE PULLED DOWN FROM HIS FEET. REMEMBER THAT THE GRAVITY DIRECTION IS DECIDED THE PLAYER'S ROTATION WHEN THE LEVEL BEGINS. YOU CAN MODIFY THIS DURING THE LEVEL BY USING **GRAVITY PADS**. GRAVITY NORMALLY HAS A NEGATIVE SIGN IN FRONT OF IT, BUT FOR THE SAKE OF SIMPLICITY I HAVE REMOVED IT HERE. YOU CAN STILL APPLY A NEGATIVE GRAVITY AND THAT WILL CAUSE THE MINUS TO CANCEL AND THE PLAYER TO FLOAT UPWARDS ($-1 * -9 = +9$)
- **GRAVITY** WITHIN PROJECT LEVELS BEHAVES EXACTLY LIKE IN THE REAL WORLD, AN ACCELERATION THAT IGNORES MASS.



PLAYER EDITOR



- IN THE **ADVANCED** SECTION YOU CAN FURTHER CUSTOMIZE THE PLAYER'S INTERACTION WITH THE WORLD, MY RECOMENDATION AS OF EARLY ACCESS IS TO LEAVE THESE SETTINGS UNCHANGED UNLESS REALLY NECESSARY AS THEY'RE PRETTY FRAGILE.
- THE THREE **CAN MOVE** TOGGLES ALLOW YOU TO LOCK MOVEMENT FOR THE PLAYER ON SPECIFIC AXES, WHICH CAN BE USEFUL FOR 2D LEVELS.
- **CAN ROTATE** SIMPLY DICTATES WHETHER THE PLAYER CAN SPIN ON ITSELF OR NOT.
- **MASS** ALLOWS YOU TO SPECIFY HOW HARD IT WILL BE FOR THE PLAYER TO MOVE, JUMP, PUSH AND BE PUSHED. JUST LIKE IN REAL LIFE HOWEVER YOUR MASS DOESN'T AFFECT HOW GRAVITY PULLS YOU. YOU WILL ALWAYS FALL WITH THE SAME GRAVITY REGARDLESS OF THE MASS YOU SET HERE.
- **AERODYNAMIC RESISTANCE** ALLOWS YOU TO DETERMINE HOW EASY OR HOW HARD IS IT FOR THE PLAYER TO MOVE WHILE MID-AIR. THE CLOSER THIS VALUE IS TO ZERO, THE HARDER IT WILL BE FOR THE PLAYER TO MOVE MID-AIR.
- **GROUND TILT TOLLERANCE** LETS YOU DECIDE THE MAXIMUM STEPNESS OF A SURFACE THAT THE PLAYER CAN WALK ON WITHOUT SLIPPING. THIS BREAKS REALLY EASILY SO YOU SHOULDN'T BE CHANGING IT UNTIL FUTURE UPDATES MAKE IT MORE STABLE. WITH THE DEFAULT VALUE, THE PLAYER CAN WALK ON SURFACES UP TO 40°-45° DEGRESS AND IT WILL START TO SLOW DOWN AFTER 15-20°.
- THE LAST SECTION REFERS TO **MULTIPLAYER SETTINGS**, AS OF NOW, YOU CAN ONLY SPECIFY WHETHER PLAYERS IN MULTIPLAYER SESSIONS **CAN COLLIDE AND PUSH EACH OTHER** OR NOT.

The screenshot shows a mobile-style interface for the Player Editor. It has a dark background with blue accents. The 'ADVANCED' section is expanded, showing several settings with toggle switches and numerical inputs. The 'MULTIPLAYER' section is partially visible below.

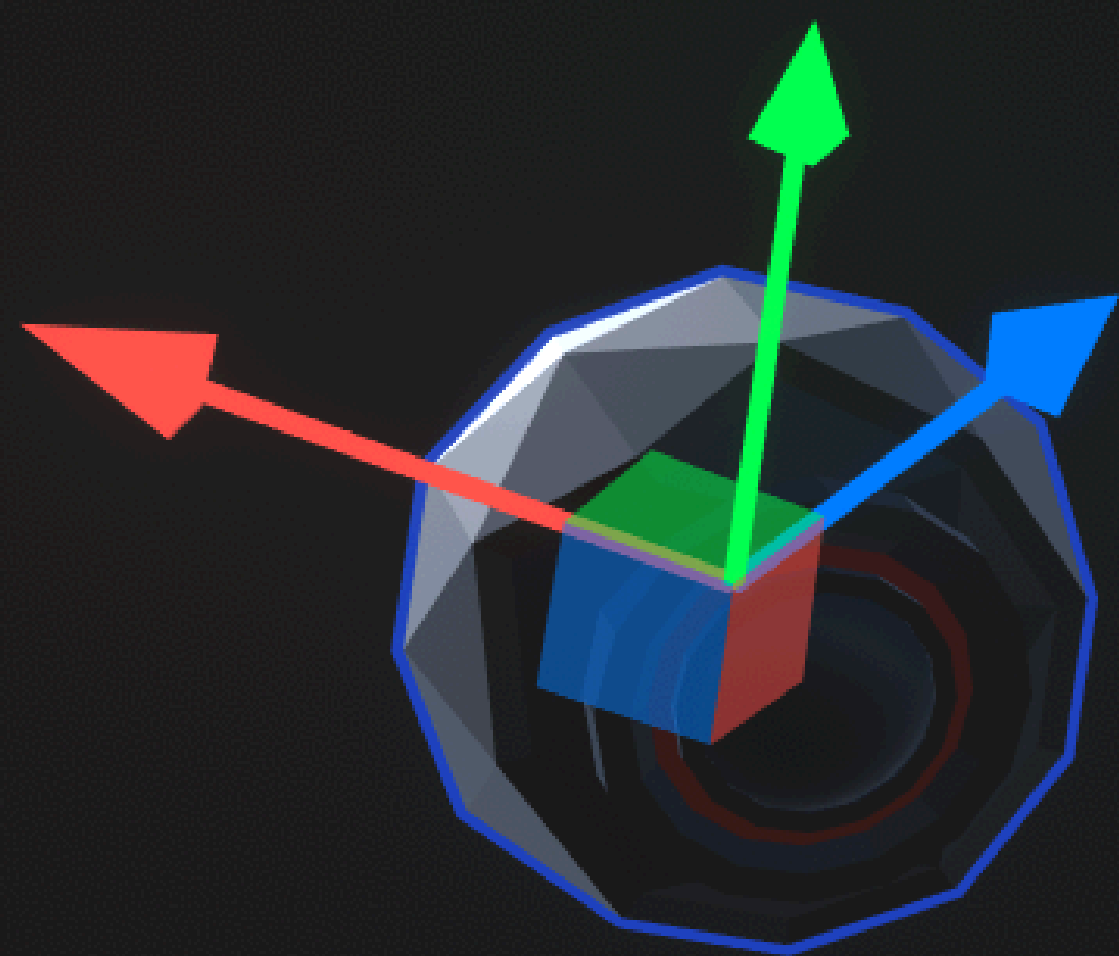
ADVANCED	
CAN MOVE ON RED AXIS	☐
CAN MOVE ON GREEN AXIS	☒
CAN MOVE ON BLUE AXIS	☒
CAN ROTATE	☐
MASS	1
AERODYNAMIC RESISTANCE	0.8
GROUND TILT TOLLERANCE	0.1
MULTIPLAYER	
CAN COLLIDE WITH OTHER PLAYERS	☒

CAMERA EDITOR



CAMERA EDITOR

- THE CAMERA EDITOR IS A SPECIAL TAB WITHIN PROJECT LEVELS THAT ALLOWS YOU TO CONTROL HOW THE CAMERA BEHAVES DURING PLAYMODE.
- IN ORDER TO MAKE IT SHOW UP, YOU JUST HAVE TO SELECT THE CAMERA:



 **CAMERA EDITOR**

VIEW



BACKGROUND COLOR

CAMERA TYPE

3D ▼

FIELD OF VIEW (FOV)

60

BEHAVIOUR

FIRST PERSON

☒

THIRD PERSON

☐

STATIC

☐

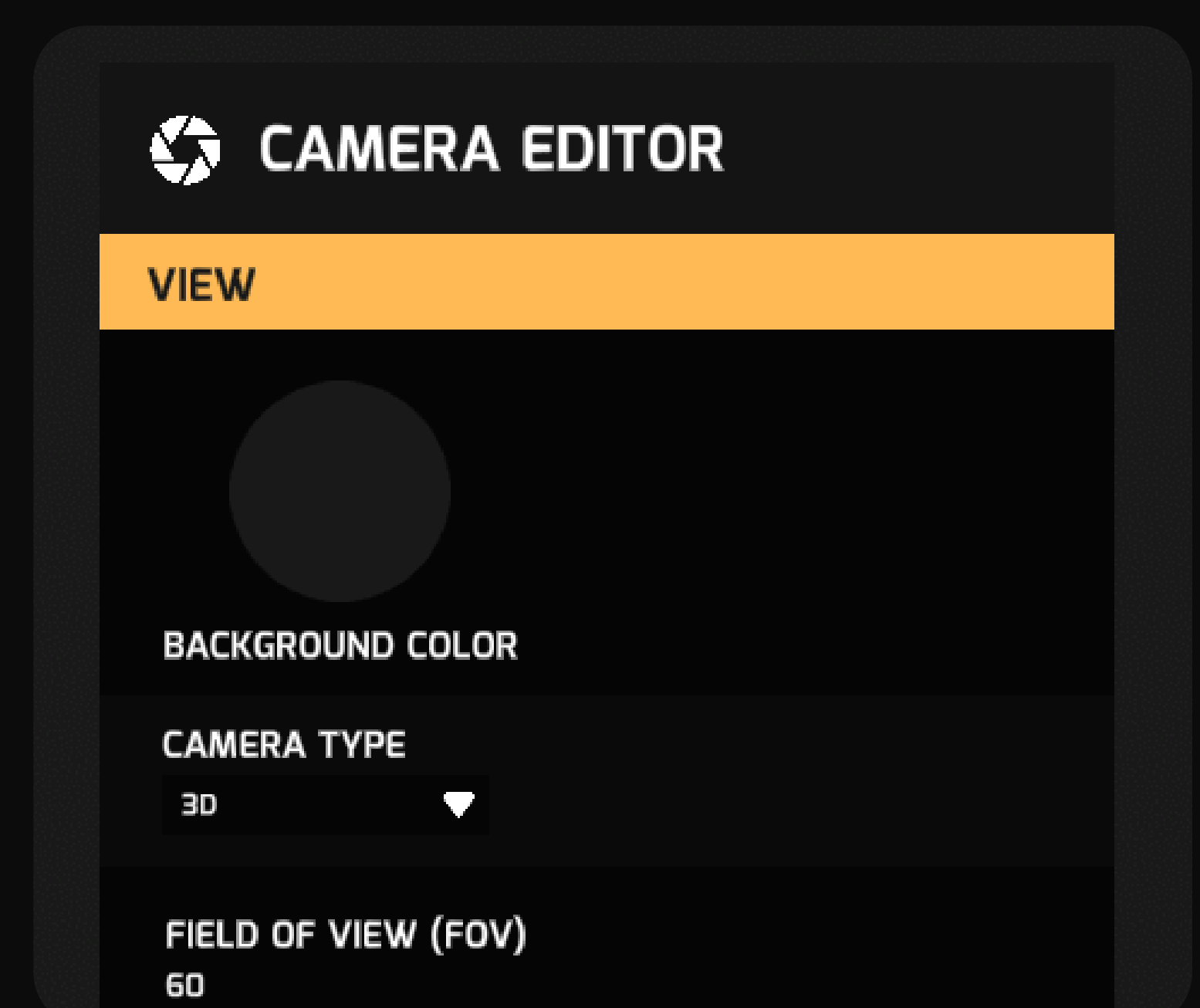
PREVIEW

- LET'S QUICKLY GO THROUGH THE VARIOUS SETTINGS. AGAIN, EVERYTHING SHOULD BE REALLY INTUITIVE, BUT IM DOING THEM ALL FOR THE SAKE OF COMPLETIONISM.

CAMERA EDITOR



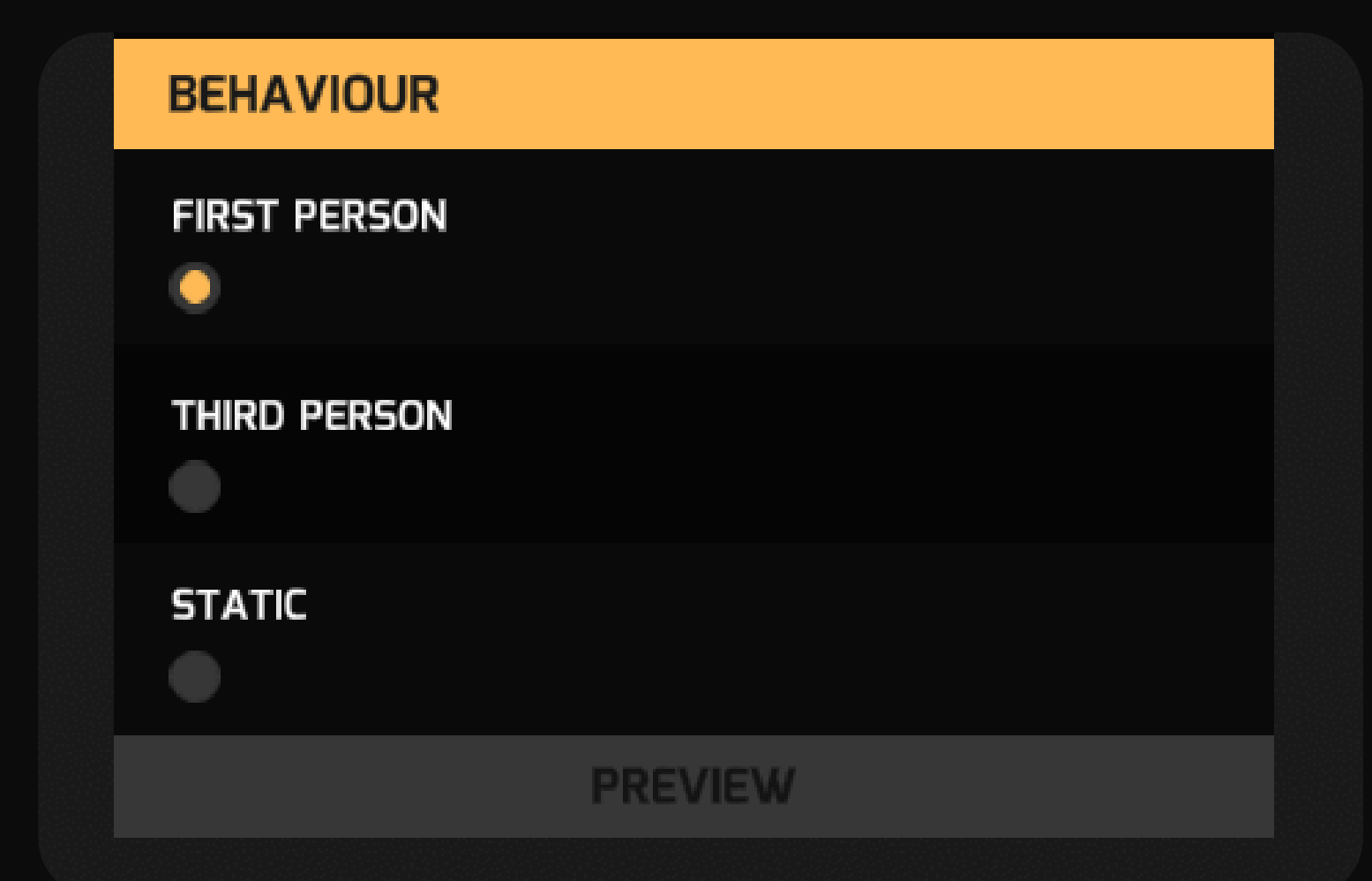
- THE **BACKGROUND COLOR** VALUE ALLOWS YOU TO ADJUST THE WORLD'S COLOR FROM THE PLAYER'S PERSPECTIVE. IF NOTHING CHANGES WHEN YOU MODIFY IT, DON'T WORRY, IT'S PERFECTLY NORMAL. ALL OF THE CAMERA SETTINGS HERE ARE VISIBLE ONLY DURING PLAYMODE OR IF YOU ENABLE **PREVIEW MODE** BY CLICKING THE BIG **PREVIEW** BUTTON AT THE BOTTOM OF THE PANEL.
- **CAMERA TYPE** ALLOWS YOU TO DECIDE IF YOU WANT THE CAMERA TO BE PERSPECTIVE OR ORTHOGRAPHIC. IT CAN BE CONFUSING SINCE I'VE DECIDED TO LABEL THEM AS **3D** AND **2D**, BUT THAT'S ESSENTIALLY WHAT YOU'LL ACHIEVE. A **PERSPECTIVE** CAMERA BEHAVES NORMALLY LIKE YOUR EYES DO IN REAL LIFE, WHILE AN **ORTHOGRAPHIC** CAMERA HAS NO DEPTH AND CAN BE USED TO CREATE 2D AND ISOMETRIC LEVELS.
- **FIELD OF VIEW** IS THE CAMERA'S FIELD OF VIEW (WOW, SHOCKING). KEEP IN MIND THAT PLAYERS WILL ACTUALLY USE THIS ONLY IF THEY'RE NOT FORCING A CUSTOM FOV FROM THEIR GAME SETTINGS.



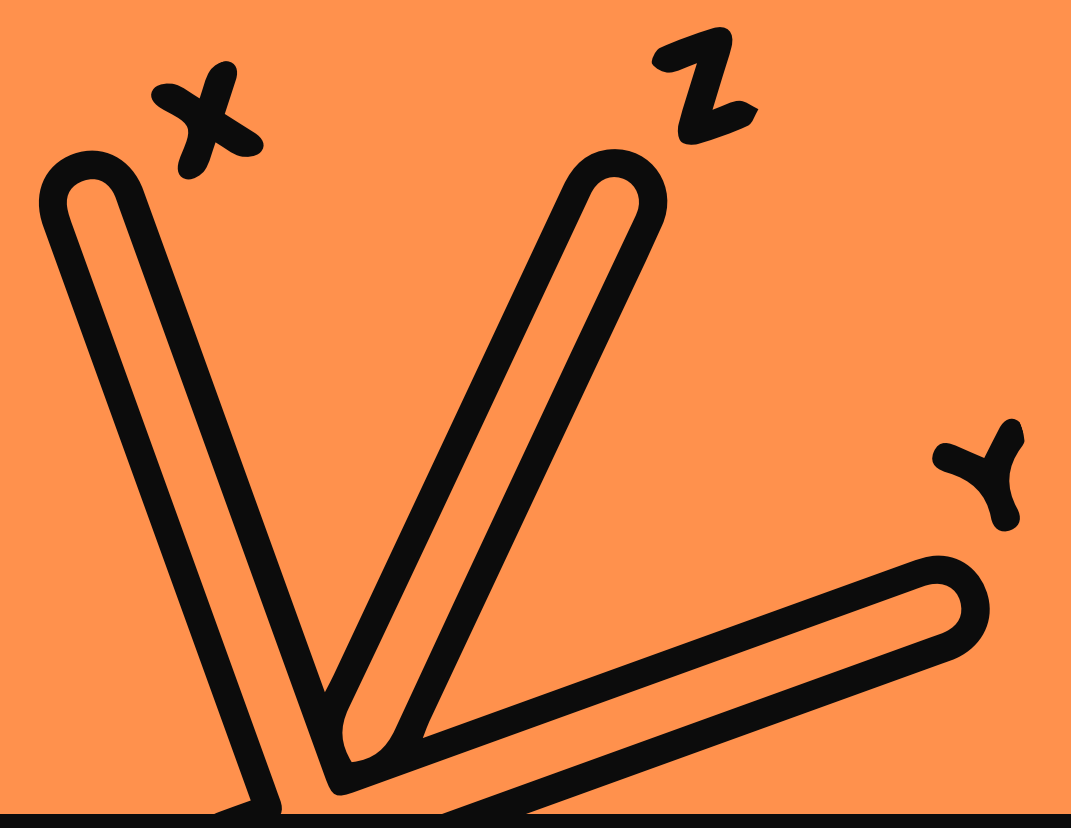
CAMERA EDITOR



- THE SECOND SECTION, **BEHAVIOUR**, ALLOWS YOU TO DECIDE HOW THE CAMERA WILL FOLLOW THE PLAYER.
- **FIRST PERSON** MEANS THE CAMERA WILL BE LOCKED ON THE HEAD OF THE PLAYER, REGARDLESS OF HIS SIZE. IN THIS MODE, MOVING THE MOUSE WILL MAKE THE PLAYER LOOK AROUND IN FIRST PERSON. IF **FIRST PERSON** IS TICKED, THE POSITION OF THE CAMERA DOESN'T MATTER, SINCE THE GAME WILL AUTOMATICALLY LOCK IT ON THE PLAYER'S HEAD.
- **THIRD PERSON** MEANS THE CAMERA WILL MAINTAIN ITS CURRENT POSITION BUT IT WILL ALSO BE "LOCKED" TO THE PLAYER, MEANING IT WILL BOTH MOVE AND ROTATE WITH IT BY KEEPING THE INITIAL OFFSET YOU GAVE IT.
- **STATIC** SIMPLY MEANS THE CAMERA WILL NEVER MOVE FROM ITS INITIAL POSITION.
- REMEMBER THAT YOU CAN MODIFY THE WAY THE CAMERA BEHAVES DURING A LEVEL BY USING TRIGGERS (PAGE 32).

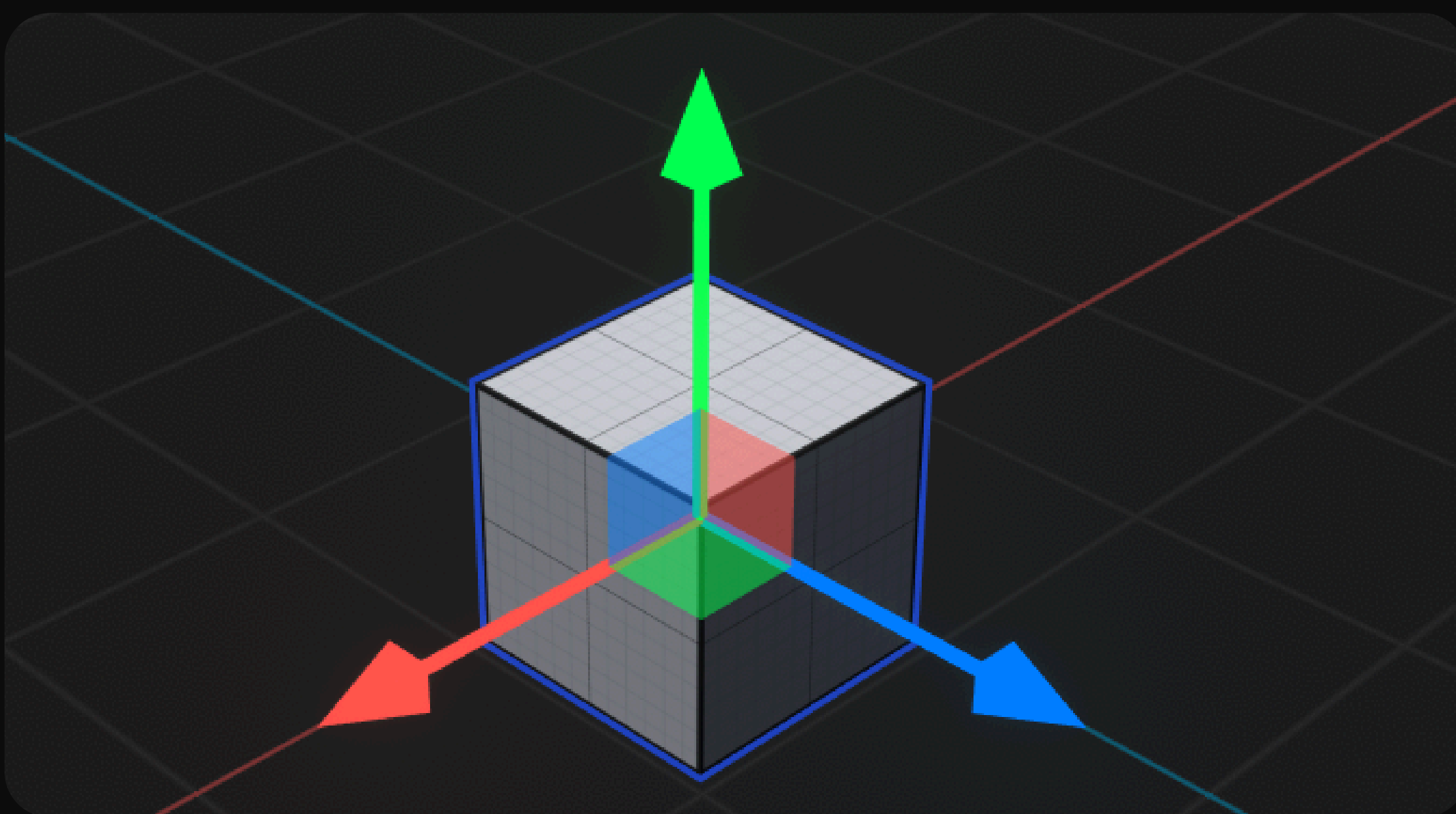


TRANSFORM GIZMO

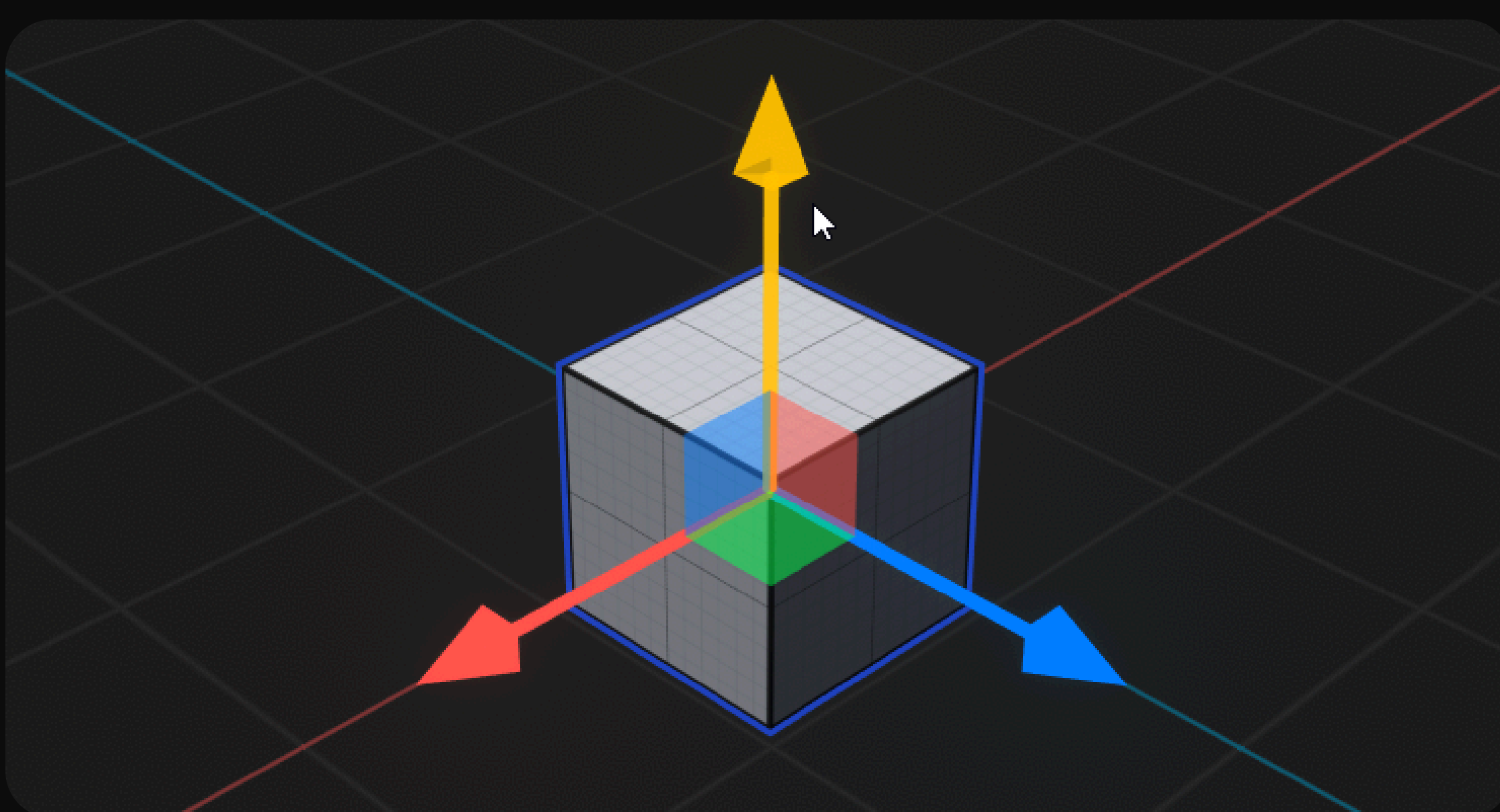


TRANSFORM GIZMO

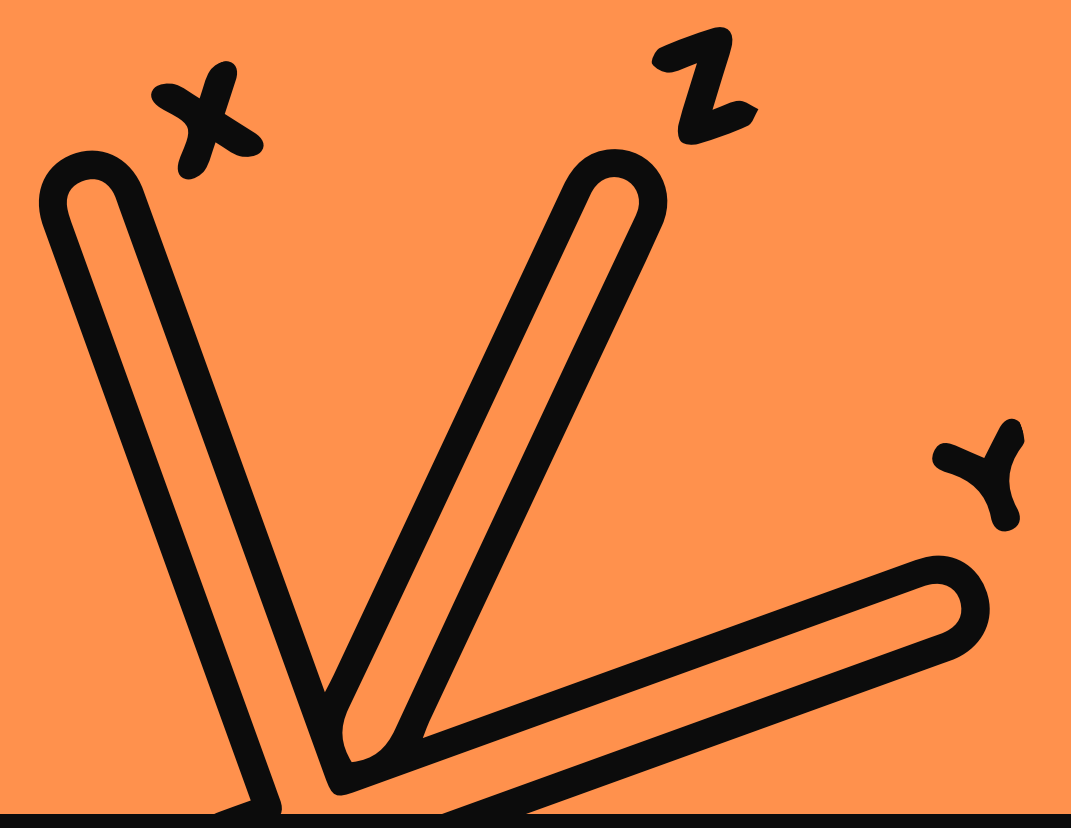
- AN INDUSTRY STANDARD IN ALL EDITORS IS A TOOL CALLED THE **TRANSFORM GIZMO**. IT USUALLY LOOKS SOMETHING LIKE THREE ARROWS (CALLED VECTORS) THAT CONVERGE AT THE CENTER OF THE OBJECT THEY'RE ATTACHED TO:



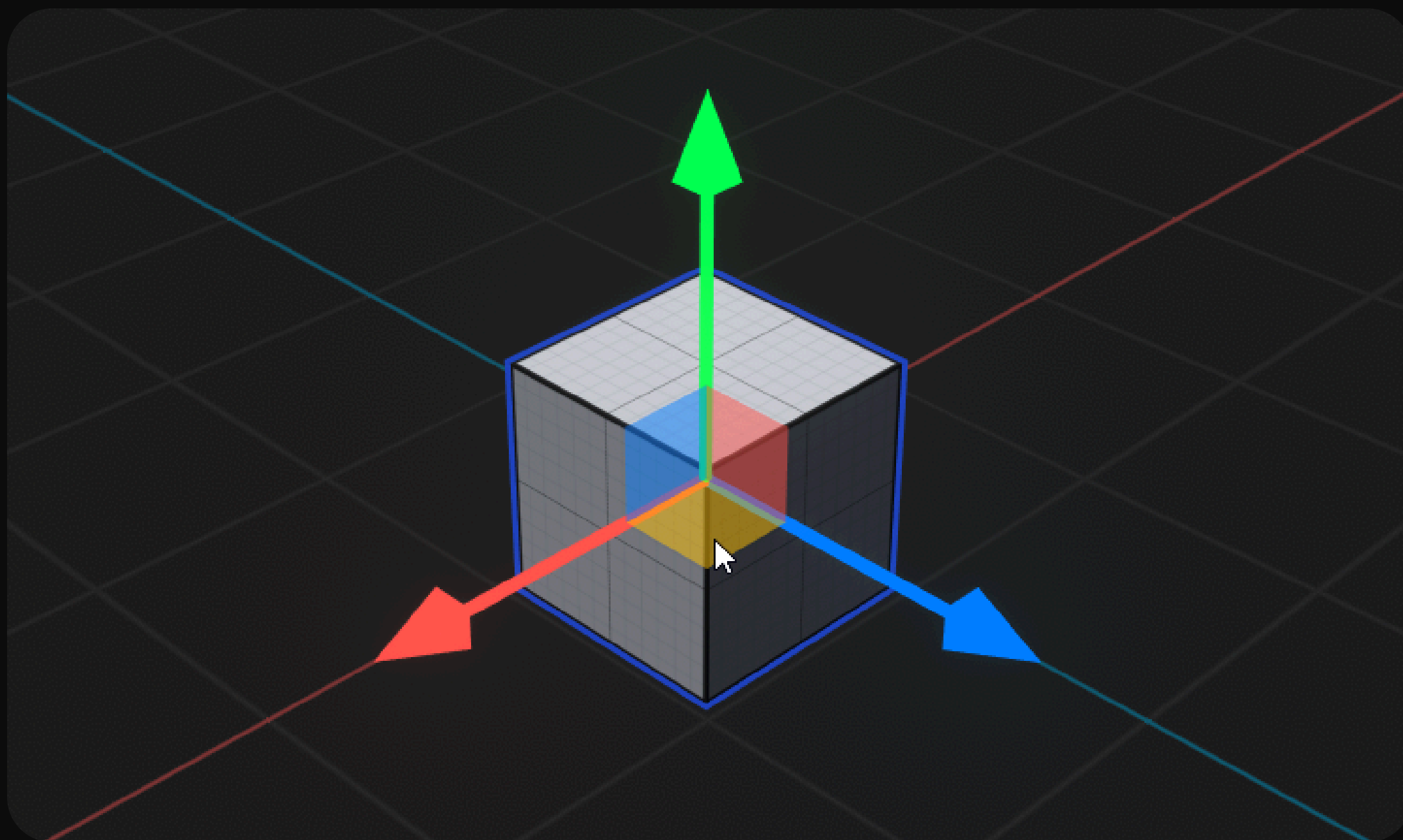
- THE ARROWS ARE NORMALLY COLORED **RED** FOR THE **X AXIS**, **GREEN** FOR THE **Y AXIS**, **BLUE** FOR THE **Z AXIS** IN ORDER TO EASILY ALLOW YOU TO UNDERSTAND IN WHICH SPACE YOU'RE WORKING.
- USING A TRANSFORM GIZMO IS EASY AND INTUITIVE: YOU JUST HAVE TO **HOVER ABOVE ONE OF THE ARROWS** WITH YOUR MOUSE UNTIL IT BECOMES **YELLOW**, AND AT THAT POINT, JUST HOLD DOWN **MOUSE 1** AND MOVE THE MOUSE AROUND TO PERFORM THE TRANSFORM:



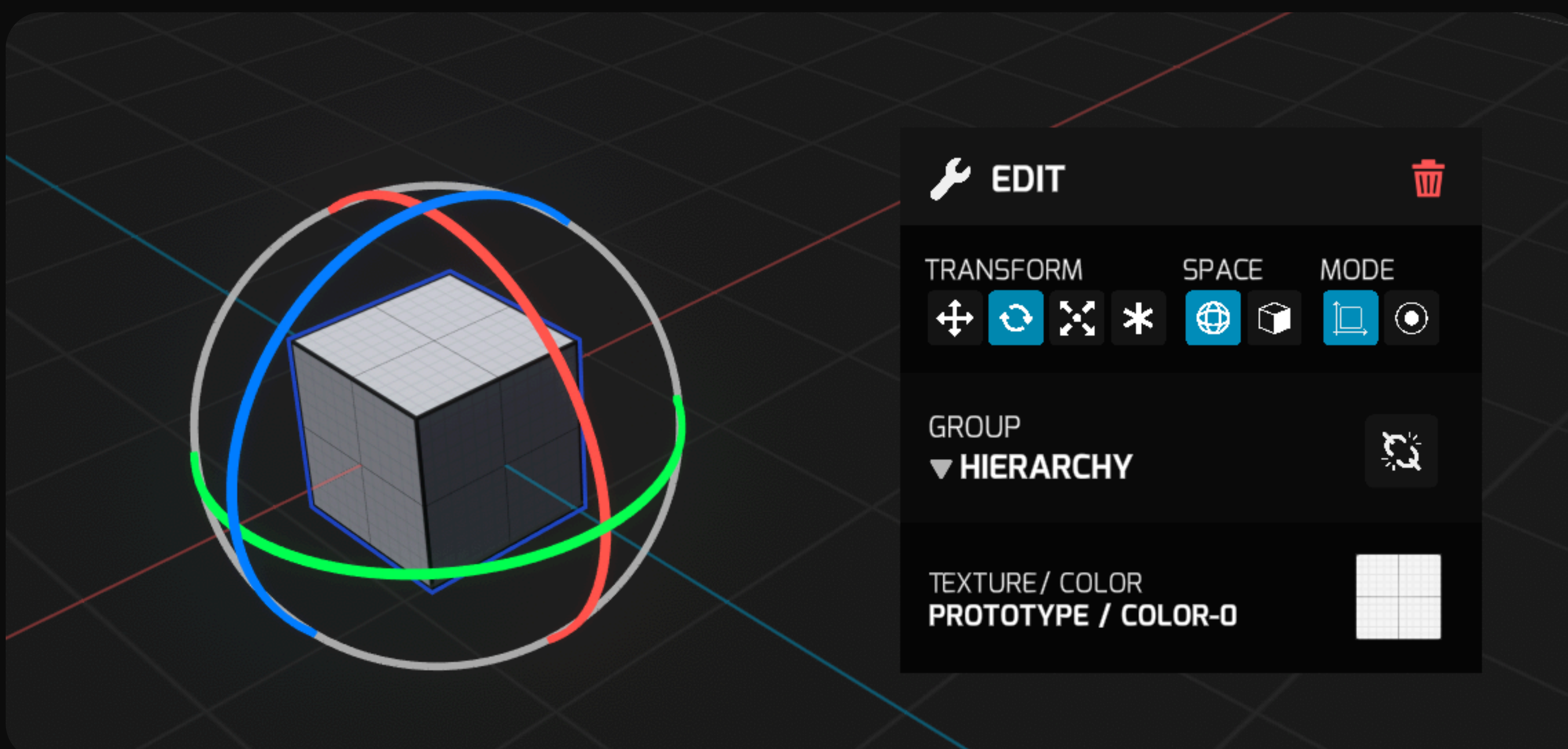
TRANSFORM GIZMO



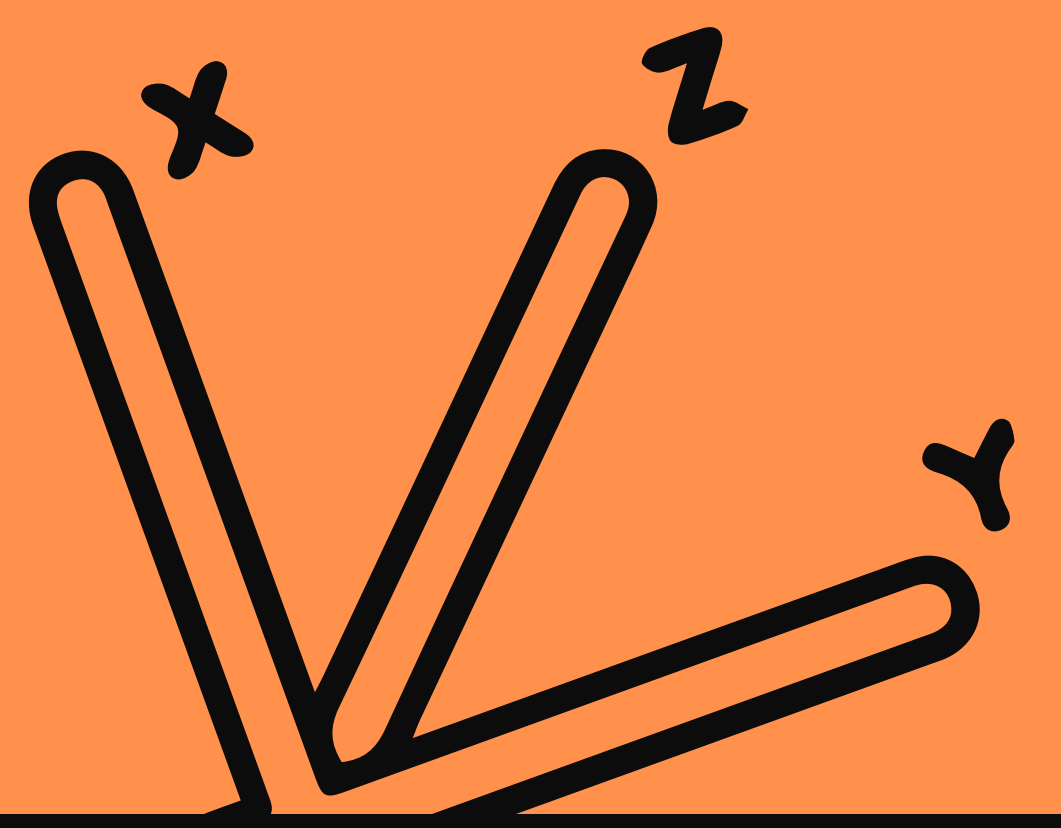
- YOU CAN ALSO USE THE **PLANES** BETWEEN THE ARROWS TO MOVE IN TWO DIMENSIONS SIMULTANEOUSLY FOR PRECISE TASKS:



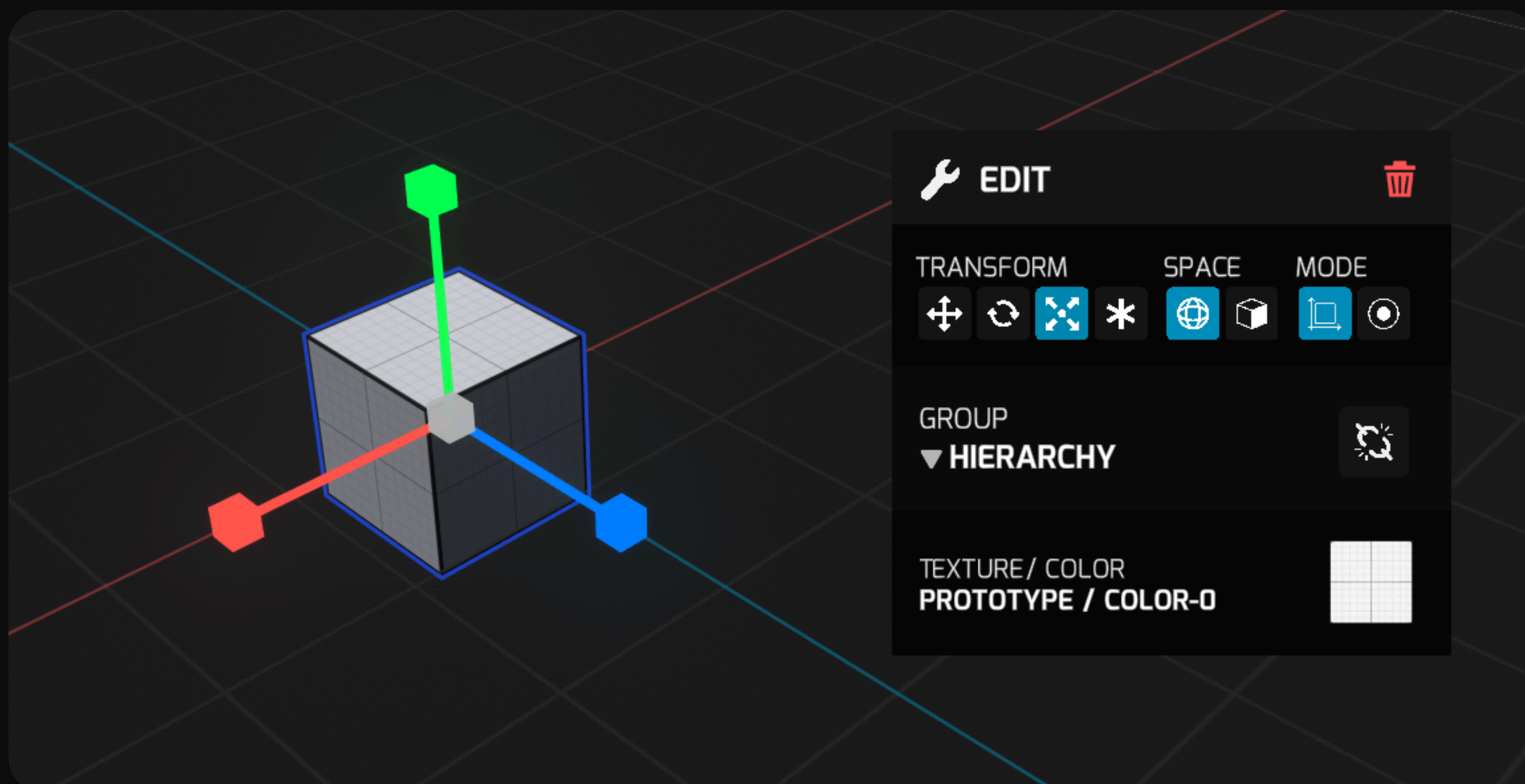
- BY DEFAULT, THE **TRANSFORM GIZMO** WILL BE IN **POSITION MODE**, MEANING THAT IT WILL ONLY ALLOW YOU TO CONTROL THE **POSITION** OF AN OBJECT. BY USING THE **ELEMENT TAB (PAGE 8)** OR SIMPLY PRESSING THE NUMBER KEYS ON YOUR KEYBOARD **FROM 1 TO 4**, YOU'LL BE ABLE TO CYCLE BETWEEN THE DIFFERENT MODES, WHICH ARE **ROTATE** (TO ROTATE OBJECTS) (KEY **2**):



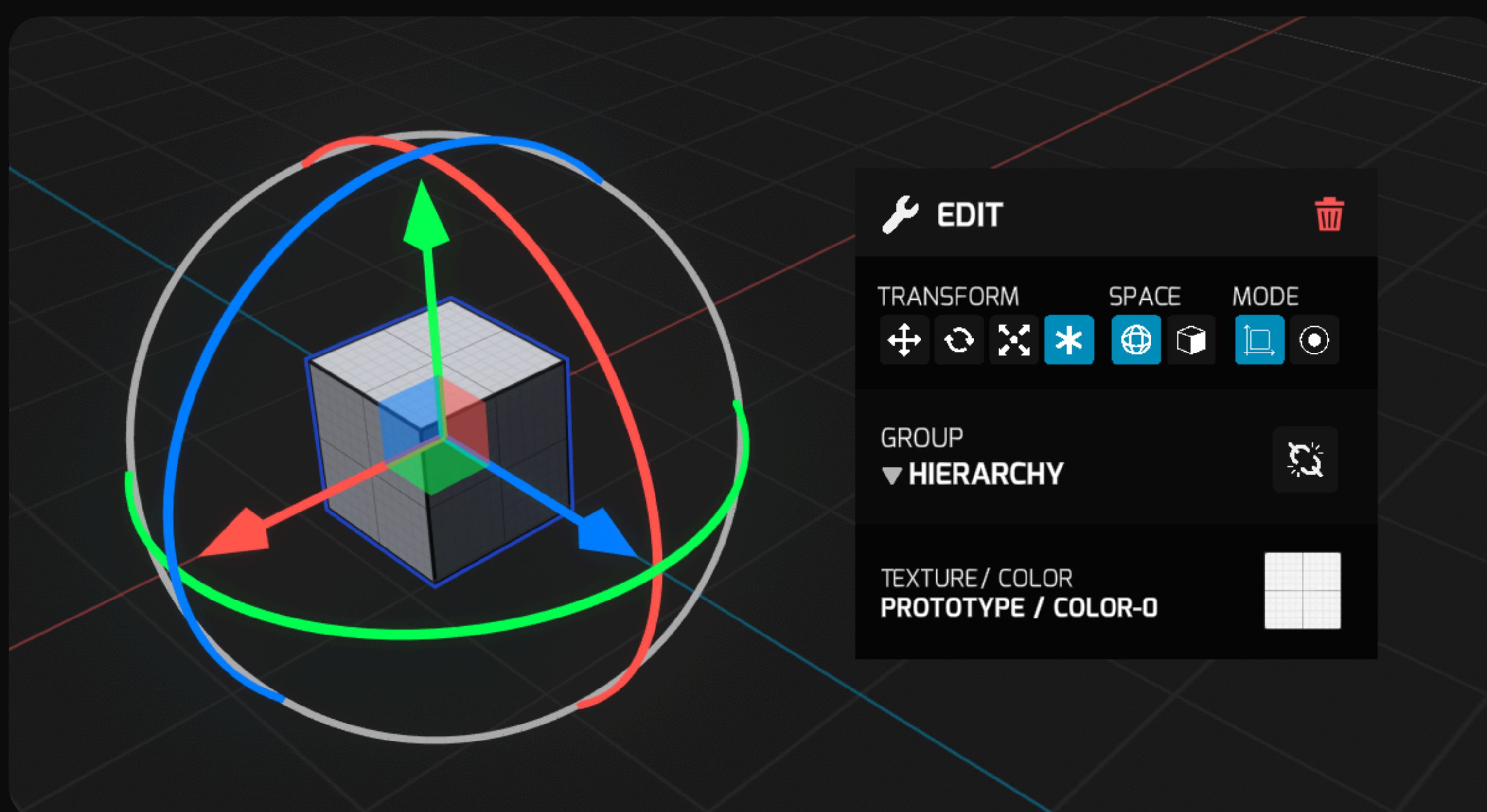
TRANSFORM GIZMO



- **SCALE** (TO CHANGE AN OBJECT'S SIZE) (KEY **3**):

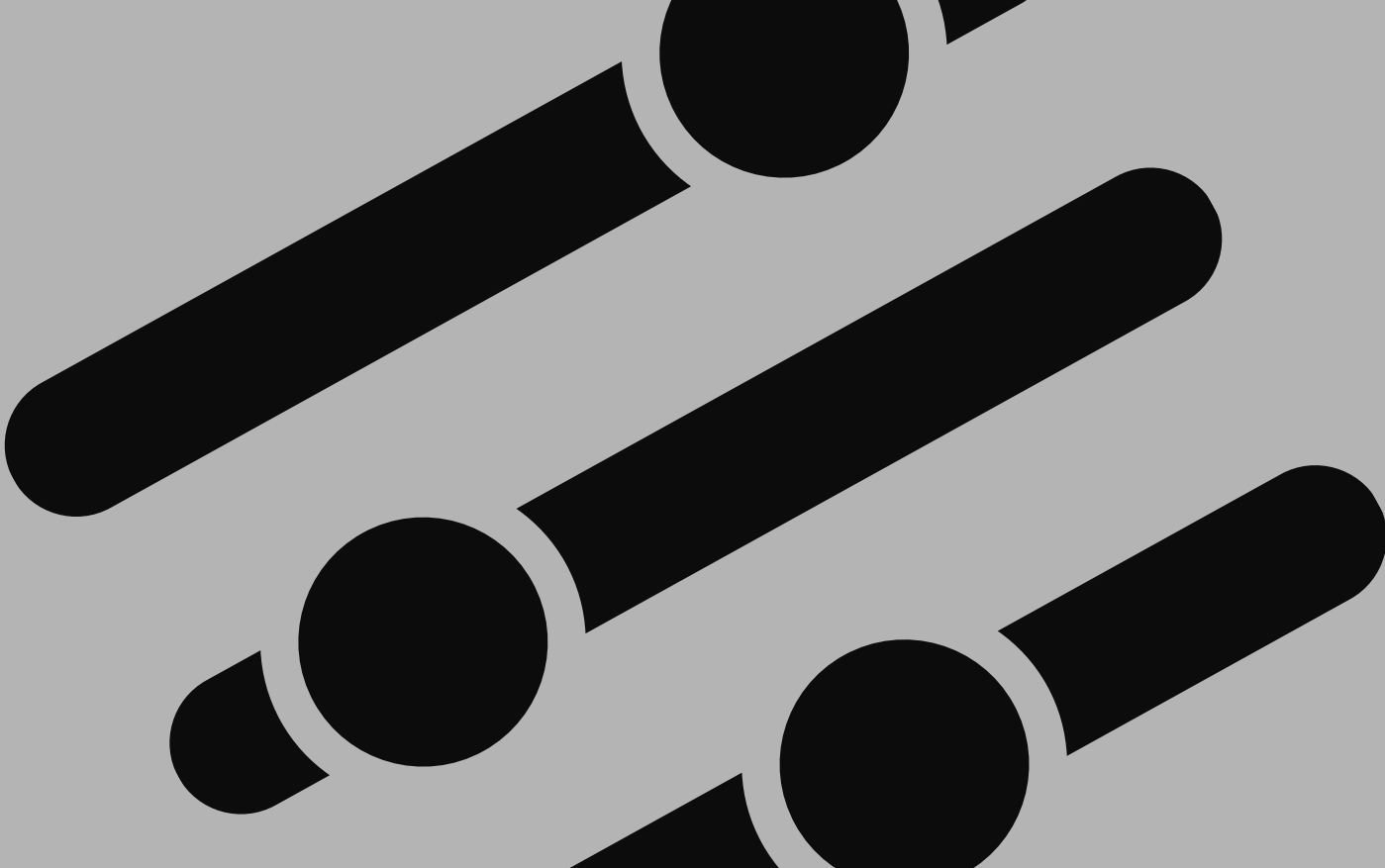


- AND **COMPLETE** (TO MODIFY ALL AT ONCE) (KEY **4**):



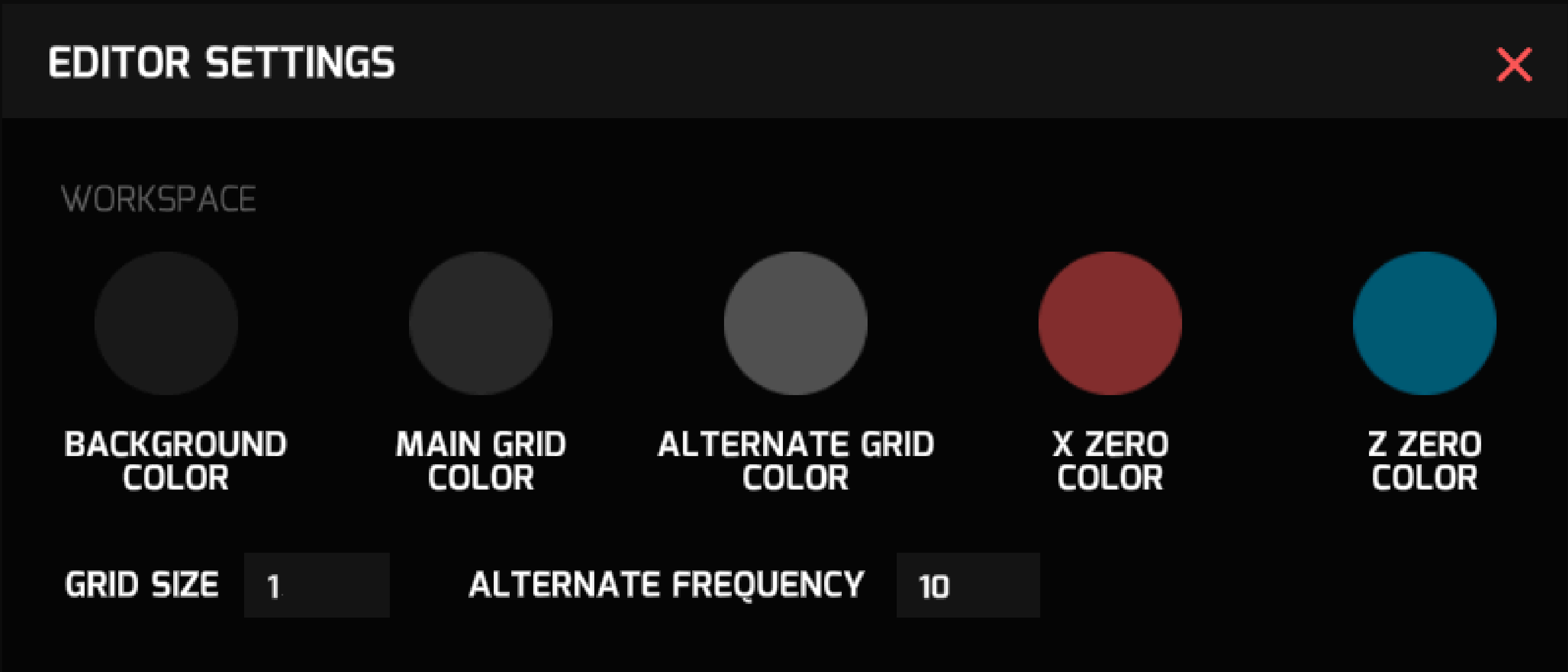
- REGARDLESS OF THE MODE YOU'RE USING, THE PRINCIPLE REMAINS THE SAME. HOVER ABOVE AN AXIS, HOLD DOWN AND MOVE THE MOUSE TO APPLY A TRANSFORM.

EDITOR SETTINGS

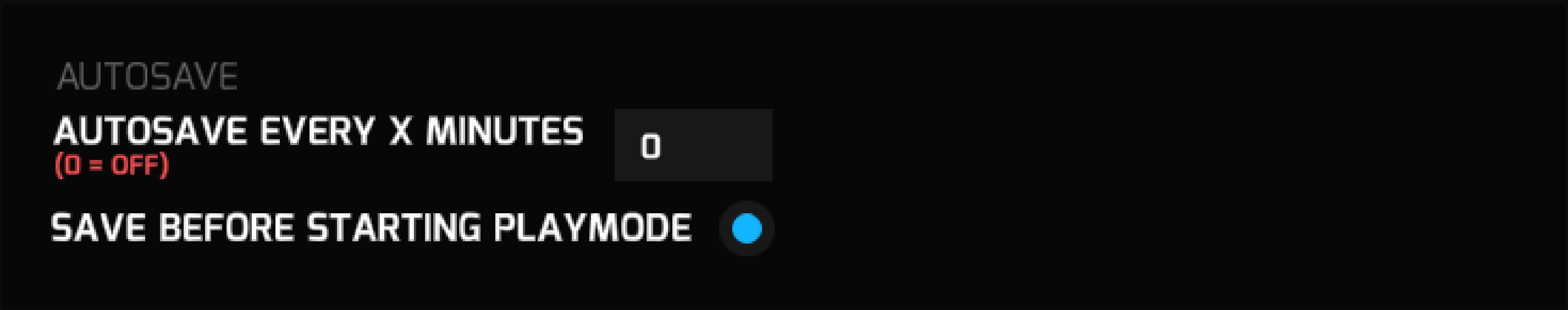


EDITOR SETTINGS

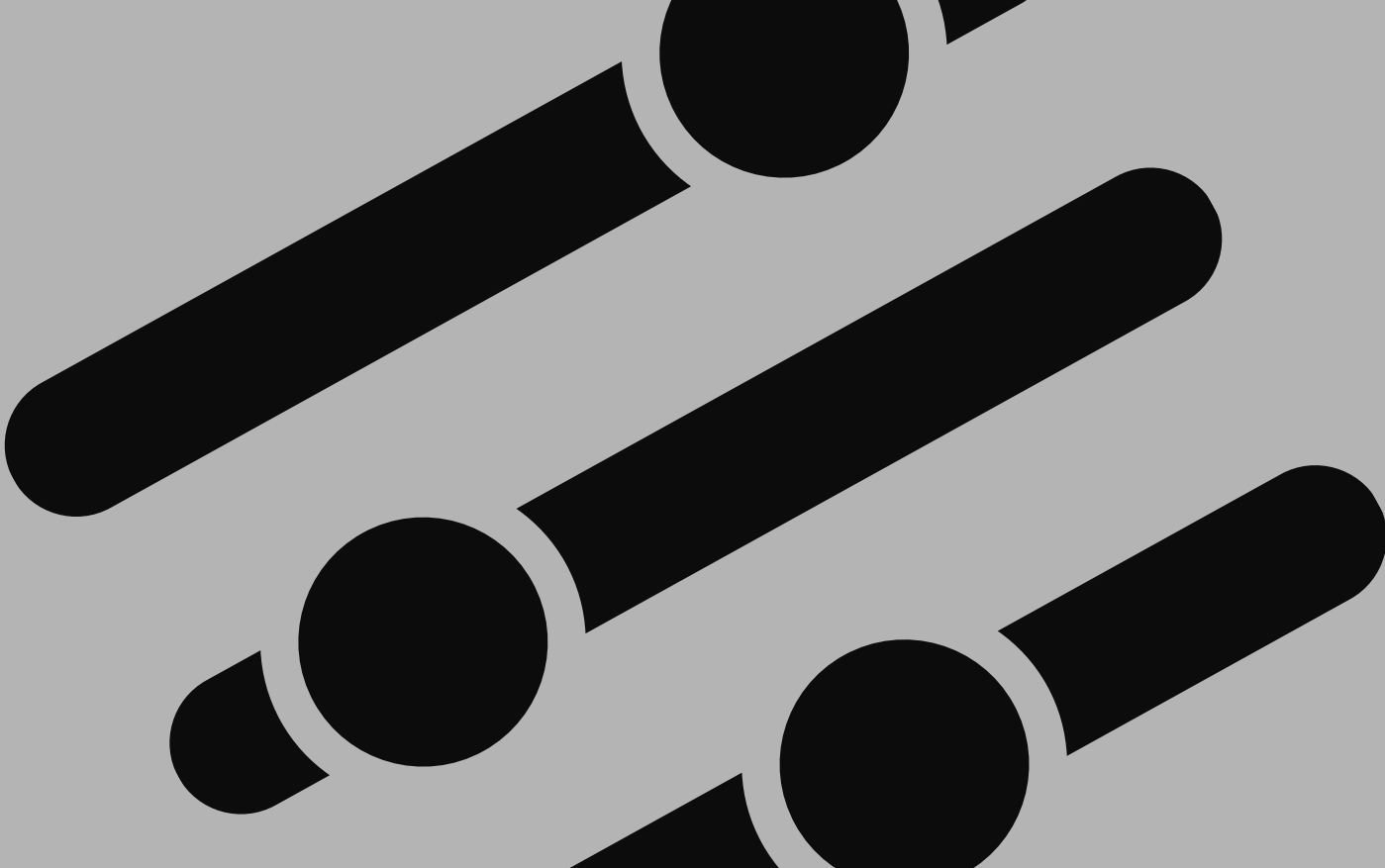
- THE EDITOR SETTINGS PANEL ALLOWS YOU TO CUSTOMIZE YOUR WORKSPACE AND GIVES YOU FULL CONTROL OVER **HOW THE CAMERA MOVES** AND **THE EDITOR BEHAVES**. LET’S EXAMINE ITS SECTIONS ONE BY ONE.
- THIS FIRST SECTION, CALLED **WORKSPACE**, ALLOWS YOU TO CUSTOMIZE THE **GRID** AND THE **WORLD’S BACKGROUND COLORS**. SIMPLY CLICK ABOVE A COLOR TO LAUNCH THE **COLOR CREATOR (PAGE 11)** AND CHANGE IT TO YOUR LIKING.
- THE LOWER VALUES ALLOW YOU TO DECIDE HOW BIG A SQUARE OF THE GRID WILL BE AND AFTER HOW MANY INTERSECTIONS THE SECONDARY (ALTERNATE) GRID WILL SHOW UP.



- THE SECOND SECTION, CALLED **AUTOSAVE**, ALLOWS YOU TO SET AN **AUTOSAVE FREQUENCY** AND LETS YOU DECIDE WHETHER THE EDITOR SHOULD **SAVE EVERYTIME YOU CLICK START**, RIGHT BEFORE ENTERING PLAYMODE.



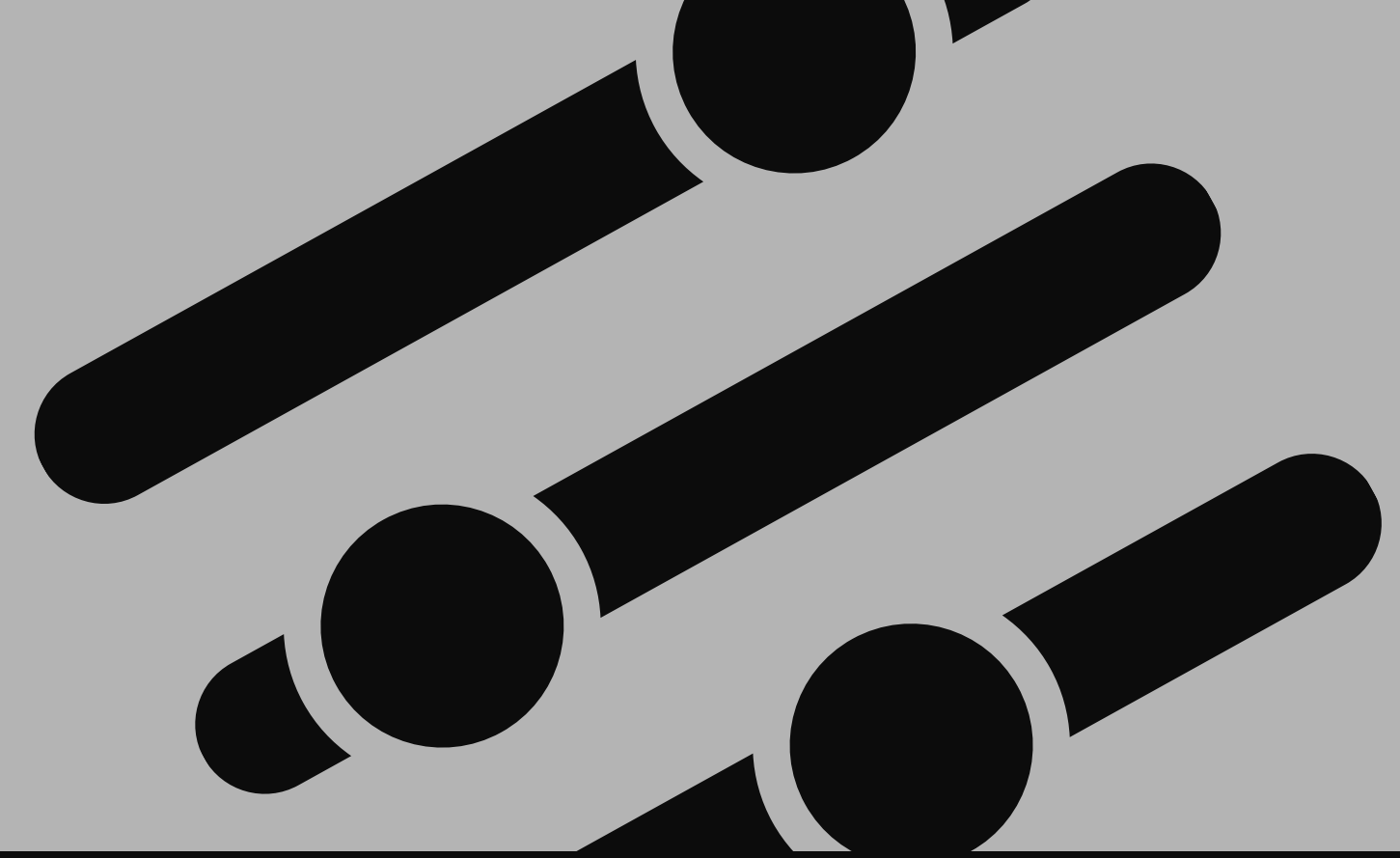
EDITOR SETTINGS



- THE **MOVEMENT** SECTION ALLOWS YOU TO CONTROL THE MOVEMENT SPEEDS OF THE EDITOR’S CAMERA DURING VARIOUS SITUATIONS.
- THE **CAMERA SPEED** AND **ZOOM SPEED** FIELDS REPRESENT THE NORMAL SPEEDS AT WHICH YOU WILL MOVE WITH **WASD** AND ZOOM IN/OUT WITH THE **MOUSE WHEEL** WITHOUT PRESSING ANY OTHER KEYS.
- THE **SLOW** SPEED VALUES ARE THOSE THAT ARE ACTIVE WHEN YOU HOLD DOWN THE **CTRL** BUTTON.
- THE **FAST** SPEED VALUES ARE THOSE THAT ARE ACTIVE WHEN YOU HOLD DOWN THE **SHIFT** BUTTON.
- THE **DRAG SPEED** IS THE SPEED AT WHICH THE CAMERA IS DRAGGED AROUND WHEN YOU HOLD DOWN THE **MOUSE WHEEL**.
- THE **CAMERA FOV** IS THE CUSTOM FOV VALUE FOR THIS SPECIFIC LEVEL. NOTE THAT THIS IS NOT THE FOV THAT PLAYERS WILL HAVE WHEN THEY PLAY YOUR LEVEL, AS THAT IS UNDER THE CAMERA SETTINGS. THIS IS ONLY THE VALUE YOU WILL SEE WHEN USING THE EDITOR ON THIS LEVEL.

MOVEMENT			
SLOW CAMERA SPEED	1	SLOW ZOOM SPEED	1
CAMERA SPEED	5	ZOOM SPEED	5
FAST CAMERA SPEED	20	FAST ZOOM SPEED	20
DRAG SPEED	0.5		
CAMERA FOV	60		

EDITOR SETTINGS



- THE **TRANSFORM SETTINGS** SECTION ALLOWS YOU TO CONTROL HOW YOU MOVE OBJECTS WITH THE **TRANSFORM GIZMO** AFTER SELECTING THEM.
- THE **ALWAYS SNAP** TOGGLE DETERMINES WHETHER THINGS YOU MODIFY WITH THE GIZMO WILL BE AFFECTED CONTINUOUSLY OR ONLY AFTER A CERTAIN THRESHOLD HAS BEEN REACHED. IT'S EASIER IF YOU SEE IT FOR YOURSELF, BUT ESSENTIALLY, INSTEAD OF MOVING CENTIMETER BY CENTIMETER, A CUBE YOU'RE DRAGGING AROUND WILL ONLY MOVE EVERY **X** METERS.
- **SNAPPING MAKES CREATING PRECISE THINGS WAY EASIER.**
- THE THREE VALUES BELOW ALLOW YOU TO SPECIFY A CERTAIN SNAP FOR EACH OF THE THREE POSSIBLE TRANSFORMS, MOVEMENT, ROTATION AND SCALE.

TRANSFORM SETTINGS

ALWAYS SNAP ☒

MOVEMENT SNAP VALUE

ROTATION SNAP VALUE

SCALE SNAP VALUE

- THE LAST SECTION IS **OBJECTS PLACEMENT**, AND IT ALLOWS YOU TO CONTROL HOW OBJECTS WILL BE SPAWNED INTO THE LEVEL.
- BY ACTIVATING **CREATE OBJECTS ON WORLD ORIGIN**, WHATEVER OBJECT YOU SPAWN WILL ALWAYS APPEAR IN THE CENTER OF THE LEVEL, SO AT (0,0,0).
- IF THIS SETTING IS NOT ACTIVATED, OBJECTS WILL SIMPLY BE SPAWNED A CERTAIN **X** METERS IN FRONT OF YOUR FACE, AND THE DISTANCE CAN BE SPECIFIED BY THE **GENERATE OBJECTS AT X METERS FROM VIEW** SETTING.

OBJECTS PLACEMENT

CREATE OBJECTS ON WORLD ORIGIN ☐

GENERATE OBJECTS AT X METERS FROM VIEW

LEVEL SETTINGS



LEVEL SETTINGS

- THE LEVEL SETTINGS PANEL ALLOWS YOU TO CUSTOMIZE YOUR LEVEL'S **GENERAL APPEARANCE AND RULES**.
- THE FIRST OPTION, **USE SUN**, SIMPLY ALLOWS YOU TO TURN OFF SUNLIGHT IN THE WHOLE WORLD, WHILE THE VALUES BELOW, **INTENSITY** AND **INCLINATION** ALLOW YOU TO MODIFY HOW THE LIGHTING ACROSS THE LEVEL LOOKS AND THE TYPE OF SHADOWS IT CASTS.
- THE **SUN COLOR** OPTION ALLOWS YOU TO MODIFY THE COLOR OF THE SUNLIGHT BY CLICKING ON IT AND PICKING A NEW COLOR.

LEVEL SETTINGS

USE SUN

SUNLIGHT INTENSITY

SUNLIGHT INCLINATION

SUN COLOR

1

50

-30

0

- **LEVEL DIFFICULTY** ALLOWS YOU TO SPECIFY THE LEVEL OF CHALLENGE THAT YOUR LEVEL CREATES, ALLOWING PLAYERS TO SEE IF A LEVEL IS TIGHT AND DEMANDING OR A CHILL EXPERIENCE.
- **TIMEFLOW** REPRESENTS THE SPEED AT WHICH TIME PASSES IN YOUR LEVEL. THIS AFFECTS PHYSICS CALCULATIONS AND TIME DEPENDANT EVENTS.
- THE **CLEAR DEBRIS AFTER X SECONDS** FEATURES ALLOWS YOU SPECIFY HOW FAST THE GAME WILL CLEAN **DEBRIS** CREATED FROM AN IMPACT, A LASER, A DRONE, ETC...

LEVEL DIFFICULTY

TIMEFLOW

CLEAR DEBRIS AFTER

EASY

1

60

(SECONDS)

LEVEL SETTINGS



- **OPEN SOURCE SAVEFILE** DETERMINES WHETHER ANYONE CAN OPEN THIS LEVEL'S SAVEFILE IN THEIR VERSION OF PROJECT LEVELS.
- IF THIS IS TICKED, ANYONE THAT COMES INTO POSSESSION OF THE FILE WILL BE ABLE TO PLACE IT IN THEIR LOCAL LEVELS FOLDER, EDIT IT AND UPLOAD IT. IF THIS IS UNTICKED, THE GAME WILL RUN A CHECK ON THE LEVEL AND BLOCK IT FROM BEING EDITED BY OTHER STEAM ACCOUNTS BY CHECKING STEAM ID.
- **FASTMODE DEATHS** SPECIFIES HOW DYING WORKS IN YOUR LEVEL.
- IF **FASTMODE DEATHS IS TICKED**, EVERYTIME A PLAYER DIES THE GAME WILL SIMPLY BRING HIM BACK TO EITHER THE START POSITION OR THE LATEST CHECKPOINT SCORED, THE LEVEL ITSELF WON'T BE RESET.
- IF **FASTMODE DEATHS IS NOT TICKED**, EVERYTIME A PLAYER DIES THE WHOLE LEVEL WILL BE RELOADED FROM ZERO, CAUSING THE OBJECTS TO RESET ASWELL.
- GENERALLY SPEAKING, IF YOUR LEVEL IS BIG (500+ OBJECTS) AND THERE'S A LOT OF CHANCES TO DIE, IT CAN BECOME FRUSTRATING FOR THE PLAYERS TO WAIT FOR THE WHOLE THING TO RELOAD EVERYTIME, SO **IT'S RECOMMENDED TO ALWAYS USE FASTMODE DEATHS** UNLESS YOU REALLY HAVE SOME SPECIFIC NEEDS.
- **KEEP IN MIND** THAT IN MULTIPLAYER **FASTMODE DEATHS** WILL ALWAYS BE ACTIVE REGARDLESS OF WHAT YOU SET HERE, DUE TO THE FACT THAT RELOADING THE WHOLE LEVEL FOR EVERY SINGLE PLAYER EVERYTIME ANY OF THEM DIES IS SIMPLY INSANE.
- **PROXIMITY VOICECHAT** SIMPLY ALLOWS YOU TO CONTROL WHETHER PLAYERS VOICES IN THE MULTIPLAYER VOICE CHAT WILL **FADE** AFTER THEY'RE NO LONGER CLOSE TO EACH OTHER OR IF THE CHAT WILL BE IN **"RADIO MODE"**, WHERE PLAYERS CAN HEAR EACH OTHER FULLY REGARDLESS OF DISTANCE.
- THIS SETTING **DOES NOT AFFECT THE EDITOR**, WHERE PLAYERS ARE ALWAYS ABLE TO HEAR EACH OTHER.

OPEN SOURCE SAVEFILE ☐ ?

FASTMODE DEATHS ☒ ?

PROXIMITY VOICECHAT ☒ ?

MULTIPLAYER



MULTIPLAYER

- I WILL NOW FOLLOW WITH A FEW BASIC INFORMATIONS ON HOW TO START AND MANAGE ALL OF THE MULTIPLAYER ASPECTS OF **PROJECT LEVELS**.
- FIRST OF ALL, **P.L.** IS BASED ON **PEER-TO-PEER MULTIPLAYER**. THIS JUST MEANS THAT THERE AREN'T ANY REAL SERVERS BUT JUST OTHER USERS THAT "OPEN" THEIR SINGLEPLAYER SESSIONS TO A GLOBAL NETWORK. THIS ALSO MEANS THAT A SESSION'S STABILITY IS HEAVILY DEPENDANT ON WHO IS HOSTING IT, ESPECIALLY IN TERMS OF CONNECTION AND COMPUTER SPECS.
- IN ORDER TO START A MULTIPLAYER SESSION, SIMPLY START ANY LEVEL IN ANY MODE (EDITOR, PLAY, HYBRID, WORKSHOP, WHATEVER...) OPEN THE PAUSE MENU (**ESC**) AND CLICK **HOST MULTIPLAYER GAME** (**MAKE SURE YOU SAVE BEFORE DOING SO!**):

MENU



RESUME



SAVE



HOST MULTIPLAYER GAME



SHOW SETTINGS

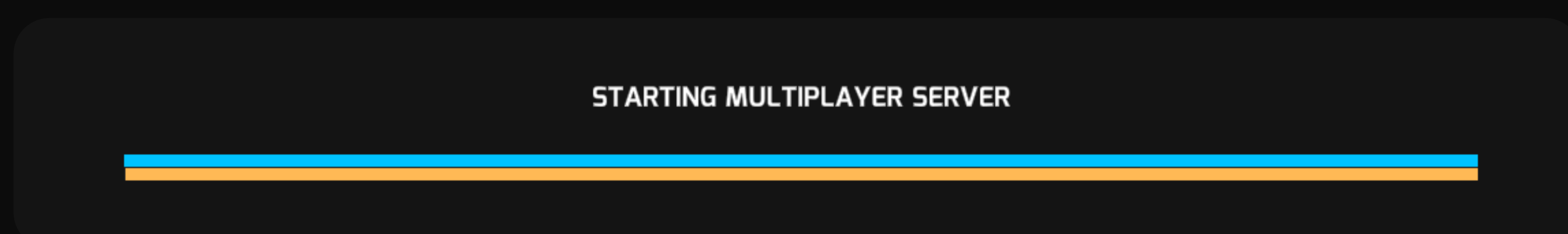


QUIT

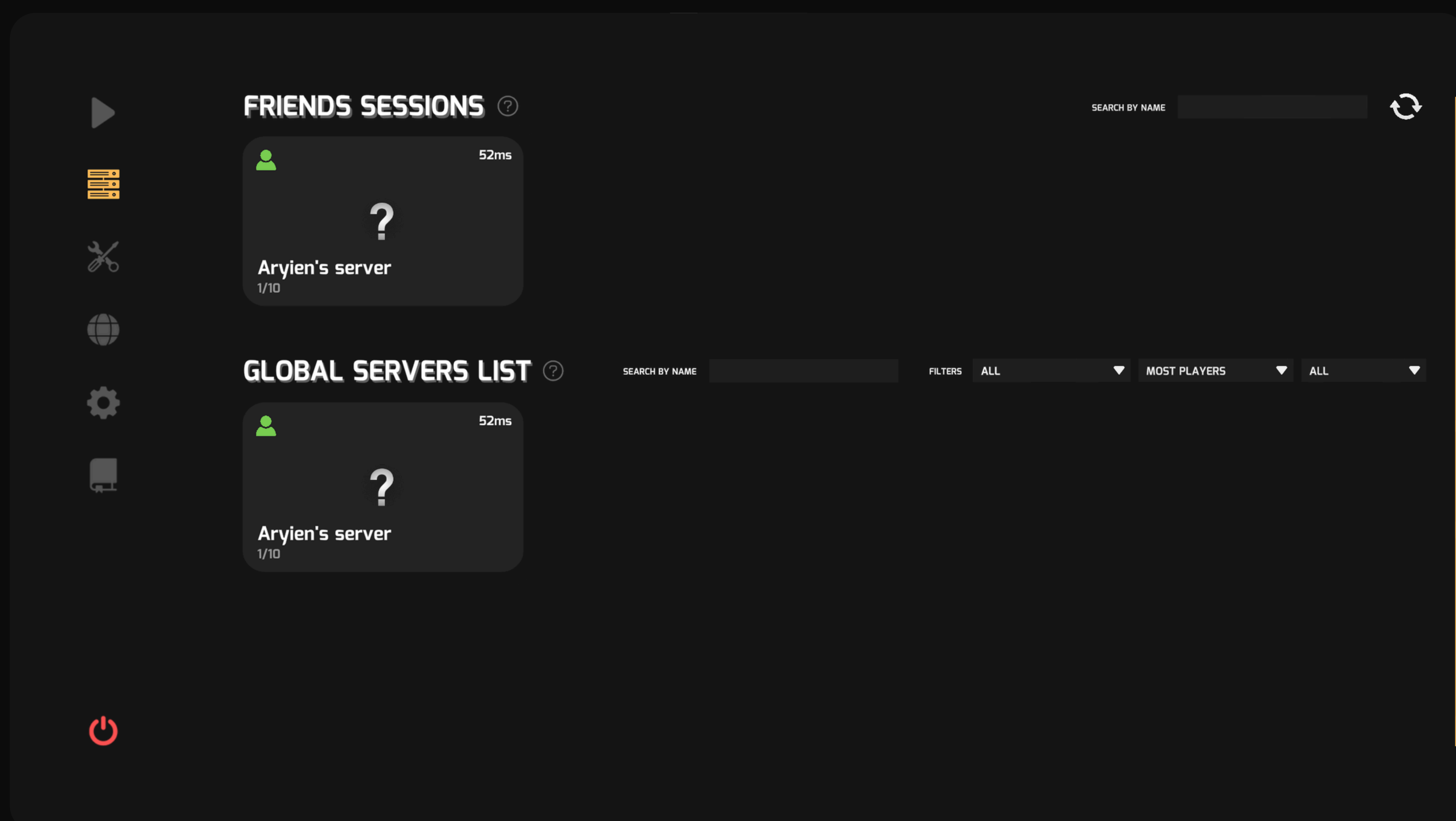
MULTIPLAYER



- THE GAME WILL DO EVERYTHING FOR YOU BY COMPLETELY RELOADING THE CURRENT LEVEL AND STARTING IT IN MULTIPLAYER MODE. YOU WILL EASILY NOTICE THAT YOU'RE STARTING A MULTIPLAYER SESSION BY THE PRESENCE OF **A SECOND YELLOW LOADING BAR** THAT REPRESENTS **SERVER & NETWORK OPERATIONS** BEING EXECUTED:



- FROM THIS POINT ONWARD YOU'RE OFFICIALLY IN A **MULTIPLAYER SESSION** AND OTHER PLAYERS **CAN ALREADY JOIN YOU** BY EITHER THEIR **STEAM OVERLAY** (JOIN GAME OR BY ACCEPTING AN INVITE) OR BY THE **MULTIPLAYER MENU** WITHIN THE GAME:

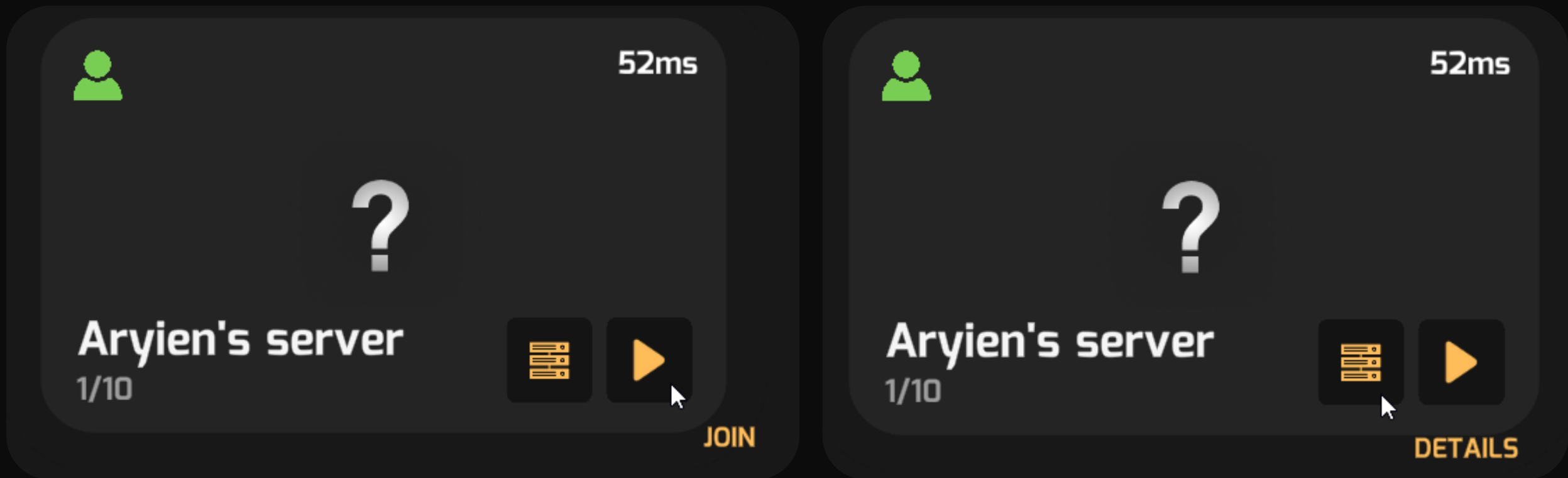


- I USED A STEAM ACCOUNT THAT'S FRIENDS WITH MINE IN THIS DEMONSTRATION, AND AS YOU CAN SEE **STEAM FRIENDS** WHO ARE CURRENTLY HOSTING A SESSION WILL BE PINNED TO THE TOP SECTION. ONLY THE FIRST 4 FRIENDS WILL BE SHOWN, SO IF YOU DON'T SEE THE SPECIFIC FRIEND YOU NEED, YOU CAN ALWAYS USE THE SEARCH BAR AND THEY WILL POP OUT.

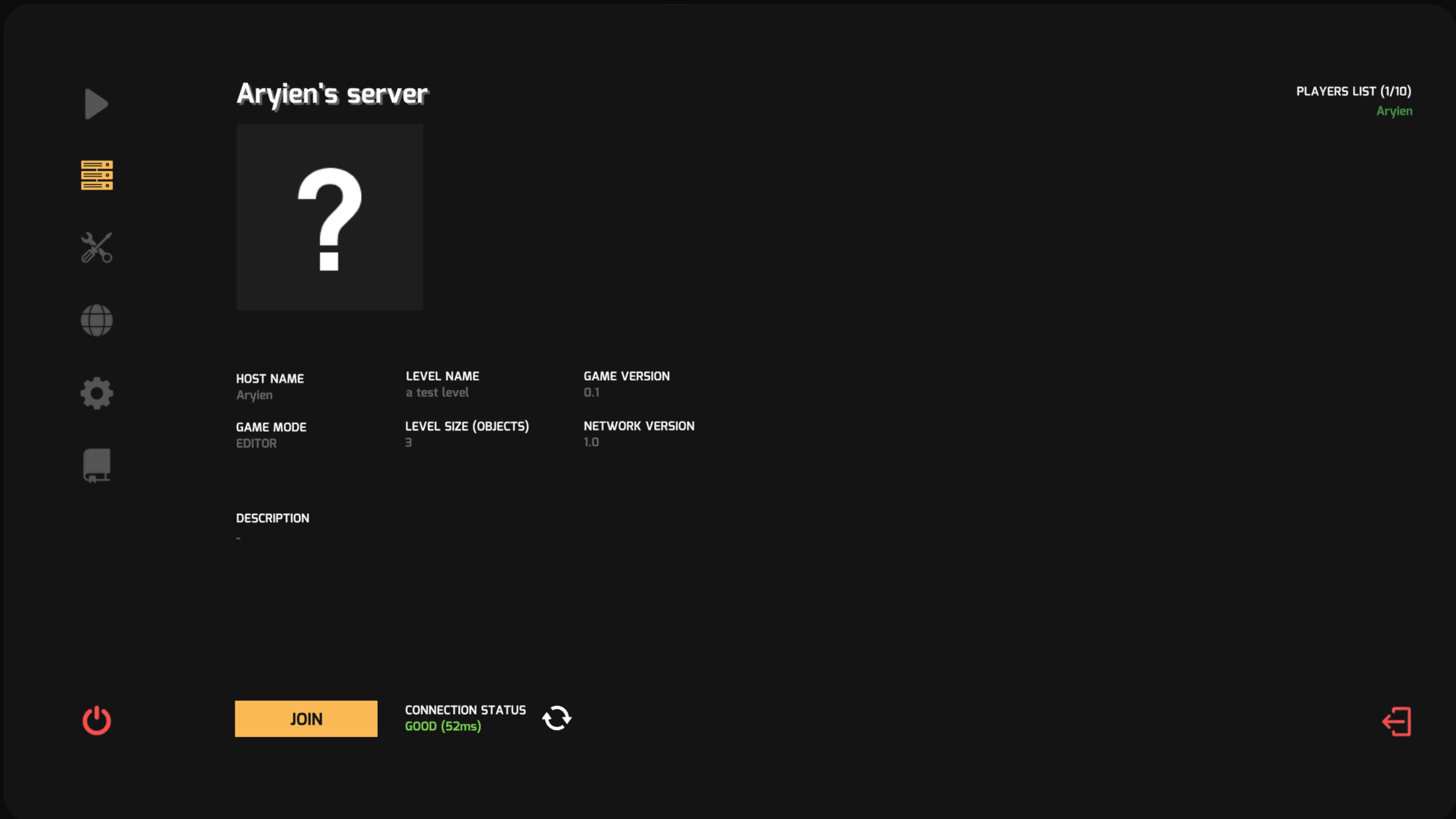
MULTIPLAYER



- ONCE YOU’VE FOUND THE SESSION YOU WANT TO JOIN, JUST HOVER YOUR MOUSE ABOVE IT AND CLICK EITHER **JOIN** TO CONNECT IMMEDIATELY OR **DETAILS** TO OPEN THE SERVER’S DETAILS PAGE:



- THE **DETAILS** PAGE ALLOWS YOU TO INSPECT A LOT OF INFO BEFORE DECIDING IF YOU WANT TO ACTUALLY JOIN A SERVER. YOU CAN CLEARLY SEE WHO CREATED THE SESSION, WHAT LEVEL IS BEING PLAYED, THE GAME’S VERSION, NETWORK VERSION (BOTH OF THESE MUST MATCH YOURS FOR THE SERVER TO BE JOINABLE), THE PLAYERS LIST, LEVEL SIZE, GAME MODE AND SESSION LATENCY:



- ONCE YOU’RE SATISFIED WITH THE INFO YOU’VE SEEN, YOU CAN CLICK THE BIG **JOIN** BUTTON TO ENTER THE SESSION, OR THE **RED** EXIT ICON TO YOUR RIGHT TO RETURN TO THE SERVERS LIST.

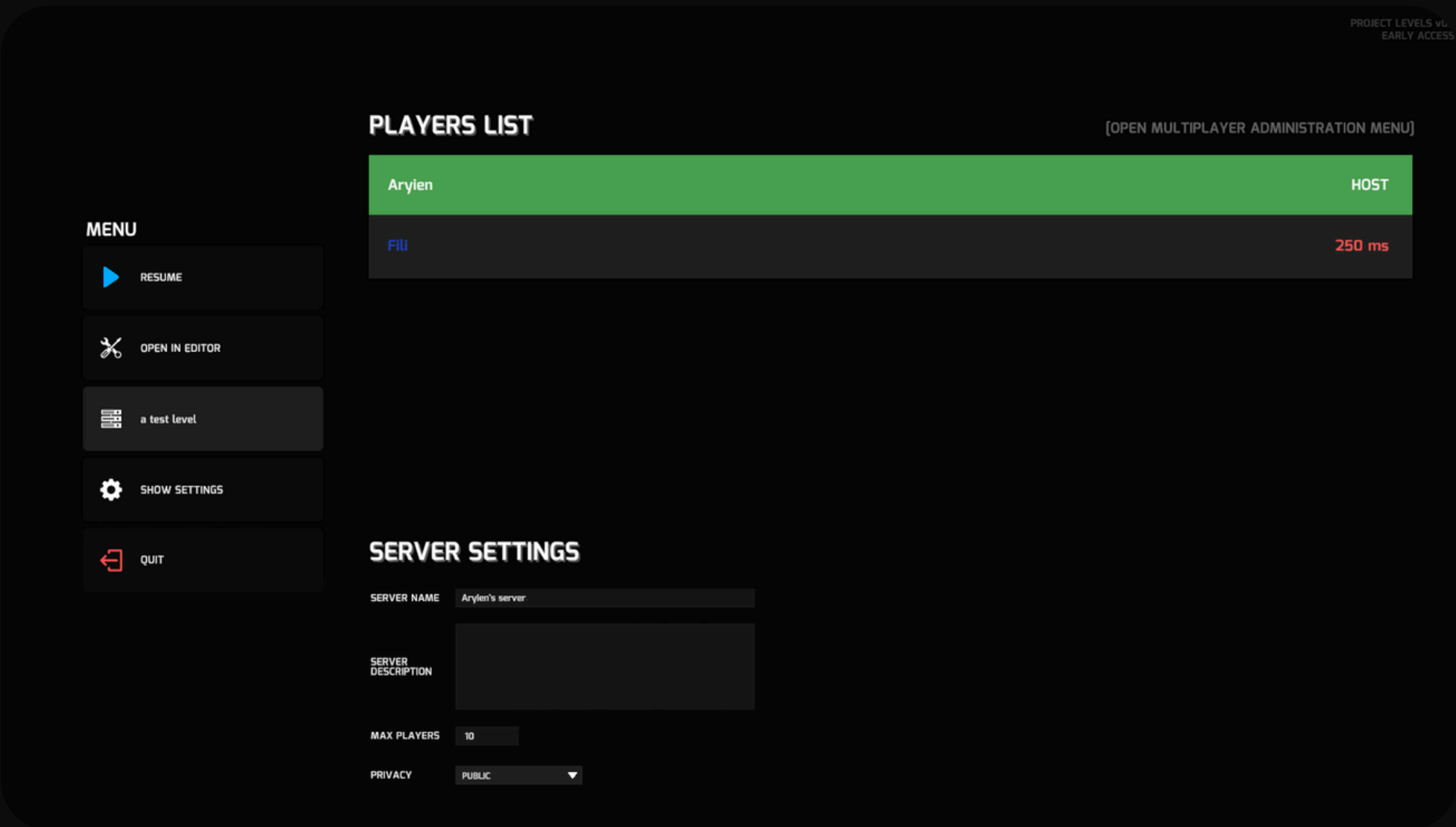
MULTIPLAYER



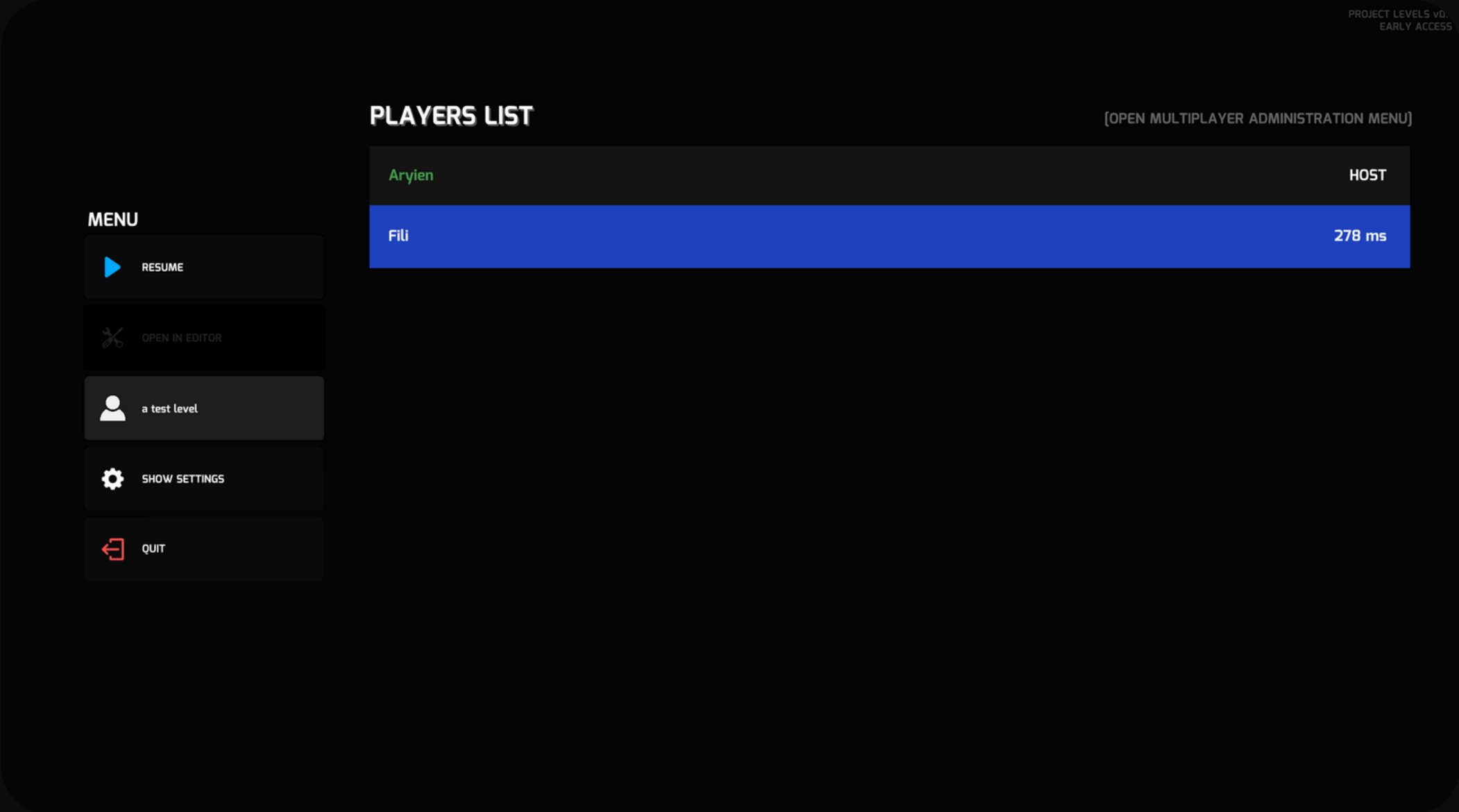
- ONCE INSIDE A SESSION, YOU WILL IMMEDIATELY NOTICE OTHER PLAYERS HANGING AROUND IN DIFFERENT “FORMS” DEPENDING ON WHAT GAMEMODE THE SESSION IS IN. IF YOU’RE IN EDITOR MODE, OTHER PLAYERS WILL APPEAR **AS SMALL CAMERAS** WITH THEIR **STEAM NAME HANGING ABOVE THEM**, WHILE IN PLAY MODE THEY’LL APPEAR **AS PLAYERS**:



- AS YOU CAN SEE, EVERY PLAYER WILL SHOW OFF THE COLOR THEY’VE CHOSEN IN THE MAIN MENU, BOTH IN EDITOR AND IN PLAYMODE.
- THE NEXT DIFFERENCE YOU’LL NOTICE IS THE PAUSE MENU (**ESC**), WHICH WILL DISPLAY A LOT MORE INFORMATION THAN BEFORE (IGNORE THE PING I WAS DOWNLOADING RE9):



- HOST PAUSE MENU



- PLAYER PAUSE MENU

MULTIPLAYER



- THE HOST OBVIOUSLY HAS ACCESS TO A WIDER RANGE OF TOOLS THAT ALLOW HIM TO CUSTOMIZE HOW OTHER PLAYERS WILL **SEE THE SESSION FROM THE SERVERS LIST**:

SERVER SETTINGS

SERVER NAME	Arylen's server.
SERVER DESCRIPTION	
MAX PLAYERS	10
PRIVACY	PUBLIC ▼

- THOSE ARE ALL PRETTY INTUITIVE, SO I'LL JUST POINT OUT A COUPLE OF THINGS:
- YOU CAN USE **RICH TEXT FORMATTING** IN BOTH THE SERVER NAME AND DESCRIPTION TO ADD COLORS, BOLD TEXT ETC... (, <color=#hex></color> ETC....)
- THE MAXIMUM NUMBER OF PLAYERS CURRENTLY SUPPORTED IS **24**, YOU CAN'T GO BEYOND THAT.
- THE MAX PLAYERS VALUE CAN'T BE SET TO A VALUE LOWER THAN THE CURRENT AMOUNT OF PLAYERS CONNECTED.
- **PUBLIC** PRIVACY = THE SERVER APPEARS IN THE GLOBAL LIST, ANYONE CAN JOIN IT
- **FRIENDS ONLY** PRIVACY = THE SERVER WON'T APPEAR IN THE GLOBAL LIST, BUT BOTH YOUR FRIENDS AND ANYONE WITH AN INVITE CAN JOIN
- **PRIVATE** PRIVACY = THE SERVER WON'T APPEAR IN THE GLOBAL LIST, NO ONE CAN JOIN WITHOUT AN INVITE FROM YOU, NOT EVEN YOUR FRIENDS.

MULTIPLAYER



- RIGHT ABOVE THE SERVER SETTINGS PANEL, BOTH THE HOST AND STANDARD PLAYERS CAN SEE THE **PLAYERS LIST**:

PLAYERS LIST

[OPEN MULTIPLAYER ADMINISTRATION MENU]

Aryien	HOST
Fili	250 ms

- THE **PLAYERS LIST** ALLOWS YOU TO CHECK ALL OF THE PLAYERS CURRENTLY CONNECTED TO THE SESSION, ALONG WITH THEIR COLOR, THEIR PING, THEIR ROLE AND A SERIES OF ACTIONS YOU CAN PERFORM ON THEM.
- HOVER ABOVE A PLAYER’S ROW TO ENABLE ALL THE POSSIBLE ACTIONS YOU CAN PERFORM ON THEM:

Fili

*

300 ms

- **HOST OPTIONS** (WHILE HOVERING)

Aryien

HOST

- **NORMAL PLAYER OPTIONS** (WHILE HOVERING)
- AGAIN, AS YOU CAN SEE, AN HOST HAS WAY MORE OPTIONS THAN A NORMAL PLAYER DOES. HOVERING ABOVE AN OPTION TELLS YOU EXACTLY WHAT IT DOES. BUT I’LL GO THROUGH A QUICK RUNDOWN ANYWAY:

MULTIPLAYER



• HOST OPTIONS

-  **ADMIN PLAYER** = MAKES THIS PLAYER AN ADMIN AND ALLOWS HIM TO PERFORM ALL HOST ACTIONS (EXCEPT ADMIN/UNADMIN) ON ALL OTHER NORMAL PLAYERS.
-  **KICK PLAYER** = FORCEFULLY DISCONNECTS THIS PLAYER FROM THE SESSION. THEY ARE ABLE TO REJOIN IMMEDIATELY.
-  **DEAFEN PLAYER** = FORCEFULLY EXCLUDES THIS PLAYER FROM VOICE AND TEXT CHATS.
-  **BAN PLAYER** = PERMANENTLY BANS THIS PLAYER FROM THIS AND ALL YOUR FUTURE SESSIONS.
-  **TELEPORT TO PLAYER** = TELEPORTS YOU TO THIS PLAYER.
-  **TELEPORT PLAYER HERE** = TELEPORTS THIS PLAYER TO YOU.
-  **REVOKE PLAYMODE EDIT MODE** = BLOCKS THIS PLAYER FROM USING **EDIT MODE** (**PAGE 18**) IN PLAYMODE.
-  **REVOKE EDITOR EDIT MODE** = BLOCKS THIS PLAYER FROM MODYFING THE LEVEL IN EDITOR MODE.

• STANDARD PLAYER OPTIONS

 **PERSONAL MUTE TEXT CHAT** = BLOCKS THIS PLAYER'S CHAT MESSAGES (JUST FOR YOU)

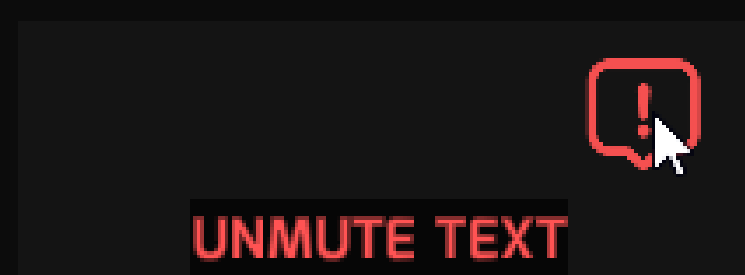
 **PERSONAL MUTE VOICE CHAT** = BLOCKS THIS PLAYER'S VOICE CHAT (JUST FOR YOU)

 **OPEN STEAM PROFILE** = VIEW THIS PLAYER'S STEAM PROFILE

- NOTE THAT OPTIONS THAT SUPPORT TOGGLE WILL GLOW RED WHEN THEY'VE BEEN ACTIVATED TO ALLOW YOU TO CHANGE YOUR MIND. TO MAKE YOU AN EXAMPLE, I'LL MUTE ARYIEN:



- AND NOW THAT HE'S MUTED, HIS MUTE OPTION IS RED, AND CLICKING IT AGAIN WILL DISABLE THE MUTE.



- THIS OF COURSE WORKS FOR ALL THE ACTIONS THAT SUPPORT TOGGLE AND UNTOGGLE.

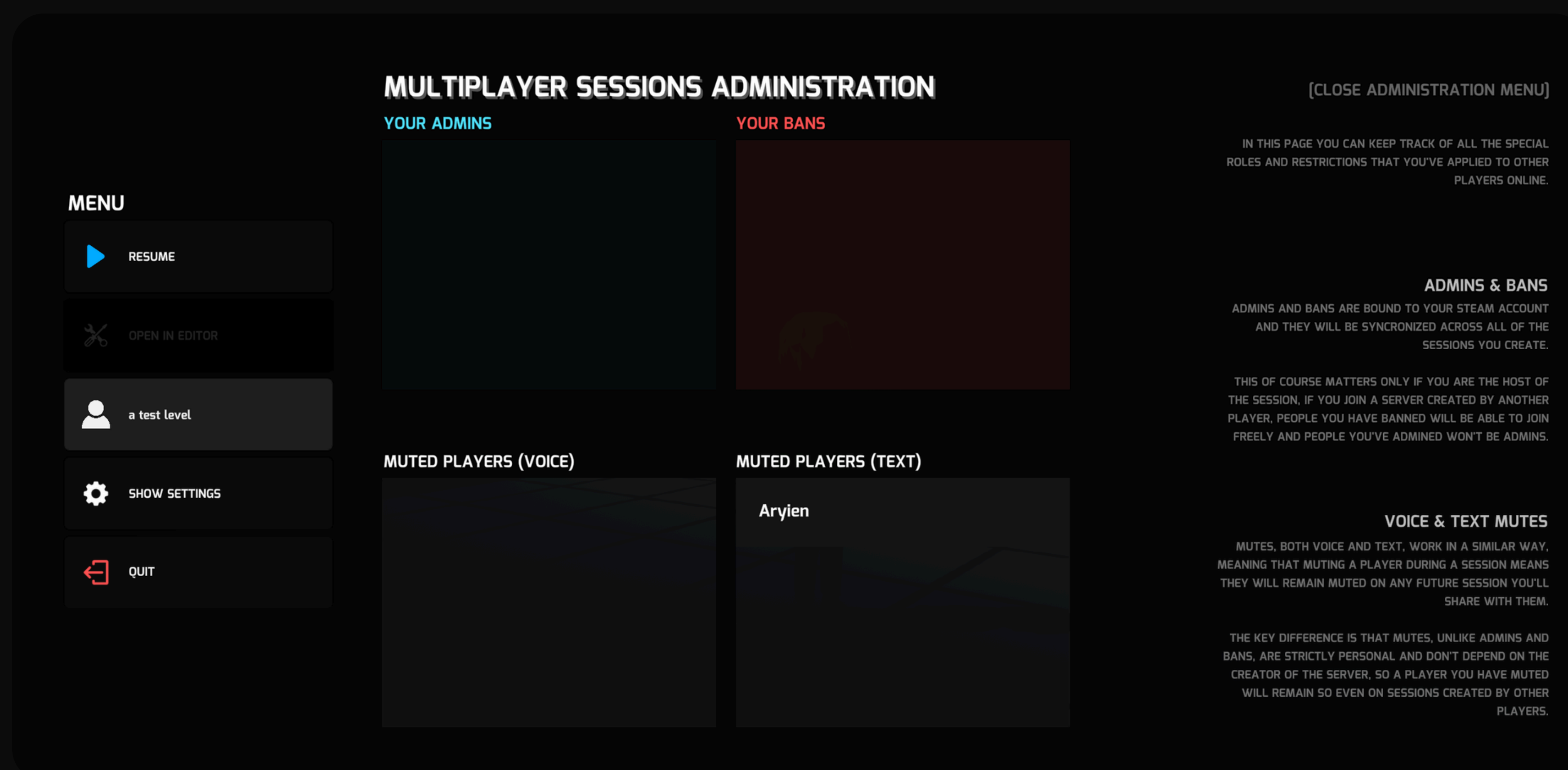
MULTIPLAYER



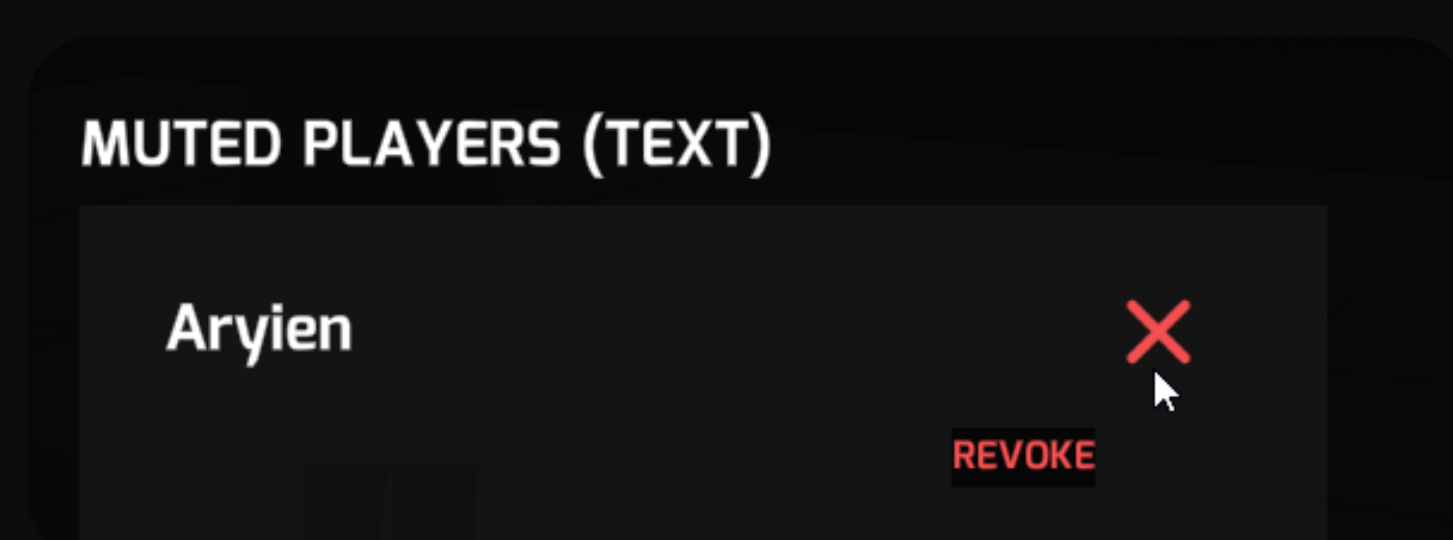
- THE VERY LAST THING TO COVER FOR THE BASIC MULTIPLAYER FEATURES IS THE **MULTIPLAYER SESSIONS ADMINISTRATION** MENU, WHICH CAN BE ACCESSED BY PRESSING THE BUTTON ABOVE THE PLAYERS LIST:



- THE MENU LOOKS SOMETHING LIKE THIS:



- YOU CAN FIND A QUICK EXPLANATION ON THE RIGHT SIDE OF YOUR SCREEN, BUT ESSENTIALLY, WHAT THIS MENU DOES IS ALLOW YOU TO MANAGE RESTRICTIONS OR ROLES YOU'VE GIVEN TO OTHER PLAYERS EVEN WHEN YOU'RE NOT IN THE SAME SESSION.
- AS YOU CAN SEE, IN MY **MUTED PLAYERS (TEXT)** SECTION THERE'S AN ENTRY FOR ARYIEN, SINCE I HAVE MUTED HIM EARLIER TO DEMONSTRATE YOU THE TOGGLE/UNTOGGLE. IF ARYIEN WASN'T ONLINE RIGHT NOW AND I WANTED TO UNMUTE HIM, I WOULD HAVE JUST NEEDED TO OPEN THIS MENU, HOVER ABOVE HIS NAME AND CLICK **REVOKE** TO CANCEL THE RESTRICTION I APPLIED TO HIM PREVIOUSLY:



MULTIPLAYER



- THE REASON THIS MENU EXISTS IS THAT WITHIN PROJECT LEVELS ALL OF YOUR **ADMINS**, **BANS** AND **VOICE/TEXT CHATS MUTES** ARE **SYNCRONIZED ACROSS ALL OF THE SESSIONS YOU WILL EVER CREATE**.
- IF I, FOR EXAMPLE, BANNED ARYIEN FROM THIS SESSION, HE WON'T BE ABLE TO JOIN A SESSION I CREATE A MONTH FROM NOW AS LONG AS I DONT REVOKE HIS BAN FROM THE **MULTIPLAYER SESSIONS ADMINISTRATION MENU**. SAME THING GOES FOR PEOPLE I ADMINED, MUTED, ETC...
- KEEP IN MIND THAT WHILE **MUTES** PERSIST IN ANY SESSION YOU JOIN, YOUR **BANS** AND **ADMINS** ARE ONLY ACTIVE IF YOU ARE THE HOST OF THE SESSION. THIS MEANS THAT YOU COULD ENCOUNTER PLAYERS YOU'VE BANNED IN OTHER PLAYERS SESSIONS AND PLAYERS YOU'VE ADMINED WON'T BE ADMINS.

SOME FINAL CONSIDERATIONS

- MULTIPLAYER IS EXTREMELY HARD, AND DOING IT REALTIME FOR AN HYBRID EDITOR/PLAYMODE SYSTEM IS EVEN HARDER, THERE ARE SURELY A LOT OF THINGS I MISSED AND PROBLEMS/BUGS LYING AROUND, SO PLEASE, WHEN YOU'LL ENCOUNTER THEM, HELP ME BY REPORTING THEM EITHER ON OUR DISCORD SERVER OR ON PROJECT LEVELS STEAM DISCUSSIONS.
- **THIS SHOULD GO WITHOUT SAYING, BUT BEHAVE YOURSELVES WITH OTHER PEOPLE WHILE IN MULTIPLAYER AND TRY TO HAVE A GOOD TIME TOGETHER.**
- **IF YOU STILL HAVE ANY QUESTIONS THAT AREN'T COVERED IN THE MANUAL, FEEL FREE TO CONTACT ME VIA DISCORD, STEAM, EMAIL, OR WHATEVER WORKS BEST. I'LL DO MY BEST TO ANSWER AS QUICKLY AS POSSIBLE. ALSO, EVEN THOUGH MY ENGLISH IS DECENT, IT'S STILL NOT MY FIRST LANGUAGE. SO IF YOU SPOT ANY GRAMMAR MISTAKES OR AWKWARDLY WRITTEN PARAGRAPHS, FEEL FREE TO POINT THEM OUT AND I'LL FIX THEM.**

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